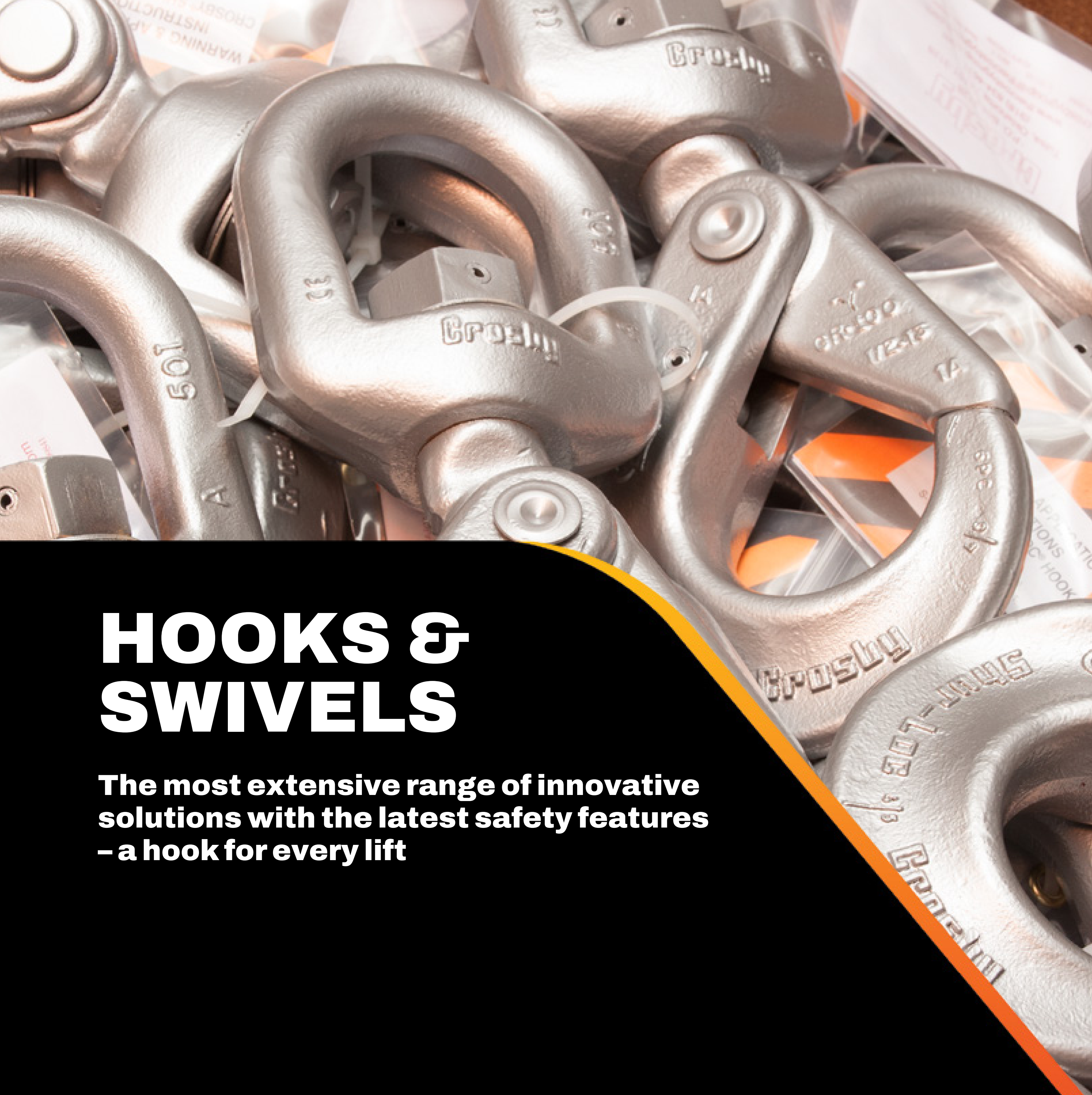
HOOKS

IMPORTED

CoreLifting.com



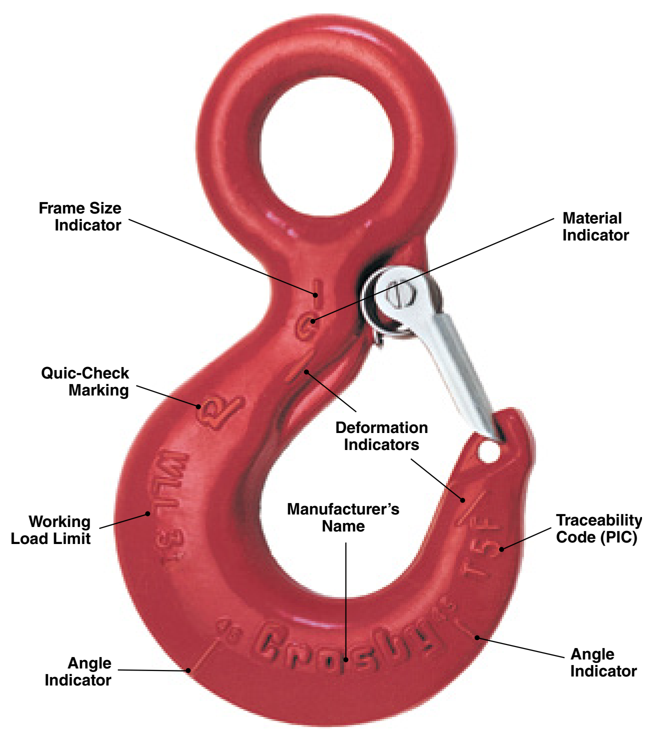
HOOKS & SWIVELS

The most extensive range of innovative solutions with the latest safety features – a hook for every lift

kitocrosby.com

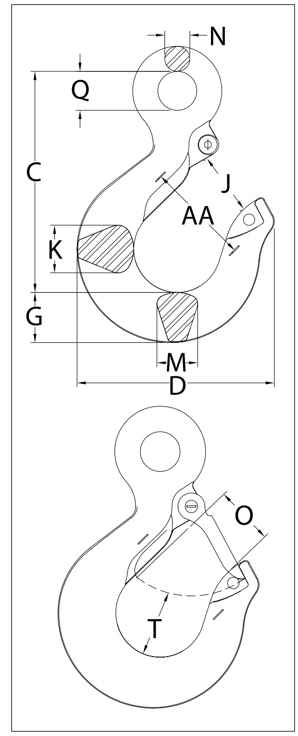
- **Application information:** Application and warning information is available for Crosby hooks. The Crosby Warning System is designed to attract the attention of the user, clearly inform the user of the factors involved in the task, and provide the user with proper application procedures. Each Crosby hook is tagged with appropriate application and warning information, thus ensuring that the information is available at the point of application.
- **Charpy impact properties:** Crosby's Quenched & Tempered® hooks have enhanced impact properties for greater toughness at all temperatures. Crosby can provide typical Charpy impact properties on selected sizes upon special request at the time of order.
- **Fatigue properties:** Typical fatigue properties are available for selected sizes. In addition, these properties will be provided upon special request for other sizes.
- **Ductility properties:** Crosby provides results of actual test values for ductility of the material. These results are measured by reduction of area and elongation. This is done for each production lot and is traceable by the Product Identification Code (PIC).
- **Tensile strengths:** Crosby provides hardness, tensile, and yield strength for each production lot of hooks, traceable by the PIC.
- **Material analysis:** Crosby can provide certified material (mill) analysis for each production lot, traceable by the PIC. Crosby, through its own laboratory, verifies the analysis of each heat of steel. Crosby purchases only special bar forging quality steel with specific cleanliness requirements and guaranteed hardenability.
- **Field inspection:** Written instructions for visual, magnaflux, and dye penetrant inspection of hooks are available from Crosby. In addition, acceptance criteria and repair procedures for hooks are available.
- **Proof testing:** If requested at the time of order, hooks can be furnished proof tested with certification. All SHUR-LOC® hooks (clevis and eye styles) are 100% proof tested with certificates.
- **Magnetic particle certification:** If requested at the time of order, hooks can be magnetic particle inspected with certification.
- **World-class certification:** Certification to world standards can be furnished upon request at the time of order. Specific standards include American Bureau of Shipping, Lloyds Register of Shipping, Det Norske Veritas, American Petroleum Institute, RINA, Nuclear Regulatory Commission, and other worldwide standards.
- **Bronze hooks:** Crosby provides bronze shank hooks for non-sparking applications.
- **QUIC-CHECK®:** Hooks incorporate markings forged into the product which address two QUIC-CHECK features:
  - Deformation Indicators: Two strategically placed marks, one just below the shank or eye and the other on the hook tip, which allows for a QUIC-CHECK measurement to determine if the throat opening has changed, thus indicating abuse or overload.
  - Angle Indicators: Indicates the maximum included angle which is allowed between two sling legs in the hook. These indicators also provide the opportunity to approximate other included angles between two sling legs.
- **McKissick® Split-Nut® Hook Retention System:** Shank hooks on crane blocks must be inspected in accordance with applicable ASME B30, CSA Z150, and other crane standards. These standards mandate the crane hook to be inspected for surface indications, damage and corrosion which could compromise the integrity of the crane block. Because of the type of environment in which these hooks are required to perform, the removal of corroded nuts from the threads can become a problem during inspections. The innovative, patented system is available on Crosby shank hooks. With four easy steps, the hook can be disassembled, inspected and put back into service in a fraction of the time of a conventional threaded nut.

Hook Identification



L-1327

- For use with wire rope. Suitable for use with Grade 100 and Grade 80 chain. Working load limit needs to be de-rated to achieve a 5:1 design factor.
- Forged alloy steel, Quenched & Tempered.
- Each hook has a Product Identification Code (PIC) for material traceability, along with the size and the name Crosby.
- 25% stronger than Grade 80.
- Eye Sling Hooks incorporate QUIC-CHECK® deformation and angle indicators. (For detailed information, see the Crosby Value Added page at the beginning of this section.)
- When secured with the proper cotter pin through the hole in the tip of hook, meets the intent of OSHA Rule 1926.1431(g) and 1926.1501(g) for personnel lifting.
- Individually Proof Tested to 2.5 times the Working Load Limit with certification.
- Fatigue rated to 20,000 cycles at 1.5 times the Working Load Limit.



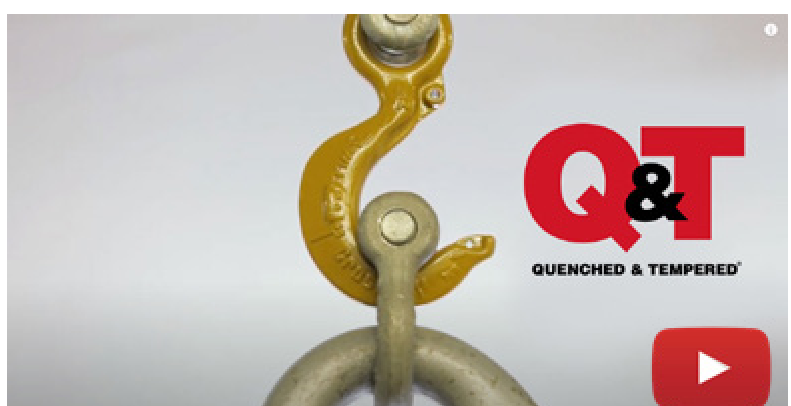
Crosby 8/10™ Fatigue Rated QUIC-CHECK® Q&T

APPLICATION AND WARNING INFORMATION
SECTION 17

L-1327 Eye Sling Hook

Grade 100 Alloy Chain Size (in)	Grade 100 Alloy Chain Size (mm)	Working Load Limit (lb)	Hook ID Code	Stock No.	Weight Each (lb)	Dimensions (in) C	D	G	J	K	M	N	O	Q	T	AA*	Replacement Latch Stock No.
-	6	3200	DA	1025860	.50	3.34	2.86	.73	.90	.63	.63	.36	.89	.75	.87	1.50	1096325
1/4-5/16	7 - 8	5700	HA	1025869	1.3	4.21	3.90	1.03	1.18	.75	.75	.50	1.15	.75	1.16	2.00	1096468
3/8	10	8800	IA	1025878	2.3	4.99	4.34	1.19	1.53	1.19	1.00	.56	1.40	.94	1.23	2.50	1096515
1/2	13	15000	JA	1025887	4.5	6.36	5.67	1.44	1.78	1.37	1.17	.72	1.67	1.12	1.88	3.00	1096562
5/8	16	22600	KA	1025896	8.4	7.43	6.78	1.88	2.38	1.66	1.44	.88	2.08	1.31	2.03	4.00	1096609
3/4	18-20	35300	KA	1025915	15.0	9.07	7.45	2.25	2.38	1.88	1.63	1.11	2.08	2.44	2.47	4.00	1096609
7/8	22-23	44100	LA	1025924	20.7	10.08	8.30	2.59	2.50	2.19	1.94	1.27	2.27	2.84	2.62	4.00	1096657
1	26	59700	NA	1025933	39.5	12.82	10.30	3.00	3.30	2.69	2.38	1.56	3.02	3.50	2.83	5.00	1096704
1 1/4	32	90400	PA	1025942	105.0	18.19	14.06	4.56	4.25	3.75	3.19	2.00	3.00	4.50	3.88	7.00	1093717

4:1 Design Factor. *Deformation indicators.



Hook Ductility: Compare an 'as forged' hook to the superior quality of a Crosby® Quenched & Tempered® hook. Watch video at **kitocrosby.com/QT**

Crosby

S-319/S-319N

- The most complete line of shank marked hooks. Available 3/4 to 300 metric tons.
- Hook Identification code marked into each hook.
- All carbon and alloy hooks are quenched and tempered.
- Quenched & Tempered.
- Available in carbon steel, alloy steel, and bronze.
- Proper design, careful forging, and precision controlled quench and tempering give maximum strength without excessive weight and bulk.
- Every Crosby Shank Hook has a pre-drilled cam which can be equipped with a latch. Simply purchase the Crosby latch assemblies. Even years after the purchase of the original hook, latch assemblies can be added.
- Type Approval Certification in accordance with ABS 2016 Steel Vessels and ABS Guide for Certification on Cranes available. Certificates available when requested at time of order and may include additional charges.
- Patented McKissick Split-Nut retention system available.
- Hooks incorporate QUIC-CHECK® deformation and angle indicators. (For detailed information, see the Crosby Value Added page at the beginning of this section.)
- Chemical analysis and tensile tests performed on each PIC to verify chemistry and mechanical properties.
- See Section 17 for guidance on minimum thread size after machining.

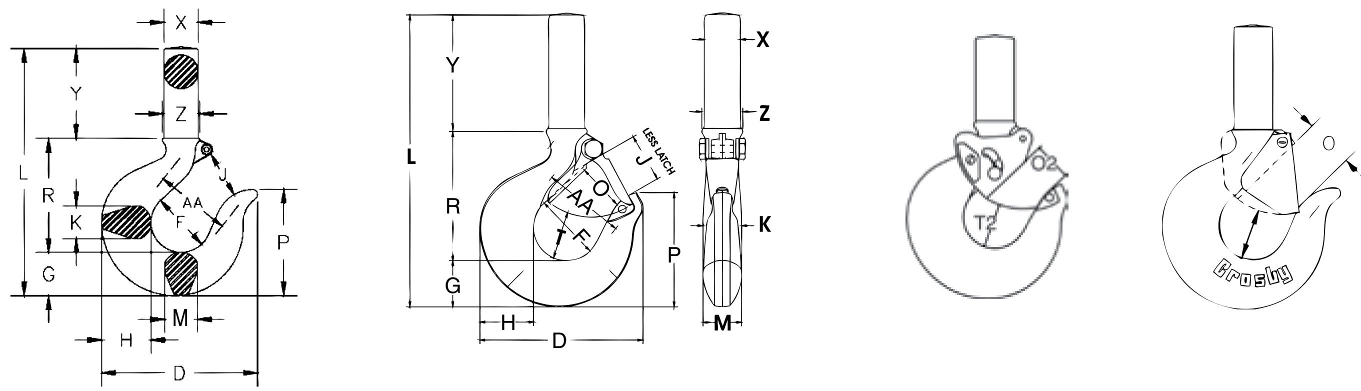
S-319 / S-319N Shank Hook

QUIC-CHECK®

APPLICATION AND WARNING INFORMATION
SECTION 17

Working Load Limit (t)			Hook ID Code	Shank Hooks Stock No.			Shank Length ‡	Weight Each (lb)	Optional Latch Kits		
Carbon	Alloy	Bronze		Carbon S-319C S-319CN	Alloy S-319A S-319AN	Bronze S-319BN			S-4320 Stock No.	PL Stock No.	SS-4055 Stock No.
0.75	1	.5	†D	1028505	1028701	1028900	Std.	.50	1096325	-	-
1	1.5	.6	†F	1028514	1028710	1028909	Std.	.75	1096374	-	-
1.5	2	1	†G	1028523	1028723	1028918	Std.	1.00	1096421	-	-
2	3	1.4	†H	1028532	1028732	1028927	Std.	1.82	1096468	-	-
3	5	2	†I	1028541	1028741	1028936	Std.	3.69	1096515	1092000	-
5	7	3.5	†J	1028550	1028750	1028945	Std.	7.25	1096562	1092001	-
7.5	11	5	†K	1028563	1028765	1028954	Std.	13.4	1096609	1092002	-
10	15	6.5	†L	1028590	1028792	1028981	Std.	21.9	1096657	1092003	-
15	22	10	†N	1028599	1028801	1028990	Std.	38.4	1096704	1092004	-
20	30	-	O	1024386	1024803	-	Std.	72	-	1093716	1090161
20	30	-	O	1024402	1024821	-	Long	85	-	1093716	1090161
25	37	-	P	1024420	1024849	-	Std.	134	-	1093717	1090189
25	37	-	P	1024448	1024867	-	Long	172	-	1093717	1090189
30	45	-	S	1024466	1024885	-	Std.	182	-	1093718	1090189
30	45	-	S	1024484	1024901	-	Long	214	-	1093718	1090189
40	60	-	T	1024509	1024929	-	Std.	268	-	1093719	1090205
40	60	-	T	1024545	1024965	-	Long	312	-	1093719	1090205
50	75	-	U	1024563	1024983	-	Std.	390	-	1093720	-
50	75	-	U	1024581	1025009	-	Long	426	-	1093720	-
-	100	-	W	-	1025027	-	Std.	610	-	1093721	-
-	100	-	W	-	1025045	-	Long	675	-	1093721	-
-	150	-	X	-	1025063	-	Std.	735	-	1093721	-
-	200	-	Y	-	1025081	-	Std.	1020	-	1093723	-

Maximum allowable Proof Load is 2 Times Working Load Limit. All carbon hooks designed with a 5:1 design factor. All alloy hooks 1 through 22t designed with a 4.5:1 design factor. All alloy hooks 30t and larger designed with a 4:1 design factor. All bronze hooks designed with a 4:1 design factor. †New 319N style hook. ‡See column "Y" on following page for actual length.



S-319 / S-319N Shank Hook

Hook ID Code	Dimensions (in)																
	D	F	G	H	J	K	L	M	O	O2 ††	P	R	T	T2 ††	X	Y	Z	AA*
†D	2.86	1.25	.73	.81	.93	.63	5.14	.63	.93 †	-	1.96	2.35	.97	-	.59	2.06	.69	1.50
†F	3.16	1.38	.84	.94	.97	.71	5.68	.71	.97 †	-	2.22	2.59	.97	-	.76	2.25	.78	2.00
†G	3.59	1.50	1.00	1.16	1.06	.88	6.35	.88	1.06 †	-	2.44	2.76	1.03	-	.88	2.59	.88	2.00
†H	4.00	1.62	1.14	1.31	1.19	.94	7.14	.94	1.16 †	-	2.78	3.16	1.16	-	.88	2.84	1.00	2.00
†I	4.84	2.00	1.44	1.63	1.50	1.31	8.63	1.13	1.36 †	1.00	3.47	3.85	1.53	1.50	1.16	3.44	1.25	2.50
†J	6.28	2.50	1.82	2.06	1.78	1.66	10.43	1.44	1.61 †	1.31	4.59	4.77	1.96	1.88	1.41	3.84	1.56	3.00
†K	7.54	3.00	2.26	2.63	2.41	1.88	12.52	1.63	2.08 †	1.81	5.25	5.88	2.47	2.25	1.81	4.38	1.94	4.00
†L	8.34	3.25	2.60	2.94	2.62	2.19	16.10	1.94	2.27 †	2.00	5.96	6.37	2.62	2.31	2.00	7.00	2.19	4.00
†N	10.34	4.25	3.01	3.50	3.41	2.69	18.15	2.38	3.02 †	2.75	6.88	8.14	2.83	2.56	2.56	7.00	2.63	5.00
O	13.62	5.00	3.62	4.62	4.00	3.00	23.09	3.00	3.25	-	8.78	9.44	3.44	-	3.12	10.00	3.12	6.50
O	13.62	5.00	3.62	4.62	4.00	3.00	31.09	3.00	3.25	-	8.78	9.44	3.44	-	3.12	18.00	3.12	6.50
P	14.06	5.38	4.56	5.00	4.25	3.62	32.12	3.00	3.00	-	11.31	12.50	3.88	-	4.00	15.00	4.00	7.00
P	14.06	5.38	4.56	5.00	4.25	3.62	41.12	3.00	3.00	-	11.31	12.50	3.88	-	4.00	24.00	4.00	7.00
S	15.44	6.00	5.06	5.50	4.75	3.72	34.12	3.25	3.38	-	12.56	14.00	4.75	-	4.19	15.00	4.19	8.00
S	15.44	6.00	5.06	5.50	4.75	3.72	43.12	3.25	3.38	-	12.56	14.00	4.75	-	4.19	24.00	4.19	8.00
T	18.50	7.00	6.00	6.50	5.75	4.44	36.06	3.91	4.12	-	14.75	15.56	5.69	-	4.50	14.50	4.50	10.00
T	18.50	7.00	6.00	6.50	5.75	4.44	47.56	3.91	4.12	-	14.75	15.56	5.69	-	4.50	26.00	4.50	10.00
U	20.62	7.75	6.69	7.25	6.50	5.25	41.16	4.25	4.88	-	16.53	19.38	6.00	-	5.00	15.00	5.00	11.50
U	20.62	7.75	6.69	7.25	6.50	5.25	49.16	4.25	4.88	-	16.53	19.38	6.00	-	5.00	23.00	5.00	11.50
W	23.00	6.81	8.59	9.88	5.88	5.50	42.12	5.50	4.50	-	17.25	18.41	7.00	-	7.00	15.00	7.00	12.00
W	23.00	6.81	8.59	9.88	5.88	5.50	48.12	5.50	4.50	-	17.25	18.41	7.00	-	7.00	21.00	7.00	12.00
X	24.38	6.75	9.12	10.94	6.00	6.00	45.75	6.00	4.50	-	18.00	18.38	7.00	-	7.25	18.00	7.25	13.00
Y	26.69	7.50	9.75	11.81	6.60	7.00	50.50	7.00	5.00	-	19.75	20.50	8.00	-	8.00	20.00	8.00	13.00

Rough as-forged dimension. Shank will not machine to this dimension. Please refer to the warnings & applications section for recommended shank diameter when machining. *Deformation indicators. †3/4t carbon through 22t alloy dimensions shown are for S-4320 Latch Kits. Dimensions for "O" frame size and larger are for PL Latch Kits. ††Dimensions are for PL-N latch kits. For the purpose of calculating D/d ratio, utilize dimension M.

L-320CN
Frame Size D-N

- Available in carbon steel and alloy steel.
- Eye hooks are load rated (marked with the Working Load Limit).
- Fatigue rated to 20,000 cycles at 1.5 times the Working Load Limit.
- Chemical analysis and tensile tests performed on each PIC to verify chemistry and mechanical properties.
- Hooks incorporate QUIC-CHECK® deformation and angle indicators. (For detailed information, see the Crosby Value Added page at the beginning of this section.)

L-320C
Frame Size O-T

APPLICATION AND WARNING INFORMATION
SECTION 17

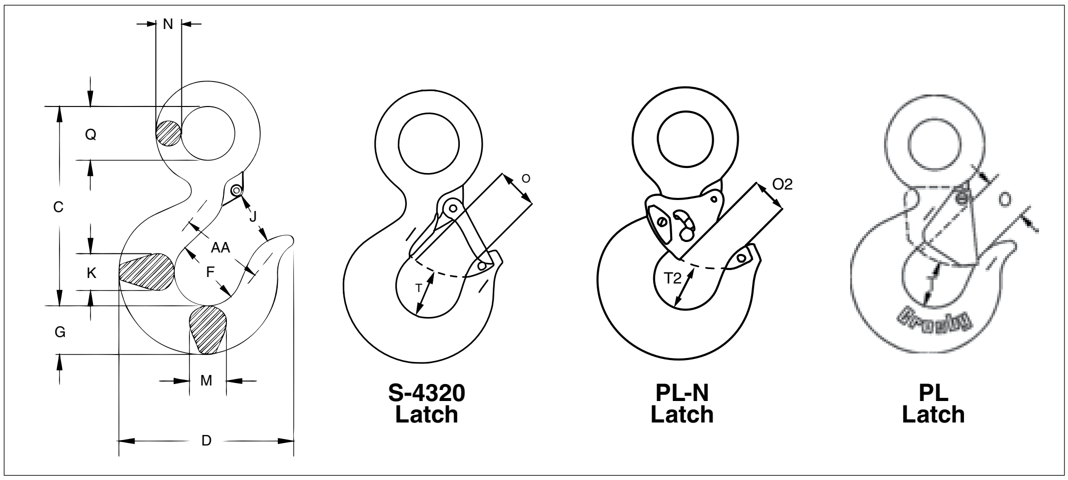
L-320N / L-320 Eye Hooks

Working Load Limit (t)			Eye Hook Stock No.				Replacement Latch Kit Stock No.		Optional Latch Kit Stock No.
Carbon	Alloy	Hook ID Code	Carbon L-320C L-320CN S.C.	Carbon GL-320CN Galv.	Alloy L-320A L-320AN S.C.	Weight Each (lb)	S-4320	SS-4055	PL
0.75	1	†D	1022205	1022208	1022380	.61	1096325	-	-
1	1.5	†F	1022216	1022219	1022391	.89	1096374	-	-
1.5	2	†G	1022227	1022230	1022402	1.44	1096421	-	-
2	3	†H	1022238	1022241	1022413	2.07	1096468	-	-
3	5	†I	1022246	1022249	1022424	4.30	1096515	-	1092000
5	7	†J	1022260	1022262	1022435	8.30	1096562	-	1092001
7.5	11	†K	1022271	1022274	1022446	15.00	1096609	-	1092002
10	15	†L	1022282	-	1022457	20.77	1096657	-	1092003
15	22	†N	1022293	-	1022468	39.50	1096704	-	1092004
20	30	O	1022302	-	1022477	60.00	-	1090161	1093716
25	37	P	1023306	-	1023565	105.00	-	1090189	1093717
30	45	S	1023324	-	1023583	148.00	-	1090189	1093718
40	60	T	1023342	-	1023609	228.00	-	1090205	1093719

All carbon hooks have a 5:1 Design Factor. Alloy eye hooks 1t through 22t have a 5:1 Design Factor. Alloy eye hooks 30t through 60t have a 4.5:1 Design Factor. For 3/4t carbon through 22t alloy eye hooks, Proof Load is 2.5 times Working Load Limit. For 20t carbon through 60t alloy eye hooks, Proof Load is 2 times Working Load Limit.

L-320AN
Frame Size D-N

L-320AN
Frame Size O-T



APPLICATION AND WARNING INFORMATION
SECTION 17

L-320N / L-320 Eye Hooks

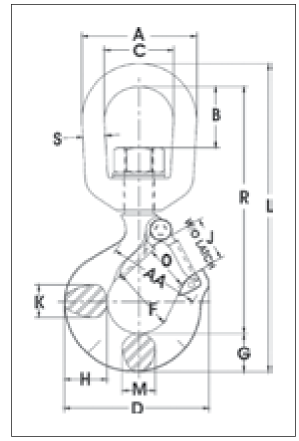
Hook ID Code*	Dimensions (in)												
	C	D	F	G	J	K	M	N	O †	O2 ††	Q	T †	T2 ††	AA**
†D	3.34	2.86	1.26	0.73	.93	.63	.63	.36	.87	-	.75	.85	-	1.50
†F	3.81	3.15	1.38	0.84	.97	.71	.71	.42	.91	-	.91	.96	-	2.00
†G	4.14	3.55	1.5	1	1	.88	.88	.5	.91	-	1.13	1.02	-	2.00
†H	4.69	3.99	1.62	1.13	1.19	.94	.94	.56	1.09	-	1.29	1.14	-	2.00
†I	5.77	4.88	2	1.44	1.53	1.46	1.33	.72	1.37	1	1.56	1.41	1.50	2.50
†J	7.37	6.26	2.5	1.81	1.75	1.66	1.53	.91	1.61	1.31	2.03	1.93	1.88	3.00
†K	9.07	8.54	3	2.25	2.41	1.88	1.63	1.11	2.21	1.81	2.44	2.32	2.25	4.00
†L	10.08	8.39	3.26	2.59	2.62	2.19	1.94	1.27	2.31	2	2.84	2.42	2.31	4.00
†N	12.53	1.38	4.26	3	3.41	2.69	2.38	1.57	3.18	2.75	3.5	2.78	2.56	5.00
O	14.06	13.63	5	3.66	4	3	3	1.75	3.73	-	3.5	3.51	-	6.50
P	18.19	14.07	5.38	4.63	4.17	4	3.19	2	3.97	-	4.5	3.92	-	7.00
S	20.12	15.44	6	5.12	4.54	4.5	3.53	2.19	3.95	-	4.94	4.77	-	8.00
T	23.72	18.51	7	6.06	5.42	5.5	3.91	2.53	5.26	-	5.69	5.72	-	10.00

*Deformation indicators. †3/4t carbon though 22t alloy dimensions shown are for S-4320 Latch Kits. Dimensions for "O" frame size and larger are for PL Latch Kits.
††Dimensions are for PL-N latch kits.

L-322CN / L-322AN

- Forged, Quenched & Tempered.
- Suitable for positioning of the hook before the load is lifted.
- Swivel hooks are load rated.
- Proper design, careful forging, and precision controlled quench and tempering gives maximum strength without excessive weight and bulk.
- Low profile hook tip designed to utilize Crosby S-4320 or PL-N latch kit.
- Hooks incorporate QUIC-CHECK® deformation and angle indicators. (For detailed information, see the Crosby Value Added page at the beginning of this section.)

Use in corrosive environment requires shank and nut inspection in accordance with ASME B30.10-1.10.4(b)(5)(c).



APPLICATION AND WARNING INFORMATION
SECTION 17

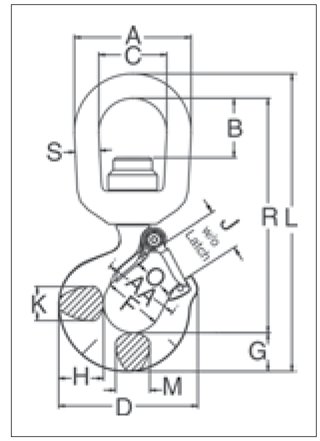
L-322CN / L-322AN Swivel Hooks with Latch

Working Load Limit (t)		Hook ID Code*	L-322CN Stock No.	L-322AN Stock No.	Weight Each (lb)	Dimensions (in)															Rep. Latch Stock No.
Carbon	Alloy					A	B	C	D	F	G	H	J	K	L	M	O †	R	S	AA*	
0.75	1	D	1048603	1048807	.75	2.00	.82	1.25	2.86	1.25	.73	.81	.93	.63	5.66	.63	.89	4.55	.38	1.50	1096325
1	1.5	F	1048612	1048816	1.25	2.50	1.31	1.50	3.15	1.38	.84	.94	.97	.71	6.71	.71	.91	5.37	.50	2.00	1096374
1.5	2	G	1048621	1048825	2.25	3.00	1.50	1.75	3.59	1.50	1.00	1.16	1.06	.88	7.75	.88	1.00	6.12	.63	2.00	1096421
2	3	H	1048630	1048834	2.30	3.00	1.50	1.75	4.00	1.62	1.13	1.31	1.19	.94	8.25	.94	1.09	6.50	.63	2.00	1096468
3	5	I	1048639	1048840	4.96	3.50	1.64	2.00	4.84	2.00	1.44	1.63	1.50	1.31	9.69	1.13	1.36	7.50	.75	2.50	1096515
5	7	J	1048648	1048859	10.29	4.56	2.29	2.50	6.28	2.50	1.81	2.06	1.78	1.66	12.47	1.44	1.61	9.63	1.00	3.00	1096562
7.5	11	K	1048657	1048868	19.40	5.00	2.44	2.75	7.54	3.00	2.25	2.63	2.41	1.88	14.75	1.63	2.08	11.37	1.13	4.00	1096609
10	15	L	1048666	1048880	23.25	5.62	2.48	3.12	8.34	3.25	2.59	2.94	2.62	2.19	16.40	1.94	2.27	12.25	1.25	4.00	1096657
15	22	N	1048675	1048889	47.00	7.10	3.76	4.10	10.34	4.25	3.00	3.50	3.41	2.69	21.34	2.38	3.02	16.71	1.50	5.00	1096704
-	30	O	-	1048898	70.50	7.10	3.76	4.10	13.62	5.00	3.61	4.63	4.00	3.00	23.25	3.00	3.62	18.01	1.50	6.50	1090161

All carbon swivel hooks have a 5:1 Design Factor and Proof Load is 2 times the Working Load Limit. Alloy swivel hooks 1t through 22t have a 4.5:1 Design Factor and Proof Load is 2.5 times the Working Load Limit. Alloy swivel hooks of 30t capacity have a 4:1 Design Factor and Proof Load is 2 times the Working Load Limit. *Deformation indicators †Dimensions for hooks 3/4t carbon through 22t alloy are for S-4320 latch kits. Dimensions for hooks 30t alloy are for 4055 latch kit.

L-3322B

- Bearing design allows hook to rotate freely under load.
- Capacities ranging from 2 through 15 metric tons.
- Forged, Quenched & Tempered.
- Low profile hook tip designed to utilize Crosby S-4320 or PL-N latch kit.
- L-3322 hooks incorporate QUIC-CHECK® deformation and angle indicators. (For detailed information, see the Crosby Value Added page at the beginning of this section.)



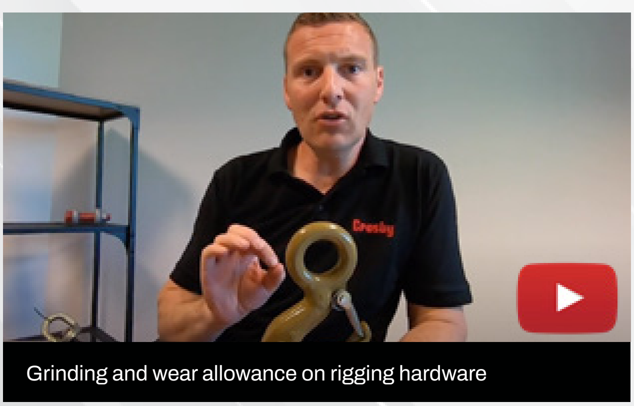
QUIC-CHECK®

APPLICATION AND WARNING INFORMATION
SECTION 17

L-3322B Swivel Hooks with Bearing

Working Load Limit (t)	Hook ID Code*	Stock No.	Weight Each (lb)	Dimensions (in)															Rep. Latch Stock No.
				A	B	C	D	F	G	H	J	K	L	M	O	R	S	AA*	
2	GA	1028609	2.5	3.00	1.50	1.75	3.59	1.50	1.00	1.16	1.06	.88	7.64	.88	1.00	6.01	.63	2.00	1096421
3	HA	1028618	3.8	3.50	1.56	2.00	4.00	1.62	1.13	1.31	1.19	.94	8.60	.94	1.09	6.72	.75	2.00	1096468
5	IA	1028627	7.0	4.00	1.56	2.25	4.84	2.00	1.44	1.63	1.50	1.31	10.32	1.13	1.36	8.00	.88	2.50	1096515
7	JA	1028636	14.0	5.00	1.94	2.75	6.27	2.50	1.81	2.06	1.78	1.66	12.84	1.44	1.61	9.90	1.13	3.00	1096562
11	KA	1028645	22.3	5.62	2.05	3.12	7.54	3.00	2.25	2.63	2.41	1.88	15.24	1.63	2.08	11.74	1.25	4.00	1096609
15	LA	1028654	36.0	7.12	3.62	4.10	8.33	3.25	2.59	2.94	2.62	2.19	18.64	1.94	2.27	14.41	1.50	4.00	1096657

4.5:1 Design Factor. Maximum allowable proof load is 2.5 times Working Load Limit. *Deformation indicators.

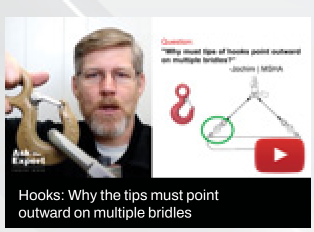

Grinding and wear allowance on rigging hardware

Hooks & Swivels Training Videos

Our experts answer some of your most common questions about hooks and swivels.

Watch these and all of our training videos at **kitocrosby.com/crosby-yt**


Crosby Shurloc® Hook inspection requirements

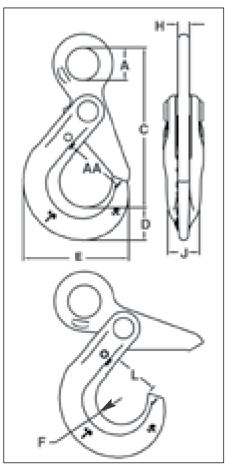

Hooks: Why the tips must point outward on multiple bridles

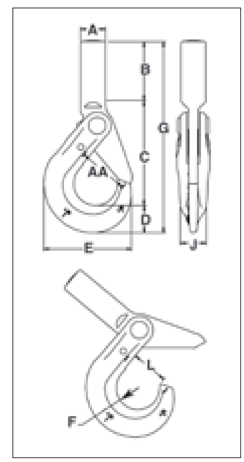
Crosby GUNNEBO® Industries

S-1316

- All SHUR-LOC® hooks have the following features:
  - Forged alloy steel, Quenched & Tempered.
  - Recessed trigger design is flush with the hook body, protecting the trigger from potential damage.
  - Easy to operate with enlarged thumb access.
  - Positive lock latch is self-locking when the hook is loaded.
  - The SHUR-LOC® hook, if properly installed and locked, can be used for personnel lifting applications and meets the intent of OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B).
  - Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
  - Contact Engineered Solutions for additional threading or Split-Nut options at thecrosbygroup.com/engineeredsolutions.
- Eye Style incorporates these added features:
  - Individually Proof Tested to 2-1/2 times the chain Working Load Limit with certification.
  - S-1316 meets the performance requirements of EN1677-3.
  - Suitable for use with Grade 100 and Grade 80 chain.
  - Designed with 'engineered flat' to connect to S-1325 chain coupler.

S-1318A





Crosby 8/10™

APPLICATION AND WARNING INFORMATION
SECTION 17

S-1316 SHUR-LOC® Eye Hook with Positive Locking Latch

Chain Size				Grade 100 Alloy Chain Working Load Limit (lb) 4:1	Working Load Limit (lb) 5:1	Weight Each (lb)	Dimensions (in)										
(in)	(mm)	Stock No.	Frame code				A	C	D	E	F	H	J	L	AA*	Replacement Trigger Kit Stock No.
-	6	1022896	D	3200	2560	.85	.78	3.95	.79	2.60	.67	.31	.63	1.14	1.50	6603010
1/4-5/16	7-8	1022914	G	5700	4560	1.80	1.08	5.31	1.10	3.50	.87	.39	.81	1.48	2.00	6603011
3/8	10	1022923	H	8800	7040	3.40	1.30	6.57	1.17	4.39	1.10	.51	.94	1.83	2.50	6603012
1/2	13	1022932	I	15000	12000	6.00	1.65	8.23	1.67	5.45	1.26	.67	1.16	2.22	3.00	6603013
5/8	16	1022941	J	22600	18000	15.1	2.20	10.06	2.04	6.56	1.50	.87	1.50	2.65	3.50	6603014
3/4	18-20	1022952	-	35300	28240	19.0	2.60	10.77	2.22	7.76	2.01	.87	2.03	3.52	5.00	6603015
7/8	22	1022943	-	42700	34160	28.0	2.87	12.49	2.45	8.75	2.27	.98	2.20	3.83	6.00	6603008
1	26	1022944	-	59700	47760	49.5	3.15	14.60	3.21	9.87	2.46	1.26	2.68	4.09	6.50	6603017

*Deformation indicators.

S-1318A SHUR-LOC® Shank Hook

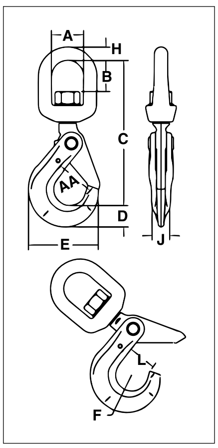
Chain Size				Grade 100 Alloy Chain Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)									
(in)	(mm)	Stock No.	Frame code			A†	B	C	D	E	F	G	J	L	AA*	Replacement Trigger Kit Stock No.
-	6	1098200	D	3200	1.00	.79	2.16	3.31	.79	2.60	.67	6.26	.63	1.16	1.50	6603010
1/4-5/16	7-8	1098209	G	5700	1.99	1.00	2.40	4.16	1.10	3.51	.87	7.66	.81	1.48	2.00	6603011
3/8	10	1098218	H	8800	3.56	1.14	2.95	5.14	1.17	4.39	1.10	9.26	.94	1.83	2.50	6603012
1/2	13	1098227	I	15000	7.00	1.34	3.35	6.31	1.67	5.49	1.26	11.33	1.16	2.22	3.00	6603013

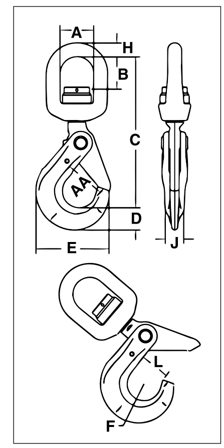
4:1 Design Factor based on Grade 100 chain. *Deformation indicators. †Dimension before machining (as forged).

S-1326

S-13326

- The S-1326 hook is a positioning device and is not intended to rotate under load. For swivel hook designed to rotate under load, use the S-13326.
- S-13326 Swivel Hook utilizes anti-friction bearing design which allows hook to rotate freely under load.
- Rated for both wire rope and for use with Grade 80/100 chain.
- Forged alloy steel, Quenched & Tempered.
- Individually Proof Tested at 2-1/2 times the chain Working Load Limit with certification.
- Recessed trigger design is flush with the hook body, protecting the trigger from potential damage.
- Easy to operate with enlarged thumb access.
- Positive lock latch is self-locking when hook is loaded.
- Trigger repair kit available (S-4316). Consists of spring, roll pin, and trigger.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- The SHUR-LOC® Hook, if properly installed and locked, can be used for personnel lifting applications and meets the intent of OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B).





Crosby 8/10™ Fatigue Rated Q&T CE

APPLICATION AND WARNING INFORMATION
SECTION 17

S-1326 SHUR-LOC® Swivel Hooks
Suitable for positioning before lifting.

Chain Size (in)	Chain Size (mm)	Frame code	Grade 100 Alloy Chain Working Load Limit (lb) 4:1 Design Factor	Working Load Limit (lb) 5:1 Design Factor	Stock No.	Weight Each (lb)	A	B	C	D	E	F	H	J	L	AA*	Replacement Trigger Kit Stock No.
-	6	D	3200	2560	1004304	1.26	1.50	1.32	6.13	.79	2.60	.67	.50	.63	1.13	1.50	6603010
1/4 - 5/16	7-8	G	5700	4560	1004313	2.62	1.75	1.59	7.60	1.10	3.50	.87	.63	.81	1.38	2.00	6603011
3/8	10	H	8800	7040	1004322	4.70	2.00	1.73	8.83	1.17	4.39	1.10	.75	.94	1.75	2.50	6603012
1/2	13	I	15000	12000	1004331	8.64	2.50	2.38	11.20	1.67	5.45	1.26	1.00	1.16	2.11	3.00	6603013
5/8	16	-	22600	18000	1004340	17.00	2.75	2.70	12.90	2.05	6.56	1.50	1.13	1.50	2.49	3.50	6603014
3/4	18 - 20	-	35300	28240	1004349	24.00	2.83	2.52	14.10	2.22	7.76	2.01	1.10	2.03	3.52	5.00	6603015
7/8	22	-	42700	34160	1004358	29.00	3.44	3.19	16.40	2.45	8.75	2.26	1.30	2.20	3.83	6.00	6603008

Dimensions (in): A through AA*.

*Deformation indicators.

S-13326 SHUR-LOC® Swivel Hooks with Bearing
Suitable for frequent rotation under load.

Chain Size (in)	Chain Size (mm)	Frame code	Grade 100 Alloy Chain Working Load Limit (lb) 4:1 Design Factor	Working Load Limit (lb) 5:1 Design Factor	Stock No.	Weight Each (lb)	A	B	C	D	E	F	H	J	L	AA*	Replacement Trigger Kit Stock No.
-	6	D	3200	2560	1004404	1.50	1.50	1.14	6.17	.79	2.60	.67	.50	.63	1.13	1.50	6603010
1/4 - 5/16	7-8	G	5700	4560	1004413	3.10	1.75	1.52	7.54	1.10	3.50	.87	.63	.81	1.44	2.00	6603011
3/8	10	H	8800	7040	1004422	5.26	2.00	1.61	8.88	1.16	4.35	1.10	.75	.94	1.83	2.50	6603012
1/2	13	I	15000	12000	1004431	11.22	2.50	2.03	11.11	1.66	5.45	1.26	1.00	1.16	2.19	3.00	6603013
5/8	16	-	22600	18000	1004440	17.32	2.75	2.25	12.90	2.05	6.56	1.50	1.13	1.50	2.61	3.50	6603014

Dimensions (in): A through AA*.

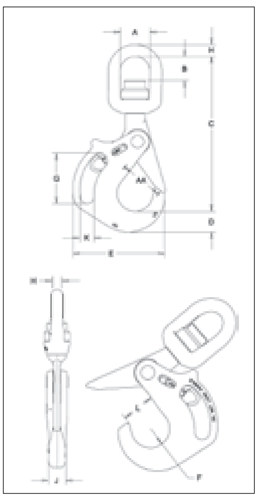
*Deformation indicators.

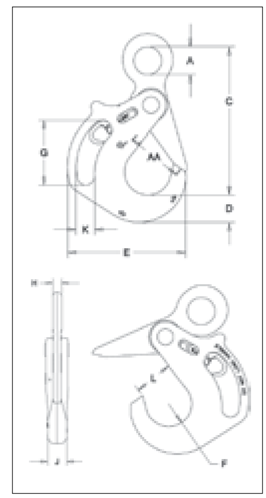
Crosby®

S-13326H

S-1316AH

- The SHUR-LOC® Handle Hook allows the user to get a confident grip on a load with ease and comfort.
- Designed with a handle opening big enough to comfortably fit a gloved hand.
- Positive lock latch is self-locking when hook is loaded.
- Rated for both wire rope and use with Grade 80/100 chain.
- S-13326H Swivel Hook utilizes anti-friction bearing design which allows hook to rotate freely under load.
- Individually Proof Tested at 2-1/2 times the chain Working Load Limit with certification.
- The replaceable pull-trigger allows the user to easily open the SHUR-LOC's positive self-locking latch.
  - Ergonomically designed for easy use and precise control.
  - Secondary side trigger is recessed to avoid inadvertent release.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- Forged alloy steel, Quenched & Tempered.
- The SHUR-LOC® hook, if properly installed and locked, can be used for personnel lifting applications and meets the intent of OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B).





APPLICATION AND WARNING INFORMATION
SECTION 17

S-13326H SHUR-LOC® Handle Swivel Hooks with Bearings

Chain Size (in)	Chain Size (mm)	Grade 100 Alloy Chain Working Load Limit (lb) 4:1*	Working Load Limit (lb) 5:1*	Frame Code	Stock No.	Weight Each (lb)	Dimensions (in) A	B	C	D	E	F	G	H	J	K	L	AA*	Replacement Trigger Kit Stock No.
5/8	16	22,600	18,080	JA	1005014	26	2.75	2.26	14.47	1.97	8.55	1.78	4.69	1.13	1.73	1.32	2.80	4.00	6603032
3/4	18/20	35,300	28,240	KA	1005023	37	3.12	2.05	15.49	2.60	9.99	1.99	4.72	1.25	2.05	1.31	3.31	5.00	6603033
7/8	22	42,700	34,160	LA	1005041	57	4.09	3.65	19.11	2.72	11.48	2.24	5.35	1.50	2.44	1.57	3.66	6.00	6603034
1	26	59,700	47,760	NA	1005050	84	5.00	4.02	21.55	3.11	12.76	2.52	6.46	1.63	2.76	1.57	4.09	6.50	6603035

4:1 Design Factor. *Deformation indicators.

S-1316AH SHUR-LOC® Handle Eye Hook

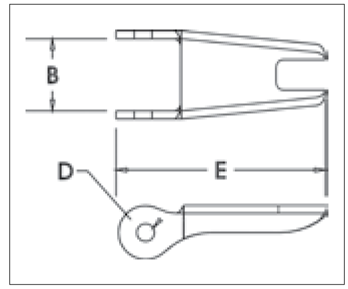
Chain Size (in)	Chain Size (mm)	Grade 100 Alloy Chain Working Load Limit (lb) 4:1*	Working Load Limit (lb) 5:1*	Frame Code	Stock No.	Weight Each (lb)	Dimensions (in) A	C	D	E	F	G	H	J	K	L	AA*	Replacement Trigger Kit Stock No.
5/8	16	22,600	18,080	JA	1023579	18	2.01	10.69	1.97	8.55	1.78	4.69	0.79	1.73	1.32	2.79	4.00	6603032
3/4	18/20	35,300	28,240	KA	1023599	29	2.76	12.03	2.60	9.99	1.99	4.73	0.87	2.05	1.31	3.35	5.00	6603033
7/8	22	42,700	34,160	LA	1023607	39	3.15	13.50	2.72	11.48	2.24	5.35	0.91	2.44	1.63	3.70	6.00	6603034
1	26	59,700	47,760	NA	1023625	60	3.54	15.59	3.27	12.76	2.52	6.46	1.18	2.76	1.58	4.09	6.50	6603035

4:1 Design Factor. *Deformation indicators.

S-4320
Replacement Latch Kit

- Heavy duty stamped latch interlocks with the hook tip.
- High cycle, long life spring.
- Can be made into a "Positive Locking" Hook when proper cotter pin is utilized.
- Latch kits shipped unassembled and individually packaged with instructions.
- Meets the intent of OSHA Rule 1926.1431(g) and 1926.1501(g)(when secured with the bolt, nut and pin) for lifting personnel.

IMPORTANT: The new S-4320 Latch Kit will not fit the old style 319, 320 and 322 hooks.

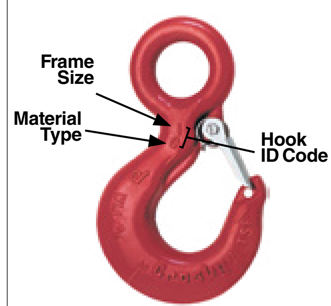


APPLICATION AND WARNING INFORMATION
SECTION 17

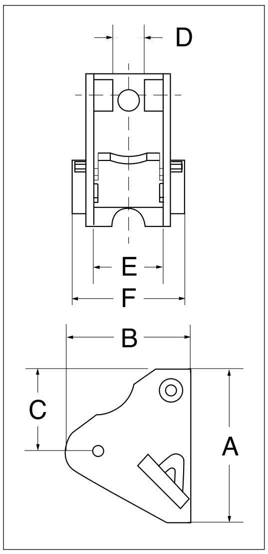
S-4320 Replacement Latch Kit for 319N, 320N, 322N, 339N, 1327 and 1339 Hooks

Hook Size (t)			Hook ID Code	Stock No.	Weight Each (lb)	Dimensions (in)		
Carbon	Alloy	Bronze				B	D	E
3/4	1	.5	D	1096325	.03	.50	.15	1.44
1	1-1/2	.6	F	1096374	.04	.54	.17	1.56
1-1/2	2	1	G	1096421	.04	.63	.17	1.66
2	3	1.4	H	1096468	.06	.66	.17	1.91
3	5	2	I	1096515	.10	.83	.20	2.31
5	7	3.5	J	1096562	.15	1.04	.20	2.88
7-1/2	11	5	K	1096609	.28	1.25	.27	3.56
10	15	6.5	L	1096657	.33	1.35	.27	3.81
15	22	10	N	1096704	.84	1.66	.39	5.18

Example of Hook ID Placement Location



PL
Latch Kits



LATCH ORDERING INSTRUCTIONS

Specify PL, PL-N or PL-O latch kit stock number from charts below.
Specify capacity of hook to which latch will be assembled.
Specify hook material (carbon or alloy).

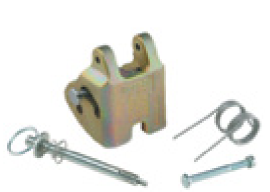
- Hot-dip galvanized.
- Heavy duty latch with easy operating features.
- Flapper lever indicates locked or unlocked position.
- Assembly instructions included with each latch.
- For additional dimensional data on eye, shank or swivel hooks, see Warnings & Applications.
- Meets the intent of OSHA Rule 1926.1431(g) and 1926.1501(g) (when secured with the bolt, nut and pin) for lifting personnel.

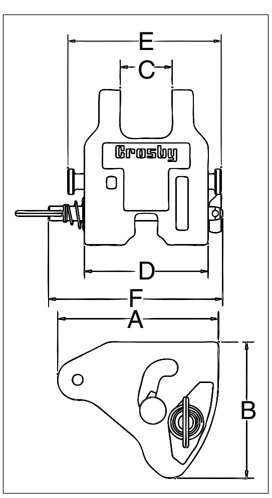
APPLICATION AND WARNING INFORMATION
SECTION 17

PL LATCH KITS

Hook Size (t)		Hook ID Code	Stock No.	Weight Each (lb)	Dimensions (in)						Replacement Spring Stock No.
Carbon	Alloy				A	B	C	D	E	F	
3	4-1/2	I	1093711	.54	2.69	2.39	2.07	0.63	1.13	1.95	-
5	7	J	1093712	.66	3.00	2.49	2.00	.63	1.38	2.20	-
7-1/2	11	K	1093713	1.00	3.63	2.46	2.38	.63	1.63	2.49	-
10	15	L	1093714	1.25	4.00	3.27	2.69	.63	1.875	3.25	-
15	22	N	1093715	2.96	5.31	4.19	2.91	.84	2.38	3.49	-
20	30	O	1093716	4.05	6.00	4.52	3.28	1.06	3.31	4.67	-
25	37	P	1093717	8.63	7.00	6.86	4.94	2.24	2.38	6.12	-
30	45	S	1093718	10.00	6.75	7.19	3.94	2.24	4.75	6.38	-
40	60	T	1093719	14.30	8.00	7.97	4.25	3.46	5.95	7.70	-
50	75	U	1093720	27.00	9.88	8.38	5.88	3.38	6.5	8.88	32468
-	100-150	W - X	1093721	33.25	10.88	10.88	6.5	3.38	7.88	10.00	32468
-	200	Y	1093723	45.00	11.88	11.19	6.38	3.38	8.75	11.27	32468
-	300	Z	1093724	55.00	12.50	12.38	7.92	3.38	10	12.25	32468

PL-N/O
Latch Kits





LATCH ORDERING INSTRUCTIONS

Specify PL, PL-N or PL-O latch kit stock number from charts below.
Specify capacity of hook to which latch will be assembled.
Specify hook material (carbon or alloy).

- Heavy duty latch with easy operating features.
- PL-N designed for Crosby 319N & 320N style hooks, PL-O designed for Crosby 319 & 320 old style hooks.
- Flapper lever indicates locked or unlocked position.
- Assembly instructions included with each latch.
- For additional dimensional data on eye, shank or swivel hooks refer to the specific product page in this section.
- Meets the intent of OSHA Rule 1926.1431(g) and 1926.1501(g) (when secured with the supplied toggle pin) for lifting personnel.

APPLICATION AND WARNING INFORMATION
SECTION 17

PL-N/O LATCH KITS

Hook Size (t)		Hook ID Code	PL-N Latch Kit Stock No.	PL-O Latch Kit Stock No.	Weight Each (lb)	Dimensions (in)					
Carbon	Alloy					A	B	C	D	E	F
3	4.5 / 5 *	I	1092000	1091900	.8	2.40	2.01	.83	2.13	2.71	3.44
5	7	J	1092001	1091901	1.3	2.94	2.50	1.00	2.52	3.19	3.83
7-1/2	11	K	1092002	1091902	2.0	3.63	3.02	1.19	2.75	3.44	4.38
10	15	L	1092003	1091903	2.8	4.00	3.39	1.34	3.19	4.00	4.50
15	22	N	1092004	1091904	4.9	5.19	4.32	1.61	3.86	4.81	5.13

*"N" style hooks are rated at 5 metric tons.

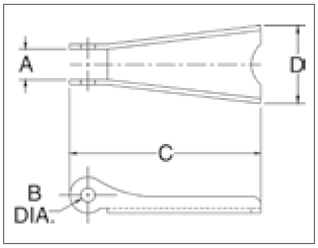
SS-4055
Latch Kits

LATCH ORDERING INSTRUCTIONS

Specify latch kit stock number.
Specify capacity of hook to which latch will be assembled.
Specify hook material (carbon or alloy).

- Stainless steel construction with cadmium plated steel nuts.
- Shipped packaged and unassembled.
- Instructions included for easy field assembly.

APPLICATION AND WARNING INFORMATION
SECTION 17



SS-4055 LATCH KITS

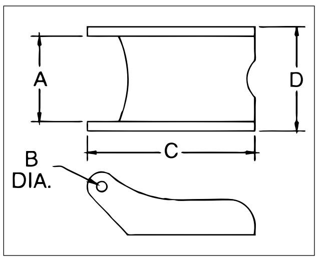
Hook Size (t)			Hook ID Code	Stock No.	Weight Each (lb)	Dimensions (in)			
Carbon	Alloy	Bronze				A	B	C	D
3/4	1	.5	D	1090027	.02	.38	.16	1.44	.59
1	1-1/2	.6	F	1090045	.02	.38	.16	1.60	.59
1-1/2 - 2	2 - 3	1.0 - 1.4	G / H	1090063	.03	.47	.19	1.84	.82
3	4-1/2	2.0	I	1090081	.06	.56	.17	2.41	1.00
5	7	3.5	J	1090107	.11	.58	.20	2.97	1.21
7-1/2 - 10	11 - 15	5.0 - 6.5	K / L	1090125	.17	.59	.27	3.66	1.50
15	22	10.0	N	1090143	.39	.83	.39	4.94	1.90
20	30	-	O	1090161	.63	.94	.52	5.88	2.56
25 - 30	37 - 45	-	P / S	1090189	1.12	2.19	.39	6.50	3.84
40	60	-	T	1090205	1.77	3.31	.52	7.88	4.12

S-4088
Alloy Hook Latch Kits

LATCH ORDERING INSTRUCTIONS

Specify latch kit stock number.
Specify capacity of hook to which latch will be assembled.
Specify hook material (carbon or alloy).

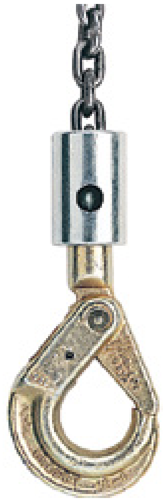
- To be used on A-327 and A-339 Grade 8 sling hooks.
- Latch kits shipped unassembled and individually packaged with instructions.



S-4088 Alloy Hook Latch Kits

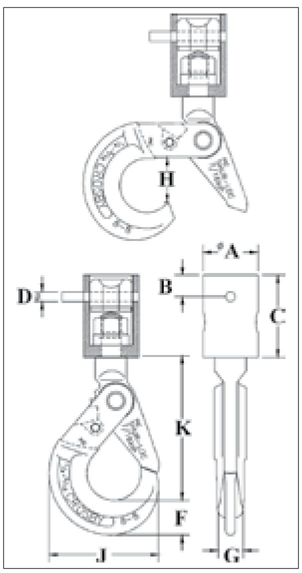
Hook Chain (in)	Stock No.	Weight Each (lb)	Dimensions (in)			
			A	B	C	D
9/32 (1/4)	1090250	.06	.78	.16	2.03	.94
3/8	1090251	.14	1.03	.19	2.69	1.25
1/2	1090252	.15	1.03	.19	3.00	1.25
5/8	1090253	.15	1.03	.19	3.25	1.25
3/4	1090254	.15	1.53	.26	4.13	1.88
7/8	1090255	.15	1.53	.26	4.66	2.00

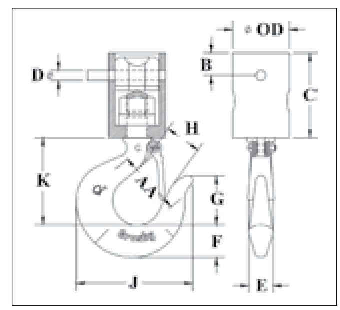
O-318



O-319

- With ball bearing swivel; attaches to chain by an alloy pin.
- Suitable for frequent rotation under load.
- HO-318 Hooks utilize Crosby SHUR-LOC® positive locking hooks. Latch is self-locking when hook is loaded.
- O-319 Hooks utilize Crosby® standard 319 Shank Hooks which incorporate QUIC-CHECK® deformation and angle indicators. (For detailed information, see the Crosby Value Added page at the beginning of this section.)
- Entire assembly is zinc plated.
- Repair kit available consisting of bearing and spring pin.





QUIC-CHECK

APPLICATION AND WARNING INFORMATION
SECTION 17

O-318 Chain Nest Hooks

Chain Size (in)	Stock No.	WLL (short Tons)*	Weight Each (lb)	Dimensions (in)									Replacement Trigger Kit Stock No.
				A	B	C	D	F	G	H	J	K	
1/4 - 9/32	1098409	1.5	1.7	1.75	.70	2.62	.31	1.10	.81	1.46	3.50	4.59	6603011
5/16 - 3/8	1098427	2.1	2.3	2.13	.70	3.19	.38	1.15	.94	1.83	4.35	5.65	6603012
3/8 - 7/16	1098445	3.8	4.2	3.00	1.00	4.38	.50	1.66	1.16	2.11	5.45	7.06	6603013
1/2 - 9/16	1098463	3.8	4.2	3.00	1.00	4.38	.63	1.66	1.16	2.11	5.45	7.06	6603013

4:1 Design Factor.

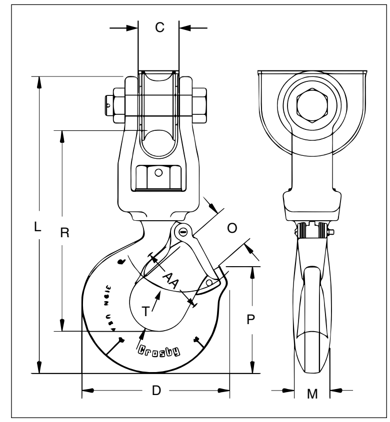
O-319 Chain Nest Hooks

Chain Size (in)	Stock No.	WLL (short Tons)*	Weight Each (lb)	Dimensions (in)											Replacement Latch Kit Stock No.
				OD	AA	B	C	D	E	F	G	H	J	K	
1/4 - 9/32	1098312	1.5	1.7	1.75	2.00	.70	2.62	.31	.75	1.00	1.53	1.00	3.62	2.69	1096421
5/16 - 3/8	1098334	2.1	2.3	2.13	2.00	.70	3.19	.38	.84	1.12	1.72	1.12	4.09	3.06	1096468
3/8 - 7/16	1098356	3.8	4.2	3.00	2.50	1.00	4.38	.50	1.12	1.44	2.12	1.34	4.84	3.78	1096515
1/2 - 9/16	1098378	3.8	4.2	3.00	2.50	1.00	4.38	.63	1.12	1.44	2.12	1.34	4.84	3.78	1096515

4:1 Design Factor. *Deformation indicators.

S-3319

- Designed for utility applications using synthetic rope.
- Suitable for positioning before lifting.
- Hook is forged alloy steel, Quenched & Tempered.
- Design of hook provides needed overhaul weight.
- Utilizes spool & shield designed to protect rope and keep rope positioned correctly on spool.
- Spool provides wider rope bearing surface resulting in an increased area for load distribution and reduces rope abrasion.



QUIC-CHECK®

APPLICATION AND WARNING INFORMATION
SECTION 17

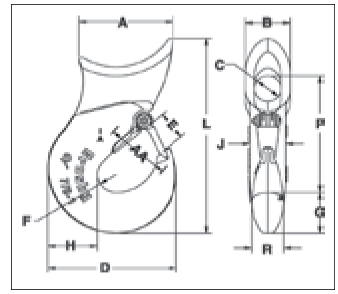
S-3319 Utility Swivel Hook

Working Load Limit (t)	Stock No.	Weight Each (lb)	Hook ID Code	Synthetic Rope Size (in)	Dimensions (in)									Replacement Latch Kit Stock No.
					C	D	L	M	O	P	R	T	AA*	
1.63	1002054	4.2	HA	9/16 - 5/8	1.09	3.99	8.75	.94	1.16	2.78	5.94	1.16	2.00	1096468
2.50	1002063	8.0	IA	3/4 - 13/16	1.31	4.84	10.56	1.13	1.41	3.47	7.06	1.53	2.50	1096515
4.50	1002072	15.0	JA	7/8 - 1-1/16	1.78	6.29	12.75	1.44	1.78	4.59	8.69	1.94	3.00	1096562

5:1 Design Factor. Maximum allowable proof load is 2 times the Working Load Limit. *Deformation indicators.

A-350L

- New style incorporates throat opening equal to or larger than old style hooks.
- Each product has a Product Identification Code (PIC) for material traceability, along with a Working Load Limit, and the name Crosby or "CG" forged into it.
- All hooks incorporate Crosby's patented QUIC-CHECK® deformation indicators to help in determining if throat opening dimension has changed.
- Each hook is equipped with a Crosby S-4320 heavy duty stamped latch with the high cycle, long life spring.
- Forged alloy steel, Quenched & Tempered.



A-350L Sliding Choker Hook

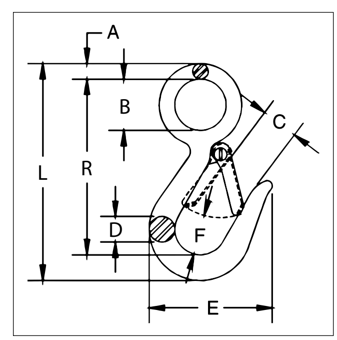
QUIC-CHECK®

Single Part Rope Size (in)	Eight Part Rope Size (in)	Stock No.	WLL (lb)	Weight Each (lb)	Dimensions (in)												Hook Frame Code	Replacement Latch Kit Stock No.
					A	B	C	D	E	F	G	H	L	P	R	AA*		
3/8	-	1011802	2500	1.0	2.06	1.13	.63	2.41	.63	.38	.84	.91	4.28	2.59	.63	1.50	DA	1096325
1/2	1/8	1011811	3800	1.4	2.25	1.31	.75	2.97	.78	.50	.97	1.06	4.97	3.09	.75	1.50	FA	1096374
† 5/8	-	1011820	5800	3.0	3.06	1.63	.75	3.56	.94	.56	1.13	1.31	6.38	3.88	1.00	2.00	GA	1096421
† 5/8	3/16	1011839	5800	2.7	3.06	1.63	1.00	3.56	.94	.56	1.13	1.31	6.38	4.00	1.13	2.00	GA	1096421
† 3/4	-	1011848	8200	4.4	3.38	2.13	1.00	4.25	1.16	.63	1.44	1.63	7.66	4.58	1.13	2.50	HA	1096468
† 3/4	1/4	1011857	8200	3.8	3.38	2.13	1.44	4.25	1.16	.63	1.44	1.63	7.66	4.78	1.13	2.50	HA	1096468
†† 7/8-1	-	1028177	15000	9.70	4.41	2.12	1.25	6.06	1.41	.88	2.00	2.33	9.55	5.72	1.50	3.00	IA	1096515

*Deformation indicators. †Determine eye diameter "C" before ordering. ††7/8-1" is cast steel.

G-3315

- Forged carbon steel, Quenched & Tempered.
- Pressed steel latches and stainless steel springs, bolts and nuts.
- For replacement latch kit, order Stock No. 9900299.
- Hook body - galvanized.
- Suitable for overhead lifting if provisions are made for meeting applicable standards and the specific lifting requirements.



G-3315 Snap Hook

Hook Size (in)	Stock No.	Working Load Limit (lb)*	Weight Each (lb)	Dimensions (in)								Replacement Latch Kit Stock No.
				A	B	C	D	E	F	L	R	
7/16	1023056	750	.23	.25	.75	.75	.44	2.25	.75	3.94	3.25	9900299
9/16	1023074	1000	.48	.34	1.12	.81	.56	2.69	.88	4.75	3.84	9900299

4:1 Design Factor.

S-377

- Forged carbon steel, Quenched & Tempered.
- The resultant load on each hook cannot exceed 1,000 lb.
- Meets the performance requirements of Federal Specification RR-C-271G, Type V, Class 6, except for those provisions required of the contractor.

S-377 Barrel Hooks

Working Load Limit Per Pair (t)	Stock No. Per Pair	Weight Each Per Pair (lb)	Dimensions (in)			
			I.D. of Eye	O.D. of Eye	Overall Length	Width of Lip
1.0	1028248	3.56	1.56	2.81	5.00	2.88

4:1 Design Factor.

A-378N Sorting Hook

- Forged alloy steel, Quenched & Tempered.
- Deep straight throat permits efficient handling of flat plates or large cylindrical shapes.

A-378N Sorting Hook

Working Load Limit at tip of Hook (T)	Working Load Limit at bottom of Hook (T)	Stock No	Weight Each (lb)	Dimensions (in)			
				I.D. of Eye	Overall Length	Opening at top of Hook	Radius at bottom of Hook
2	7.5	1028033	6.42	1.38	9.69	2.81	.625

4:1 Design Factor.

Crosby® HOIST HOOKS

WARNINGS & APPLICATION INSTRUCTIONS

⚠ WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **See OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B) for personnel hoisting by cranes and derricks, and OSHA Directive CPL 2-1.36 - Interim Inspection Procedures During Communication Tower Construction Activities. A Crosby 319, L-320 or L-322 hook with a PL latch attached and secured with a bolt, nut and cotter pin (or toggle pin) may be used for lifting personnel. A Crosby 319N, L-320N or L-322N hook with an S-4320 latch attached and secured with cotter pin or bolt, nut and pin; or a PL-N latch attached and secured with toggle pin may be used for lifting personnel. A hook with a Crosby SS-4055 latch attached shall NOT be used for personnel lifting.**
- **See OSHA Directive CPL 2-1.36 - Crosby does not recommend the placement of lanyards directly into the positive locking Crosby hook when hoisting personnel. Crosby requires that all suspension systems (vertical lifelines / lanyard) shall be gathered at the positive locked hook by use of a master link, or a bolt-type shackle secured with cotter pin.**
- **Threads may corrode and/or strip and drop the load.**
- **Remove securement nut to inspect or to replace L-322, S-3316, and S-3319 bearing washers (2).**
- **Hook must always support the load. The load must never be supported by the latch.**
- **Never apply more force than the hook's assigned Working Load Limit (WLL) rating.**
- **Read and understand these instructions before using hook.**

QUIC-CHECK® Hoist hooks incorporate markings forged into the product which address two (2) **QUIC-CHECK®** features:

1. **Deformation Indicators –** Two strategically placed marks, one just below the shank or eye and the other on the hook tip, which allows for a **QUIC-CHECK®** measurement to determine if the throat opening has changed, thus indicating abuse or overload.
2. **To check**, use a measuring device (i.e., tape measure) to measure the distance between the marks. The marks should align to either an inch or half-inch increment on the measuring device. If the measurement does not meet criteria, the hook should be inspected further for possible damage.
3. **Angle Indicators** – Indicates the maximum included angle which is allowed between two (2) sling legs in the hook. These indicators also provide the opportunity to approximate other included angles between two sling legs.

QUIC-CHECK®

IMPORTANT SAFETY INFORMATION - READ & FOLLOW

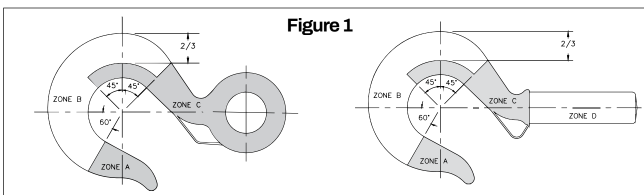
A visual periodic inspection for cracks, nicks, wear, gouges and deformation as part of a comprehensive documented inspection program, should be conducted by trained personnel in compliance with the schedule in ASME B30.10.

- For hooks used in frequent load cycles or pulsating loads, the hook and threads should be periodically inspected by Magnetic Particle or Dye Penetrant (Note: Some disassembly may be required).
- Never use a hook whose throat opening has been increased, or whose tip has been bent more than 10 degrees out of plane from the hook body, or is in any other way distorted or bent. Note: A latch will not work properly on a hook with a bent or worn tip.

APPLICATIONS & WARNINGS

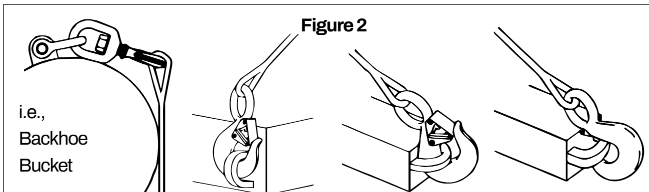
- Never use a hook that is worn beyond the limits shown in Figure 1.
- Any crack in a hook is reason to take it out of service. Hooks with a nick or gouge can be repaired only by a qualified person by grinding lengthwise, following the contour of the hook, provided that the reduced dimension is within the limits shown in Figure 1. Contact Crosby Engineering to evaluate any crack.
- Never repair, alter, rework, or reshape a hook by welding, heating, burning, or bending.
- Never side load, back load, or tip load a hook.(Side loading, back loading and tip loading are conditions that damage and reduce the capacity of the hook.) (See Figure 2)
- Eye, Shank and Swivel hooks are designed to be used with wire rope or chain. Clevis hooks are design to be used with chain. Efficiency of assembly may be reduced when used with synthetic material.
- Do not swivel the L-322, S-3316, or S-3319 swivel hooks while supporting a load. These hooks are distinguishable by hex nuts and flat washers.
- The L-3322 swivel hook is designed to rotate under load. The L-3322 is distinguishable from the L-322 by use of a round nut designed to shield bearing.
- The frequency of bearing lubrication on the L-3322 depends upon frequency and period of product use as well as environmental conditions, which are contingent upon the user's good judgment.
- The use of a latch may be mandatory by regulations or safety codes; e.g., OSHA, MSHA, ANSI/ASME B30, Insurance, etc. (Note: When using latches, see instructions in "Understanding The Crosby Group Warnings" for further information.)
- Always make sure the hook supports the load (See Figure 3). The latch must never support the load (See Figure 4).
- When multileg slings are placed in the base (bowl/saddle) of the hook, the maximum included angle between sling legs shall be 90 deg. The maximum sling leg angle with respect to the hook centerline for any rigging arrangement shall be 45 degrees. A collector ring, such as a link or shackle, should be used to maintain in-line load when more than two legs are placed in a hook or for angles greater than 45 degrees with respect to hook centerline. When more than two legs are placed in the hook bunching of the legs shall be avoided.
- See ASME B30.10 "Hooks" for additional information.

Figure 1



ZONE A: REPAIR NOT REQUIRED
ZONE B: 10% OF ORIGINAL DIMENSION
ZONE C: 5% OF ORIGINAL DIMENSION
ZONE: D: SEE MINIMUM THREAD SIZE CHART

Figure 2



Side Load WRONG | Side Load WRONG | Back Load WRONG | Tip Load WRONG

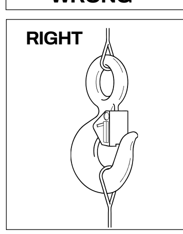


Figure 3

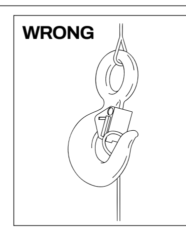


Figure 4

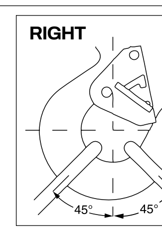
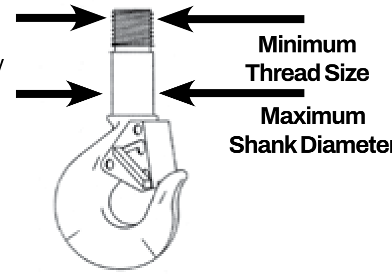


Figure 5

This application/warning information apply to Crosby products only.

READ AND UNDERSTAND THESE INSTRUCTIONS BEFORE USING HOOKS
IMPORTANT – BASIC MACHINING AND THREAD INFORMATION

- Wrong thread and/or shank size can cause stripping and loss of load.
- The maximum diameter is the largest diameter, after cleanup, that could be expected after allowing for straightness, pits, etc.
- All threads must be Class 2 or better.
- The minimum thread length engaged in the nut should not be less than one (1) thread diameter. Install a properly sized retention device to secure the nut to the hook shank after the nut is properly adjusted at assembly. Nut retention devices such as set screws or roll pins are suitable for applications using anti-friction thrust bearings or bronze thrust washers. If the hook is intended for other applications that introduce a higher torque into the nut, a more substantial retaining device may be required.
- Hook shanks are not intended to be swaged on wire rope or rod.
- Hook shanks are not intended to be drilled (length of shank) and internally threaded.
- Crosby can not assume responsibility for, (A) the quality of machining, (B) the type of application, or (C) the means of attachment to the power source or load.
- Consult the Crosby Hook Identification & Working Load Limit Chart (See below) for the minimum thread size for assigned Working Load Limits (WLL).†
- Remove from service any Hook which has threads corroded more than 20% of the nut engaged length.



CROSBY HOOK IDENTIFICATION & WORKING LOAD LIMIT CHART†

Hook Identification			Working Load Limit (t)					Frame Size	Maximum Shank Diameter after Machining (in)	Minimum Thread Size	
319C 319CN L-320C L-320CN L-322C L-322CN	319AN L-320A L-320AN L-322A L-322AN 3319 L-3322B	319BN	319C 319CN L-320C L-320CN L-322C L-322CN	319A 319AN L-320A L-320AN L-322A L-322AN L-3322B	319BN	S-3319	S-3316			319C 319CN (Carbon)	319A 319AN (Alloy)
DC	DA	DB	.75	1	.5	—	—	D	.53	1/2 - 13unc	1/2 - 13 unc
FC	FA	FB	1	1.5	.6	—	.45	F	.62	5/8 - 11unc	5/8 - 11 unc
GC	GA	GB	1.5	2	1	—	—	G	.66	5/8 - 11unc	5/8 - 11 unc
HC	HA	HB	2	3	1.4	1.63	.91	H	.81	3/4 - 10unc	3/4 - 10 unc
IC	IA	IB	3	*4.5 / 5	2.0	2.5	—	I	1.03	7/8 - 9unc	7/8 - 9 unc
JC	JA	JB	5	7	3.5	4.5	—	J	1.27	1-1/8 - 7unc	1-1/8 - 7 unc
KC	KA	KB	7.5	11	5.0	—	—	K	1.52	1-1/4 - 7unc	1-3/8 - 6 unc
LC	LA	LB	10	15	6.5	—	—	L	1.75	1-5/8 - 8un	1-5/8 - 8 un
NC	NA	NB	15	22	10	—	—	N	2.00	2 - 8un	2 - 8 un
OC	OA	—	20	30	—	—	—	O	2.50	2-1/4 - 8un	2-1/4 - 8 un
PC	PA	—	25	37	—	—	—	P	3.50	2-3/4 - 8un	2-3/4 - 8 un
SC	SA	—	30	45	—	—	—	S	3.50	3 - 8un	3 - 8 un
TC	TA	—	40	60	—	—	—	T	4.00	3-1/4 - 8un	3-1/2 - 8 un
UC	UA	—	50	75	—	—	—	U	4.50	3-3/4 - 8un	4 - 4 unc
—	WA	—	—	100	—	—	—	W	6.12	—	4-1/2 - 8 un
—	XA	—	—	150	—	—	—	X	6.38	—	5-1/2 - 8 un
—	YA	—	—	200	—	—	—	Y	7.00	—	6-1/4 - 8 un
—	ZA	—	—	300	—	—	—	Z	8.62	—	7-1/2 - 8 un

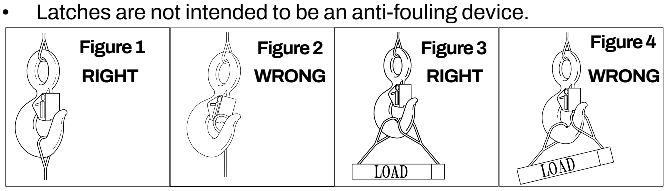
* 319AN, L-320AN, L-3322 and L-322AN are rated at 5 tons.

† Working Load Limit - The maximum mass or force which the product is authorized to support in general service when the pull is applied in-line, unless noted otherwise, with respect to the centerline of the product. This term is used interchangeably with the following terms: 1. WLL, 2. Rated Load Value, 3. SWL, 4. Safe Working Load, 5. Resultant Safe Working Load.

Warning and Application Instructions For Crosby® Hook Latch Kit

IMPORTANT SAFETY INFORMATION - READ & FOLLOW

- Always inspect hook and latch before using.
- Never use a latch that is distorted or bent.
- Always make sure spring will force the latch against the tip of the hook.
- Always make sure hook supports the load. The latch must never support the load (See Figures 1 & 2).
- When placing two (2) sling legs in hooks, make sure the angle between the legs is less the 90° and if the hook or load is tilted, nothing bears against the bottom of this latch (See Figures 3 & 4).
- Latches are intended to retain loose sling or devices under slack conditions.
- Latches are not intended to be an anti-fouling device.



⚠ WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **See OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B) for personnel hoisting for cranes and derricks. Only a Crosby or McKissick hook with a PL Latch attached and secured with bolt, nut and cotter (or Crosby Toggle Pin) or a Crosby hook with a S-4320 Latch attached and secured with a cotter pin, or a Crosby SHUR-LOC® hook in the locked position may be used for any personnel hoisting. A hook with a Crosby SS-4055 latch attached shall NOT be used for personnel lifting.**
- **Hook must always support the load. The load must never be supported by the latch.**
- **DO NOT use this latch in applications requiring non-sparking.**
- **Read and understand these instructions before using hook and latch.**

This application/warning information apply to Crosby products only.

McKissick® HOIST HOOKS

WARNINGS & APPLICATION INSTRUCTIONS

L-320N Series

S-319 Series

L-322 Series Positioning Only

L-3322B Series

⚠ WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **See OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B) for personnel hoisting by cranes and derricks, and OSHA Directive CPL 2-1.36 - Interim Inspection Procedures During Communication Tower Construction Activities. A Crosby 319, L-320 or L-322 hook with a PL latch attached and secured with a bolt, nut and cotter pin (or toggle pin) may be used for lifting personnel. A Crosby 319N, L-320N or L-322N hook with an S-4320 latch attached and secured with cotter pin or bolt, nut and pin; or a PL-N latch attached and secured with toggle pin may be used for lifting personnel. A hook with a Crosby SS-4055 latch attached shall NOT be used for personnel lifting.**
- **See OSHA Directive CPL 2-1.36 - Crosby does not recommend the placement of lanyards directly into the positive locking Crosby hook when hoisting personnel. Crosby requires that all suspension systems (vertical lifelines / lanyard) shall be gathered at the positive locked load hook by use of a master link, or a bolt-type shackle secured with cotter pin.**
- **Threads or Split-Nut may corrode and/or strip and drop the load.**
- **Remove securement nut to inspect or to replace S-322 and S-3319 bearing washers (2).**
- **Hook must always support the load. The load must never be supported by the latch.**
- **Never apply more force than the hook's assigned Working Load Limit (WLL) rating.**
- **Read and understand these instructions before using hook.**

QUIC-CHECK® Hoist hooks incorporate markings forged into the product which address two (2) QUIC-CHECK® features:

Deformation Indicators - Two strategically placed marks, one just below the shank or eye and the other on the hook tip, which allows for a **QUIC-CHECK®** measurement to determine if the throat opening has changed, thus indicating abuse or overload.

QUIC-CHECK®

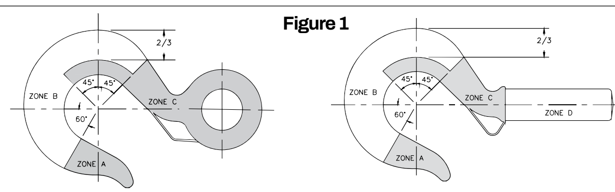
To check, use a measuring device (i.e., tape measure) to measure the distance between the marks. The marks should align to either an inch or half-inch increment on the measuring device. If the measurement does not meet criteria, the hook should be inspected further for possible damage.

Angle Indicators - Indicates the maximum included angle which is allowed between two (2) sling legs in the hook. These indicators also provide the opportunity to approximate other included angles between two sling legs.

IMPORTANT SAFETY INFORMATION - READ & FOLLOW

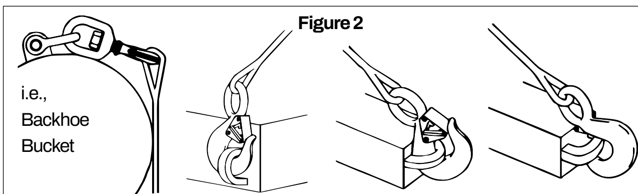
- A visual periodic inspection for cracks, nicks, wear, gouges and deformation as part of a comprehensive documented inspection program, should be conducted by trained personnel in compliance with the schedule in ASME B30.10.
- For hooks used in frequent load cycles or pulsating loads, the hook and threads should be periodically inspected by Magnetic Particle or Dye Penetrant. (Note: Some disassembly may be required.)
- Never use a hook whose throat opening has been increased, or whose tip has been bent more than 10 degrees out of plane from the hook body, or is in any other way distorted or bent.
- **Note: A latch will not work properly on a hook with a bent or worn tip.**
- Never use a hook that is worn beyond the limits shown in Figure 1.
- Any crack in a hook is reason to take it out of service. Hooks with a nick or gouge can be repaired only by a qualified person by grinding lengthwise, following the contour of the hook, provided that the reduced dimension is within the limits shown in Figure 1. Contact Crosby Engineering to evaluate any crack.
- Remove from service any hook which has threads corroded more than 20% of the nut engagement length.
- Never repair, alter, rework, or reshape a hook by welding, heating, burning, or bending.
- Never side load, back load, or tip load a hook.(Side loading, back loading and tip loading are conditions that damage and reduce the capacity of the hook.) (See Figure 2)
- Eye hooks, shank hooks and swivel hooks are designed to be used with wire rope or chain. Efficiency of assembly may be reduced when used with synthetic material.
- Do not swivel the L-322 or S-3319 swivel hooks while supporting a load. These hooks are distinguishable by hex nuts and flat washers.
- The L-3322 swivel hook is designed to rotate under load. The L-3322 is distinguishable from the L-322 by use of a round nut designed to shield bearing.
- The frequency of bearing lubrication on the L-3322 depends upon frequency and period of product use as well as environmental conditions, which are contingent upon the user's good judgment.
- The use of a latch may be mandatory by regulations or safety codes; e.g., OSHA, MSHA, ASME B30, Insurance, etc.. (Note: When using latches, see instructions in "Understanding: The Crosby Group Warnings" for further information.)
- Always make sure the hook supports the load (See Figure 3). The latch must never support the load (See Figure 4).
- When multileg slings are placed in the base (bowl/saddle) of the hook, the maximum included angle between sling legs shall be 90 deg. The maximum sling leg angle with respect to the hook centerline for any rigging arrangement shall be 45 degrees. A collector ring, such as a link or shackle, should be used to maintain in-line load when more than two legs are placed in a hook or for angles greater than 45 degrees with respect to hook centerline. When more than two legs are placed in the hook bunching of the legs shall be avoided.
- Reference Crosby's Hoist Hook Warning and Application Information for basic machining and minimum thread size.
- See ASME B30.10 "Hooks" for additional information.

Figure 1



ZONE A: REPAIR NOT REQUIRED
ZONE B: 10% OF ORIGINAL DIMENSION
ZONE C: 5% OF ORIGINAL DIMENSION
ZONE: D: SEE MINIMUM THREAD SIZE CHART

Figure 2



Side Load WRONG | Side Load WRONG | Back Load WRONG | Tip Load WRONG

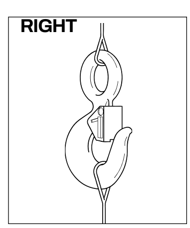


Figure 3

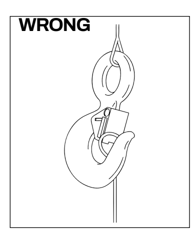


Figure 4

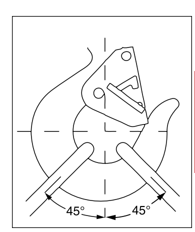


Figure 5

Removal of Split-Nut assembly (Reference Figure A):

- Remove vinyl cover.
- Remove spring retaining ring.
- Slide steel keeper ring off split nuts ⚠ CAUTION **emoval of keeper ring will allow split nut halves to fall from hook shank).**
- Remove split nut halves.

Inspection of split nut assembly and hook shank interface area (Reference Figure B):

- Inspect hook shank and split nut for signs of deformation on and adjacent to the load bearing surfaces.
- Inspect outside corner of hook shank load bearing surface to verify the corner is sharp.
- Verify retaining ring groove will allow proper seating of the retaining ring.
- Inspect retaining ring for corrosion or deformation. Remove from service any retaining ring that has excessive corrosion or is deformed.
- Use fine grit emery or crocus cloth to remove any corrosion from machined hook shank and split nut assembly.
- Follow inspection recommendations listed in this document under IMPORTANT SAFETY INFORMATION.
- If corrosion is present on the nut / shank interface area and deterioration or degradation of the metal components is evident, further inspection is required.
  - The use of a feeler gauge is required to properly measure the maximum allowable gap width between the split nut inside diameters and shank outside diameters.
  - With one split nut half seated against the hook shank, push the nut to one side and measure the maximum gaps as shown in Figure B. The hook should be measured in four places, 90-degrees apart.
  - Repeat above inspection procedure with other half of split nut.
  - Remove from service any hook and split nut assembly that exhibits a gap greater than 0.030".

Installation of split nut assembly (Reference Figure A):

- Coat hook shank and inside of split nut with an anti-seize compound or heavy grease.
- Install split nut halves onto shank. The flanged bottom of the split nut should be closest to the hook shoulder.
- Slide steel keeper ring over split nut halves. Verify the split nut halves properly seat against the load bearing surface of the hook shank and the steel keeper ring seats against the flange of the split nut.
- Install retaining ring onto split nut halves. Verify the retaining ring seats properly in the retaining ring groove on the outside diameter of the split nut assembly.
- Install vinyl cover over split nut and hook shank assembly.
- Verify all fasteners are correctly installed.
- Always use Genuine Crosby replacement parts.

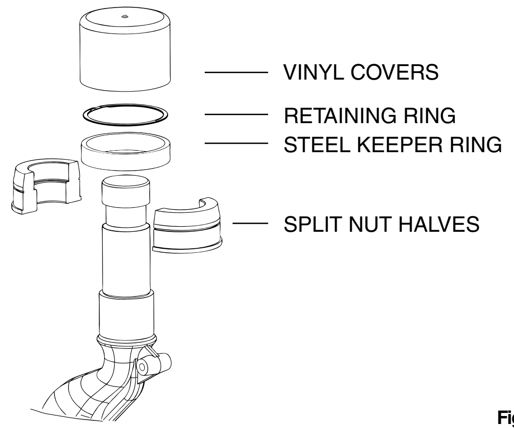


Figure A

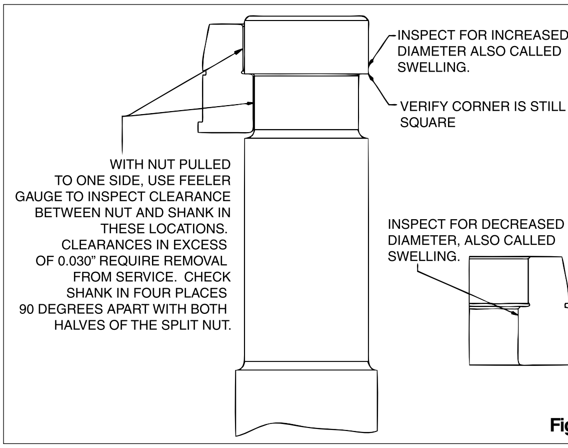


Figure B

Warning and Application Instructions For McKissick® Hook Latch Kit

IMPORTANT SAFETY INFORMATION - READ & FOLLOW

- Always inspect hook and latch before using.
- Never use a latch that is distorted or bent.
- Always make sure spring will force the latch against the tip of the hook.
- Always make sure hook supports the load. The latch must never support the load (See Figures 1 & 2).
- When placing two (2) sling legs in hooks, make sure the angle between the legs is less the 90° and if the hook or load is tilted, nothing bears against the bottom of this latch (See Figures 3 & 4).
- Latches are intended to retain loose sling or devices under slack conditions.
- Latches are not intended to be an anti-fouling device.

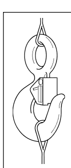


Figure 1
RIGHT

Figure 2
WRONG

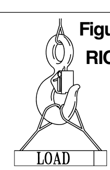


Figure 3
RIGHT

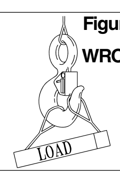


Figure 4
WRONG

⚠ WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **See OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B) for personnel hoisting for cranes and derricks. Only a Crosby or McKissick hook with a PL Latch attached and secured with bolt, nut and cotter (or Crosby Toggle Pin) or a Crosby hook with a S-4320 Latch attached and secured with a cotter pin, or a Crosby SHUR-LOC® hook in the locked position may be used for any personnel hoisting. A hook with a Crosby SS-4055 latch attached shall NOT be used for personnel lifting.**
- **Hook must always support the load. The load must never be supported by the latch.**
- **Do not use this latch in applications requiring non-sparking.**
- **Read and understand these instructions before using hook and latch.**

This application/warning information apply to Crosby products only.

S-4320 HOOK LATCH KIT

WARNINGS & APPLICATION INSTRUCTIONS

(For Crosby 319N, 320N, and 322N, S-1327, and A-1339 Hooks)

Important Safety Information - Read & Follow

- Always inspect hook and latch before using.
- Never use a latch that is distorted or bent.
- Always make sure spring will force the latch against the tip of the hook.
- Always make sure hook supports the load. The latch must never support the load (See Figures 1 & 2).
- When placing two (2) sling legs in hook, make sure the angle between the legs is less than 90° and if the hook or load is tilted, nothing bears against the bottom of this latch (See Figures 3 & 4).
- Latches are intended to retain loose sling or devices under slack conditions.
- Latches are not intended to be an anti-fouling device.
- When using latch for personnel lifting, select proper cotter pin (See Figure 5). See Step 7 below for proper installation instructions.
  - Never reuse a bent cotter pin.
  - Never use a cotter pin with a smaller diameter or different length than recommended in Figure 5.
  - Never use a nail, a welding rod, wire, etc., in place of recommended cotter pin.
  - Always ensure cotter pin is bent so as not to interfere with sling operation.
  - Periodically inspect cotter pin for corrosion and general adequacy.

⚠ WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **Hook must always support the load. The load must never be supported by the latch.**
- **See OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B) for Personnel Hoisting by Crane or Derricks. A Crosby S-319N, S-320N, S-322N, S-1327, and A-1339 Hook with an S-4320 latch attached (when secured with cotter pin) may be used for lifting personnel.**
- **An S-4320 Latch is only to be used with a Crosby S-319N, S-320N, S-322N, S-1327, and A-1339 Hook.**
- **DO NOT use this latch in applications requiring non-sparking.**
- **Read and understand these instructions before using hook and latch.**

Right

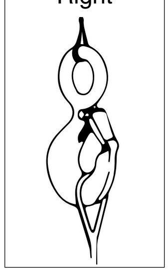


Figure 1

Wrong

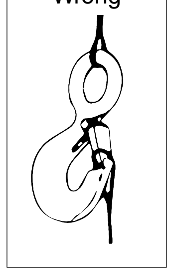


Figure 2

Right

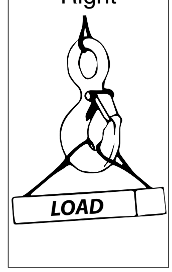


Figure 3

Wrong

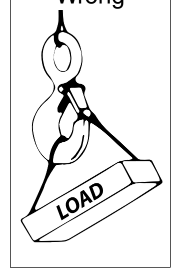
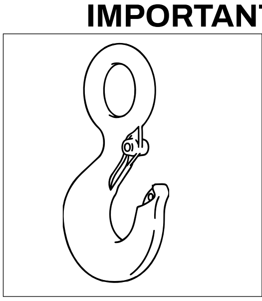


Figure 4

Hook Identification Code	Recommended Cotter Pin Dimensions (mm)	
	Diameter	**Length**
D	3.19	19.1
F	3.19	19.1
G	3.19	25.4
H	4.76	31.8
I	6.35	38.1
J	23.8	50.8
K	23.8	50.8
L	9.53	76.2
N	9.53	76.2

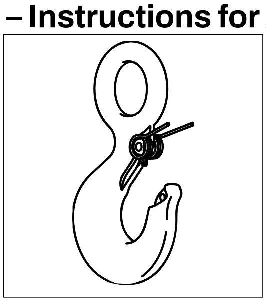
† The current SS-4055 latch kit and the PL latch will not fit new 319N, 320N, or 322N hooks. They will continue to be offered in both styles to service existing hooks. Important – The new S4320 latch kit will not fit the old 319, 320, or 322 hooks.

IMPORTANT – Instructions for Assembling S-4320 Latch on Crosby 320N Hooks



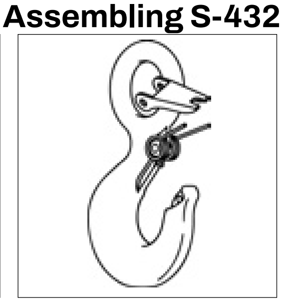
Step 1

1. Place hook at approximately a 45 degree angle with the cam up.



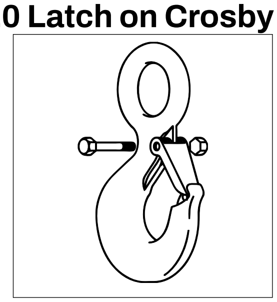
Step 2

2. Position coils of spring over cam with legs of spring pointing toward point of hook and loop of spring positioned down and lying against the hook.



Step 3

3. Position latch to side of hook points. Slide latch onto spring legs between lockplate and latch body until latch is partially over hook cam. Then depress latch and spring until latch clears point of hook.

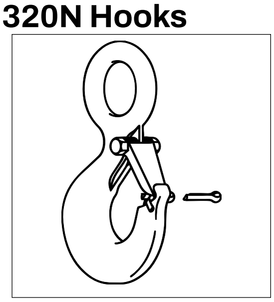


Steps 4, 5, & 6

4. Line up holes in latch with hook cam.

5. Insert bolt through latch, spring, and cam.

6. Tighten self-locking nut on one end of bolt.

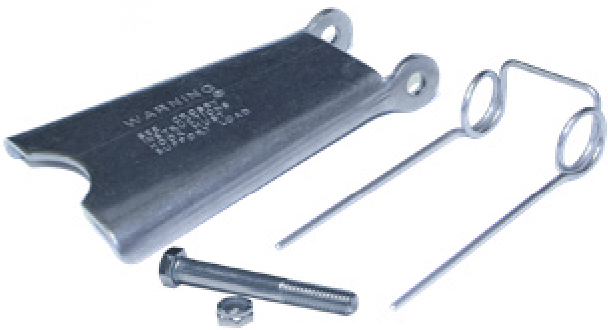


Step 7 – For Personnel Lifting

7. With latch in closed position and rigging resting in bowl of hook, insert cotter pin through hook tip and secure by bending prongs.

Crosby® HOOK LATCH KIT

WARNINGS & APPLICATION INSTRUCTIONS



SS-4055
(Stainless Steel)

IMPORTANT SAFETY INFORMATION - READ & FOLLOW

- Always inspect hook and latch before using.
- Never use a latch that is distorted or bent.
- Always make sure spring will force the latch against the tip of the hook.
- Always make sure hook supports the load. The latch must never support the load (See Figures 1 & 2).
- When placing two (2) sling legs in hook, make sure the angle between legs is small enough and the legs are not tilted so that nothing bears against the bottom of the latch (See Figures 3 & 4).
- Latches are intended to retain loose sling or devices under slack conditions.
- Latches are not intended to be an anti-fouling device.

⚠ WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **See OSHA Rule 1926.1431(g)(1)(i)(A) and 1962.1501(g)(4)(iv)(B) A hook and this style latch must not be used for lifting personnel.**
- **Hook must always support the load. The load must never be supported by the latch.**
- **Read and understand these instructions before using hook and latch.**

RIGHT

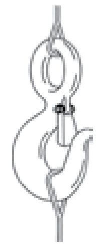


Figure 1

WRONG

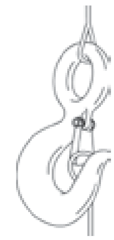


Figure 2

RIGHT

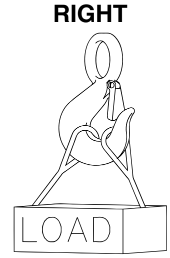


Figure 3

WRONG

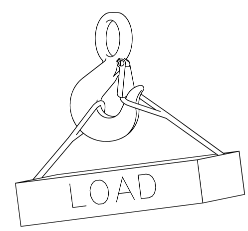
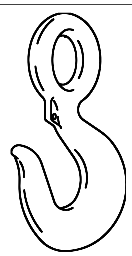


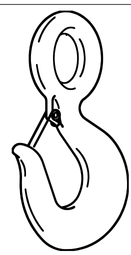
Figure 4

IMPORTANT – Instructions for Assembling Model SS-4055 Latch on Crosby Hooks



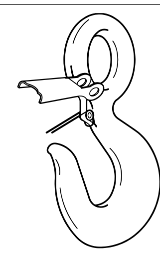
Step 1

1. Place hook at approximately a 45 degree angle with the cam up.



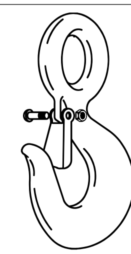
Step 2

2. Position coils of spring over cam with tines of spring pointing toward point of hook and loop of spring positioned down and lying against the hook.



Step 3

3. Position latch over tines of spring with ears partially over hook cam. Swing latch to one side of hook, point and depress latch and spring until latch clears point of hook.



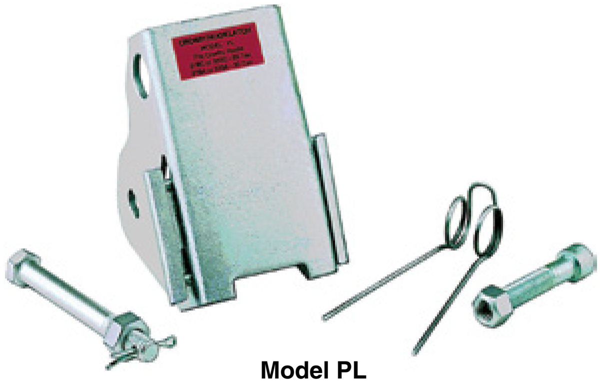
Steps 4, 5, & 6

4. Line up holes in latch with hook cam.
5. Insert bolt through latch, spring, and cam.
6. Tighten self-locking nut on one end of bolt.

This application/warning information apply to Crosby products only.

Crosby® MODEL PL HOOK LATCH KIT

WARNINGS & APPLICATION INSTRUCTIONS



Model PL

IMPORTANT SAFETY INFORMATION - READ & FOLLOW

(Pat. USA & Canada)

- Always inspect hook and latch before using.
- Never use a latch that is distorted or bent.
- Always make sure spring will force the latch against the tip of the hook.
- Always make sure hook supports the load. The latch must never support the load (See Figures 1 & 2).
- When placing two (2) sling legs in hook, make sure the angle between the legs is less than 90° and if the hook or load is tilted, nothing bears against the bottom of this latch (See Figures 3 & 4).
- Latches are intended to retain loose sling or devices under slack conditions.
- Latches are not intended to be an anti-fouling device.

⚠ WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **See OSHA Rule 1926.1431(g)(1)(i)(A) and 1926.1501(g)(4)(iv)(B)for Personnel Hoisting by Cranes or Derricks. A Crosby or McKissick Hook with a positive Locked PL or S-4320 Latch may be used to Lift Personnel.**
- **Hook must always support the load. The load must never be supported by the latch.**
- **DO NOT use this latch in applications requiring non-sparking.**
- **Read and understand these instructions before using hook and latch.**

Right

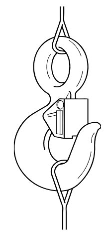


Figure 1

Wrong

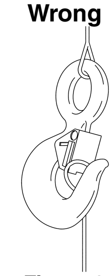


Figure 2

Right

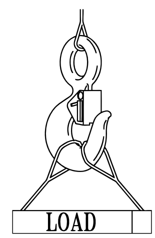


Figure 3

Wrong

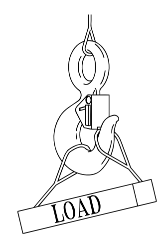
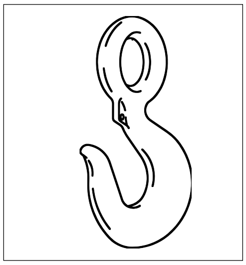


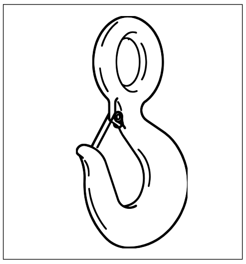
Figure 4

IMPORTANT - Instructions for Assembling Model PL Latch on Crosby or McKissick Hooks



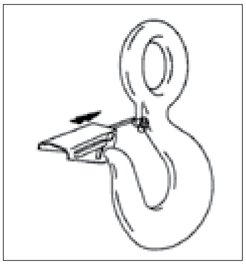
Step 1

1. Place hook at approximately a 45 degree angle with the cam up.



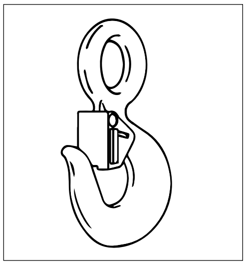
Step 2

2. Position coils of spring over cam with legs of spring pointing toward point of hook and loop of spring positioned down and lying against the hook.



Step 3

3. Position latch to side of hook points. Slide latch onto spring legs between lockplate and latch body until latch is partially over hook cam. Then depress latch and spring until latch clears point of hook.

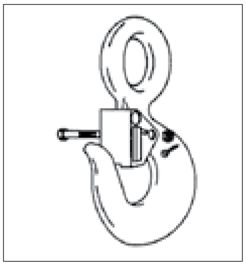


Steps 4, 5, & 6

4. Line up holes in latch with hook cam.

5. Insert bolt through latch, spring, and cam.

6. Tighten self-locking nut on one end of bolt.



Step 7 — For Personnel Lifting

7. With latch in closed position and rigging resting in bowl of hook, insert bolt through latch and secure with nut and cotter pin.When bolt, nut and cotter pin are not being used, store them in a designated place upon the personnel platform.

Crosby® SHUR-LOC® HOOKS

WARNING & APPLICATION INSTRUCTIONS

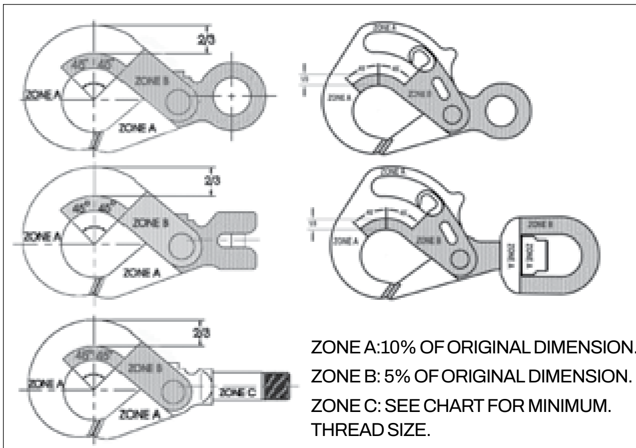
Important Safety Information - Read and Follow

- A visual periodic inspection for cracks, nicks, wear, gouges and deformation as part of a comprehensive documented inspection program, should be conducted by trained personnel in compliance with the schedule in ASME B30.10.
- For hooks used in frequent load cycles, pulsating loads, or severe duty as defined by ASME B30.10, the hook and threads should be periodically inspected by Magnetic Particle or Dye Penetrant (Note: Some disassembly may be required).
- Never use a hook whose throat opening has been increased 5%, not to exceed 1/4",(6mm) or shows any visible apparent bend or twist from the plane of the unbent hook, or is in any other way distorted or bent. **NOTE: A latch will not work properly on a hook with a bent or worn tip.**
- Never use a hook that is worn beyond the limits shown in Figure 1.
- Remove from service any hook with a crack, nick, or gouge. Hooks with a nick, or gouge shall be repaired by grinding lengthwise, following the contour of the hook, provided that the reduced dimension is within the limits shown in Figure 1. Contact Crosby Engineering to evaluate any crack.
- Never repair, alter, rework, or reshape a hook by welding, heating, burning, or bending.
- Never side load, back load or tip load a hook. Side loading, back loading and tip loading are conditions that damage and reduce the capacity of the hook (See Figure 2).
- S-1326A can be used for limited rotations under load (infrequent, noncontinuous).
- Efficiency of synthetic sling material may be reduced when used in eye or bowl of hook.
- Always make sure the hook supports the load (See Figure 3).

Do not use hook tip for lifting (See Figure 4).

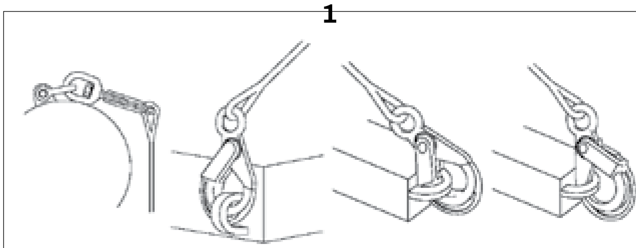
⚠WARNING

- **Loads may disengage from hook if proper procedures are not followed.**
- **A falling load may cause serious injury or death.**
- **Positive locking latch will unlock when trigger is depressed. Never use hook unless hook and latch are fully closed and locked.**
- **Keep body parts clear of pinch point between hook tip and hook latch when closing.**
- **Keep hand(s) from between throat of hook and sling or other device.**
- **Do not use hook tip for lifting.**
- **Do not use hook handle for lifiting.**
- **Do not rig the finger pull open, place objects in the finger pull area, or in any way inhibit complete and full operation of the finger pull mechanism.**
- **Shank threads may corrode and/or strip and drop the load.**
- **Remove securement nut to inspect threads for corrosion or to replace S-1326A bearing washers (2) and or S-13326 thrust bearing.**
- **Never apply more force than the hook's assigned Working Load Limit (WLL) rating.**
- **See OSHA Rule 1926.1431(g) and 1926.1501(g) for personnel hoisting by cranes or derricks. A Crosby 1318A, 1326A, 13326, 1316A, or 1317A hook may be used for lifting personnel.**
- **Use only genuine Crosby parts as replacements.**
- **Read and understand these instructions before using hook.**



ZONE A:10% OF ORIGINAL DIMENSION.
ZONE B: 5% OF ORIGINAL DIMENSION.
ZONE C: SEE CHART FOR MINIMUM. THREAD SIZE.

FIGURE 1

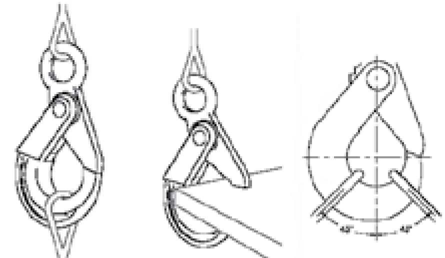


Side Load Wrong	Side Load Wrong	Back Load Wrong	Tip Load Wrong

FIGURE 2

- When placing two (2) sling legs in hook, make sure the angle from vertical to the leg nearest the hook tip is not greater than 45 degrees, and the included angle between the legs does not exceed 90 degrees* (See Figure 5).
- See ASME B30.10 “Hooks” for additional information.

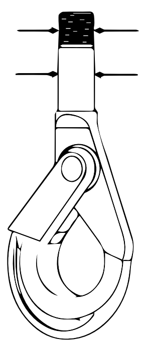
*For two legged slings with angles greater than 90°, use an intermediate link such as a master link or bolt type shackle to collect the legs of the slings. The intermediate link can then be placed over the hook to provide an en línea load on the hook. This approach must also be used when using slings with three or more legs.



RIGHT FIGURE 3 | WRONG FIGURE 4 | RIGHT FIGURE 5

Important Basic Machining and Thread Information: Read and Follow

- Wrong thread and/or shank size can cause stripping and loss of load.
- The maximum diameter is the largest diameter, after cleanup, that could be expected after allowing for straightness, pits, etc.
- All threads must be Class 2 or better.
- The minimum thread length engaged in the nut should not be less than one (1) thread diameter.
- Hook shanks are not intended to be swaged on wire rope or rod.
- Hook shanks are not intended to be drilled (length of shank) and internally threaded.
- Crosby cannot assume responsibility for, (A) the quality of machining, (B) the type of application, or (C) the means of attachment to the power source or load.
- Consult the Crosby Hook Identification & Working Load Limit Chart (See below) for the minimum thread size for assigned Working Load Limits (WLL).†
- Remove from service any Hook which has threads corroded more than 20% of the nut engaged length.



Crosby® Hook Identification & Working Load Limit Chart†

S-1316A & S-1317A Only Grade 100 Chain			S-1318A, S-1326A					S-1318A Only ††		
Chain Size		Working Load Limit (lb)** 4:1	Grade 100 Chain			Wire Rope XXIP Mechanical Splice		Maximum Shank Diameter		Minimum Thread Size (in)
			Chain Size		Working Load Limit (lb)** 4:1	Wire Rope Size (in)	Working Load Limit (lb)* 5:1			
(in)	(mm)		(in)	(mm)				(in)	(mm)	
—	6	3200	—	6	3200	5/16	2200	.72	18	5/8 - 11 UNC
1/4	7	4300	1/4	7 - 8	4300	7/16	4200	.94	24	5/8 - 11 UNC
5/16	8	5700	5/16	8	5700	7/16	4200	.94	24	3/4 - 10 UNC
3/8	10	8800	3/8	10	8800	1/2	5600	1.06	27	3/4 - 10 UNC
1/2	13	15000	1/2	13	15000	5/8	8600	1.19	30	1-1/8 - 7 UNC
5/8	16	22600	5/8	16	22600	7/8	16600	1.38	35	1-3/8 - 6 UNC
3/4	18/20	35300	3/4	18-20	35300	1	22000	—	—	—
7/8	22	42700	7/8	22	42700	1-1/8	26500	—	—	—
1	26	59700	1	26	59700	1-1/4	32500	—	—	—

* Ultimate Load is 5 times the Working Load Limit based on XXIP Wire Rope.

** Ultimate Load is 4 times the Working Load Limit based on Grade 100 Chain.

+ Working Load Limit - The maximum mass of force which the product is authorized to support in general service when the pull is applied in-line, unless noted otherwise, with respect to the centerline of the product. This term is used interchangeably with the following terms: 1. WLL, 2. Rated Load Value, 3. SWL, 4. Safe Working Load, 5. Resultant Safe Working Load.

Technical Information

The following information aims to give advice and explain the most common questions in order to ensure safe and proper use of lifting equipment.

It is of the utmost importance that this information is known to the user, and in accordance with the Machinery Directive 2006/42/EC this information must be delivered to the customer.

See website or user instructions for assembly instructions.

Meets listed current specifications and standards at time of publication of this catalog.

All G80 and G100 Alloy Chains, and Alloy components meet or exceed the safety standards as prescribed by ASME B30.9 and OSHA 1910-184 for slings. Always comply with applicable International, National, Federal and local regulations as they govern worksite activity. Understand all governing laws and safety standards before any products are used. Contact your International, National, Federal and local standards and regulations organizations for reference assistance.

Extreme Environments

The in-service temperature affects the WLL as follows:

Temperature (°F)	Reduction of WLL			
	Gunnebo Grade 10 (400) chain	Crosby Grade 10 & Gunnebo Grade 10 (200) chain	Crosby & Gunnebo Grade 10 components	Crosby & Gunnebo Grade 8 chain & components
-40 to + 392 °F	0 %	0 %	0 %	0 %
+392 to + 572 °F	10 %	Not allowed	10 %	10 %
+572 to + 752 °F	25 %	Not allowed	25 %	25 %

Upon return to normal temperature, the sling reverts to its full capacity within the above temperature range. Chain slings should not be used above or below these temperatures. Note: A chain sling with Grade 10 (100) chain must not be used in temperatures above 392°F.

- Chain and components must not be used in alkaline (>pH10) or acidic conditions (<pH6).
- Comprehensive and regular examination must be carried out when used in severe or corrosive inducing environments.
- In uncertain situations consult your Gunnebo Industries dealer.

Surface Treatment

Note: Hot-dip galvanizing or plating is not allowed outside the control of the manufacturer.

Protect Yourself and Others

- Before each use the chain sling should be checked for obvious damage or deterioration.
- Know the weight of the load, the center of gravity and ensure it is ready to move and no obstacles will obstruct the lift.
- Check the conformity of the load with the WLL of the ID tag for the specific working configuration. Never use a sling without a legible valid ID tag!
- Prepare the landing site.
- Never overload a sling and avoid shock loading.
- Never use an improper sling configuration.
- Never use a worn out or damaged sling.
- Never ride on the load.
- Never walk or stand under a suspended load.
- Take into consideration that the load may swing or rotate.
- Watch your feet and fingers while loading/unloading.
- Always ensure that your back is clear.

General Advice

- Ensure that the sling is precisely as ordered.
- Ensure that the manufacturers certificate is in order.
- A metal I.D. Tag must always be attached to a chain sling, showing serial number, size, reach, rated capacity at angle of lift and manufacturer.
- Ensure that all details of the chain sling are recorded.
- Ensure that the staff using the chain sling has received the appropriate information and training.

Asymmetrical Loading Conditions

For unequally loaded chain legs we recommend that the WLL are determined as follows:

- 2-leg slings calculated as the corresponding 1-leg sling
- 3 and 4-leg slings calculated as the corresponding 1-leg sling. (If it is certain that 2-legs are equally carrying the major part of the load, it can be calculated as the corresponding 2-leg sling.)

Correct Use

Machining and threading specifications for BKT shank hook

- BKT self-locking hook shank machining limits are defined and are given in TABLE 2 and these limits are required for WLL's given. Failure to comply can result in stripped threads and loss of load. Hook shank threads shall end with a thread relief. Hook shank shall not be swaged to wire rope or rod. Hook shank shall not be drilled and internally threaded.

- Gunnebo Industries cannot assume responsibility for:

  1. Machining quality,
  2. Application,
  3. Attachment to power source or load

Table 2a

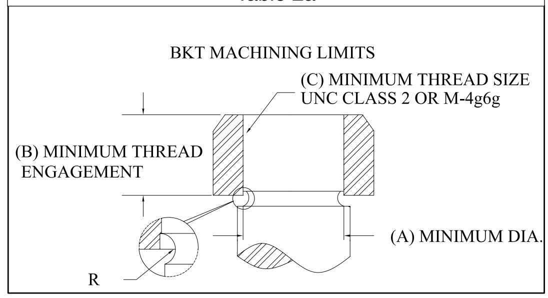


Table 2b				
English				
Trade Size		(A)	(B)	(C) Min. Thread
MM	IN	Dia.	Len.	Class 2
5/6	7/32	.430	.563	9/16-12 UNC
7/8	9/32	.485	.625	5/8-11 UNC
10	3/8	.600	.750	3/4-10 UNC
13	1/2	.820	1.00	1-8 UNC
16	5/8	1.048	1.25	1-1/4-7 UNC

Table 2c				
Metric				
Table Size		(A)	(B)	(C) Min. Thread
MM	IN	Dia.	Len.	Class 4g6g
5/6	7/32	11	14	M14x2
7/8	9/32	13	16	M16x2
10	3/8	16	20	M20x2.5
13	1/2	20	24	M24x3
16	5/8	25	30	M30x3.5

Safe use of self-locking hook

- Alloy steel BK self-locking hooks may be used to rig personnel platforms when lift system is in full compliance with OSHA 1926.1501(g) and passing the applicable inspection criteria.

- Loads shall be centered in the base (bowl/ saddle) of hook to prevent point loading of the hook (See Figure 1a, 1b & 1c).

- Hooks shall not be used in such a manner as to place a side load or back load on the hook (See Figure 2a & 2b).

- When using a device to close the throat opening of the hook, care shall be taken that the load is not carried by the closing device (See Figure 3a & 3b).

- Hands, fingers and body shall be kept from between hook and load.

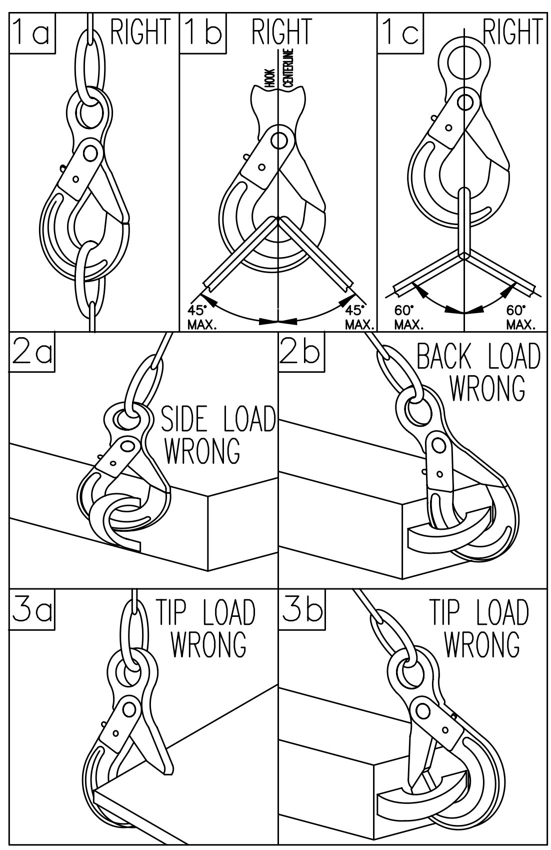
- The use of a hook with a latch does not preclude the inadvertent detachment of a slack sling or a load from the hook. Visual verification of proper hook engagement is required in all cases.

- Self-locking hooks shall be locked during use.

- When a hook is equipped with a latch, the latch should not be restrained from closing during use.

- Self-locking hooks shall not be rigged with more than two (2) sling legs in the hook saddle and sling leg angles shall not be greater than 45° from hook centerline (Figure 1b).

- Self-locking hooks shall be rigged with a master ring or shackle when three (3) or more sling legs are used or sling leg angles exceed 45° from hook centerline (Figure 1c).



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FORGED HOOKS

Miller's forged hooks are produced to the following comprehensive, international standards:

- DIN 15400- Materials, properties, capacities, stresses
- DIN 15401- Single hooks, forged
- DIN 15402- Double hooks, forged
- DIN 7540 - Eye hooks class 8 (grade 80), forged

DIN

DIN is the German Institute for Standardization (Deutsches Institut für Normung) and has been based in Berlin since 1917. DIN has historically developed the detailed and exacting standards used in German engineering and is the body that represents Germany in international standards organizations.

Forged Shank Hooks

Hooks are identified by hook number and material. Each hook number maintains identical dimensions across a range of different materials with its load capacity dependent on the material used. Shank hooks are forged and heat-treated for optimal strength and toughness and are available in single and double configurations in standard working loads to 1,100 tons. Standard design factor is 5:1.

Hook forgings are available in three increasingly stronger carbon and alloy steel materials:

DIN class P: Fine-grained carbon steel, St-E355/St-E420, similar to ASTMA573 Gr. 65
DIN class T: Structural low alloy steel, 34CrMo4 or 34CrNiMo6, similar to SAE 4135/4340
DIN class V: Super alloy steel, 34CrNiMo6 or 30CrNiMo8, similar to SAE 4340/4337

Forged hooks are also available to order, in bronze alloy or stainless steel for non-sparking or other special applications. Stainless steel hooks are forged from ANSI 304 or 316L stainless steel and bronze hooks are forged from a non-sparking aluminum-nickel bronze alloy (CuAl10Ni/C95500).

In the succeeding selection tables, working load limits (WLL) are indicated for each hook number depending on the material selected.

Standard safety latches are included with all hook forgings. Heavy duty positive locking latches are available upon request. See the special section of this catalog for instructions on how to order replacement latches.

Extended length shanks are available upon request.

All hooks are supplied with certifications by serial number for mechanical and chemical properties including Charpy testing, and include 100% ultrasonic and magnetic particle non-destructive examination. Hooks include markings of fixed distances ("y" dimension in tables) which allow confirmation that no deformation has occurred in use or during proof testing. Class V hooks are API-2C compliant.

Machined Hooks with Nut, Suspensions and Blocks

Miller can provide forged hooks fully machined and threaded to customer requirements, matching nut included and ready for assembly into the customer's structure. We can also provide the complete hook suspension with trunnion and thrust bearing, and since Miller is a block manufacturer, we can provide a complete hook block assembly for any crane type. Miller has a full range of standard blocks for mobile cranes and produces custom blocks to order for overhead cranes.

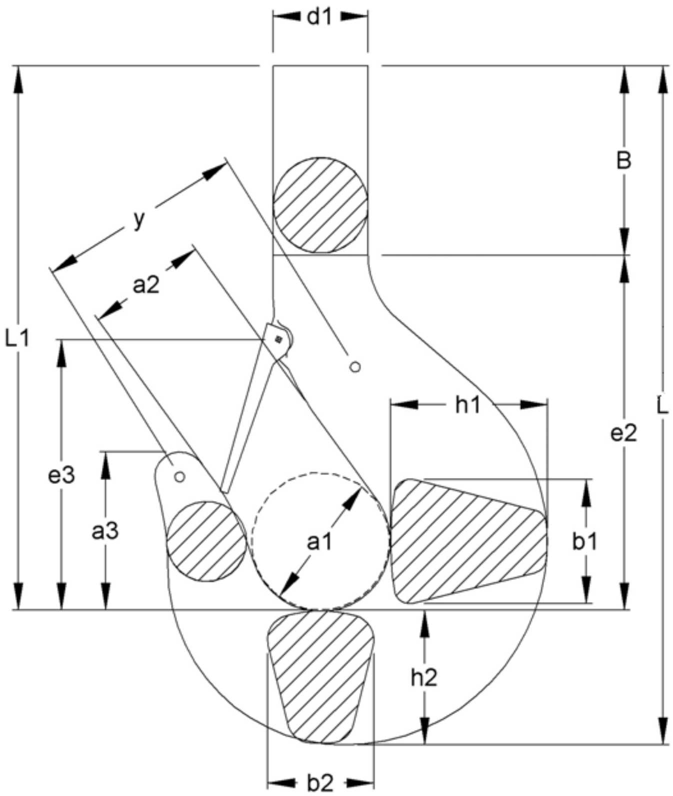
Forged Heavy-Duty Eye Hooks

Miller's DIN 7540 Grade 80 Eye Hook features the traditional eye-ring of circular cross section allowing the maximum degree of motion in the connection and is available standard in working loads up to 400 metric tons (440 short tons). These heavy duty eye hooks are forged from 34CrNiMo6V alloy steel, include a safety latch, and can be adapted for ROV use. The standard design factor is 4:1. See detailed eye hook tables following the shank hook tables.

MILLER®

FORGED HOOKS- SINGLE HOOKS DIN 15401

Imperial Dimensions



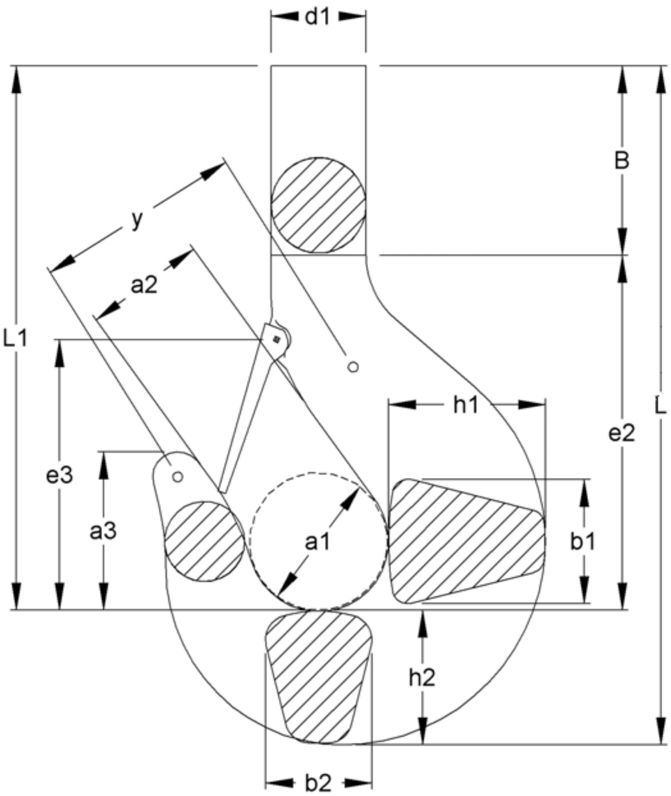
Model Number	Capacity Short Tons Carbon Class P	Capacity Short Tons Alloy Class T	Capacity Short Tons Super Alloy Class V	Capacity Short Tons STAIN-LESS STEEL	a1	a2	a3	B	b1	b2	d1	e2	e3	h1	h2	L	L1	y	Weight Lbs.
GS 1,6	3.5	6	6.9	3	2.21	1.77	2.52	3.07	1.77	1.50	1.42	5.75	4.65	2.21	1.89	10.72	8.83		10
GS 2,5	6	9	11	4.4	2.48	1.97	2.84	3.39	2.09	1.77	1.65	6.58	5.20	2.64	2.29	12.25	9.97		14
GS 4	9	13.8	18	6.9	2.80	2.21	3.15	3.74	2.48	2.09	1.89	7.49	5.83	3.15	2.64	13.87	11.23		19
GS 5	11	18	22	9	3.15	2.48	3.55	4.06	2.80	2.36	2.09	8.47	6.50	3.55	2.96	15.48	12.53		27
GS 6	14	22	28	11	3.55	2.80	3.98	5.52	3.15	2.64	2.36	9.46	7.29	3.94	3.35	18.32	14.97	5.12	38
GS 8	18	28	35	14	3.94	3.15	4.45	5.91	3.55	2.96	2.64	10.56	8.27	4.41	3.74	20.21	16.47	5.71	53
GS 10	22	35	44	18	4.41	3.55	5.00	6.54	3.94	3.35	2.96	11.27	8.71	4.93	4.18	21.99	17.81	6.30	75
GS 12	28	44	55	22	4.93	3.94	5.63	8.23	4.41	3.74	3.35	12.45	9.93	5.52	4.65	25.33	20.69	7.09	121
GS 16	35	55	69	28	5.52	4.41	6.30	9.38	4.93	4.18	3.74	14.07	11.03	6.30	5.20	28.64	23.44	7.88	170
GS 20	44	69	88	35	6.30	4.93	7.09	10.24	5.52	4.65	4.18	15.96	13.00	7.09	5.91	32.11	26.20	8.87	247
GS 25	55	88	110	44	7.09	5.52	7.96	11.03	6.30	5.20	4.65	17.93	14.18	7.88	6.70	35.66	28.96	10.05	353
GS 32	69	110	138	55	7.88	6.30	8.87	11.82	7.09	5.91	5.20	20.09	15.76	8.83	7.49	39.40	31.91	11.43	485
GS 40	88	138	176	69	8.83	7.09	9.93	13.32	7.88	6.70	5.91	22.34	17.61	9.85	8.35	44.01	35.66	12.61	683
GS 50	110	176	220	88	9.85	7.88	11.23	13.99	8.83	7.49	6.70	25.02	19.11	11.03	9.30	48.30	39.01	13.99	948
GS 63	138	220	276	110	11.03	8.83	12.61	16.15	9.85	8.35	7.49	27.97	21.67	12.41	10.44	54.57	44.13	15.76	1323
GS 80	176	276	353	138	12.41	9.85	14.11	18.44	11.03	9.30	8.35	31.6	23.56	13.99	11.82	61.86	50.04	17.73	1896
GS 100	220	353	441	176	13.99	11.03	15.84	20.21	12.41	10.44	9.26	35.54	27.11	15.76	13.20	68.95	55.75	19.90	2690
GS 125	276	441	551	n/a	15.76	12.41	17.73	22.46	13.99	11.82	10.44	40.19	29.55	17.73	14.78	77.42	62.65	22.46	3836
GS 160	353	551	705	n/a	17.73	13.99	19.90	25.41	15.76	13.20	11.82	45.11	32.51	19.70	16.75	87.27	70.53	25.22	5467
GS 200	441	705	882	n/a	19.70	15.76	22.26	30.38	17.73	14.78	13.20	50.24	35.46	22.06	18.72	99.41	80.69	28.37	7540
GS 250	705	882	1102	n/a	22.83	17.71	25.00	34.44	19.68	16.73	14.76	56.29	38.58	24.80	20.86	111.6	90.75	39.96	10582

•Design factor is 5:1 •For dimensional tolerances and extended shank options see page 70 •Capacities listed are per DIN 15400 drive group 1Am, which generally reflect loading conditions for mobile crane applications. The 1Am drive group can generally be approximated to CMAA Specification No. 70, Service Class B. For application-specific information consult the relevant standard or contact Miller for assistance.

MILLER®

FORGED HOOKS- SINGLE HOOKS DIN 15401

Metric Dimensions

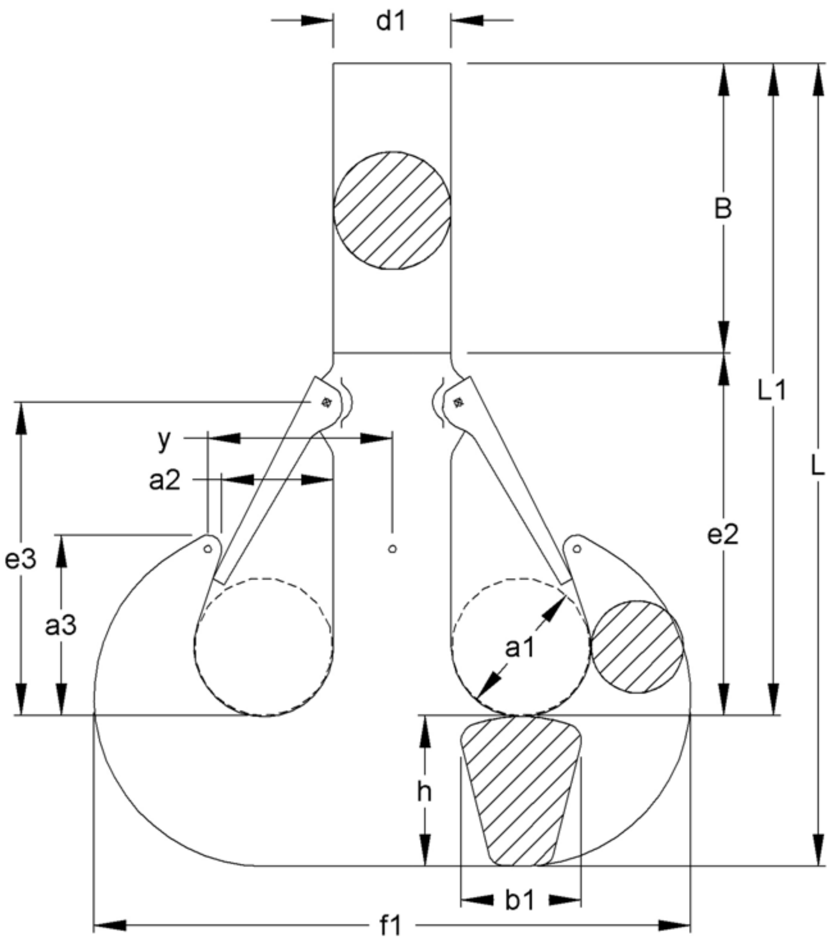


Model Number	Capacity Metric Tons Carbon Class P	Capacity Metric Tons Alloy Class T	Capacity Metric Tons Super Alloy Class V	Capacity Metric Tons STAIN-LESS STEEL	a1	a2	a3	B	b1	b2	d1	e2	e3	h1	h2	L	L1	y	Weight Kg
GS 1,6	3.2	5	6.3	2.5	56	45	64	78	45	38	36	146	118	56	48	272	224		4.5
GS 2,5	5	8	10	4	63	50	72	86	53	45	42	167	132	67	58	311	253		6.3
GS 4	8	12.5	16	6.3	71	56	80	95	63	53	48	190	148	80	67	352	285		8.8
GS 5	10	16	20	8	80	63	90	103	71	60	53	215	165	90	75	393	318		12.3
GS 6	12.5	20	25	10	90	71	101	140	80	67	60	240	185	100	85	465	380	130	17.1
GS 8	16	25	32	12.5	100	80	113	150	90	75	67	268	210	112	95	513	418	145	24
GS 10	20	32	40	16	112	90	127	166	100	85	75	286	221	125	106	558	452	160	34
GS 12	25	40	50	20	125	100	143	209	112	95	85	316	252	140	118	643	525	180	55
GS 16	32	50	63	25	140	112	160	238	125	106	95	357	280	160	132	727	595	200	77
GS 20	40	63	80	32	160	125	180	260	140	118	106	405	330	180	150	815	665	225	112
GS 25	50	80	100	40	180	140	202	280	160	132	118	455	360	200	170	905	735	255	160
GS 32	63	100	125	50	200	160	225	300	180	150	132	510	400	224	190	1000	810	290	220
GS 40	80	125	160	63	224	180	252	338	200	170	150	567	447	250	212	1117	905	320	310
GS 50	100	160	200	80	250	200	285	355	224	190	170	635	485	280	236	1226	990	355	430
GS 63	125	200	250	100	280	224	320	410	250	212	190	710	550	315	265	1385	1120	400	600
GS 80	160	250	320	125	315	250	358	468	280	236	212	802	598	355	300	1570	1270	450	860
GS 100	200	320	400	160	355	280	402	513	315	265	235	902	688	400	335	1750	1415	505	1220
GS 125	250	400	500	n/a	400	315	450	570	355	300	265	1020	750	450	375	1965	1590	570	1740
GS 160	320	500	640	n/a	450	350	505	645	400	335	300	1145	825	500	425	2215	1790	640	2480
GS 200	400	640	800	n/a	500	400	565	771	450	375	335	1275	900	560	475	2523	2048	720	3420
GS 250	640	800	1000	n/a	580	450	635	875	500	425	375	1430	980	630	530	2835	2305	1015	4800

•Design factor is 5:1 •For dimensional tolerances and extended shank options see page 70 •Capacities listed are per DIN 15400 drive group 1Am, which generally reflect loading conditions for mobile crane applications. The 1Am drive group can generally be approximated to CMAA Specification No. 70, Service Class B. For application-specific information consult the relevant standard or contact Miller for assistance.

FORGED HOOKS- DUPLEX HOOK DIN 15402

Imperial Dimensions

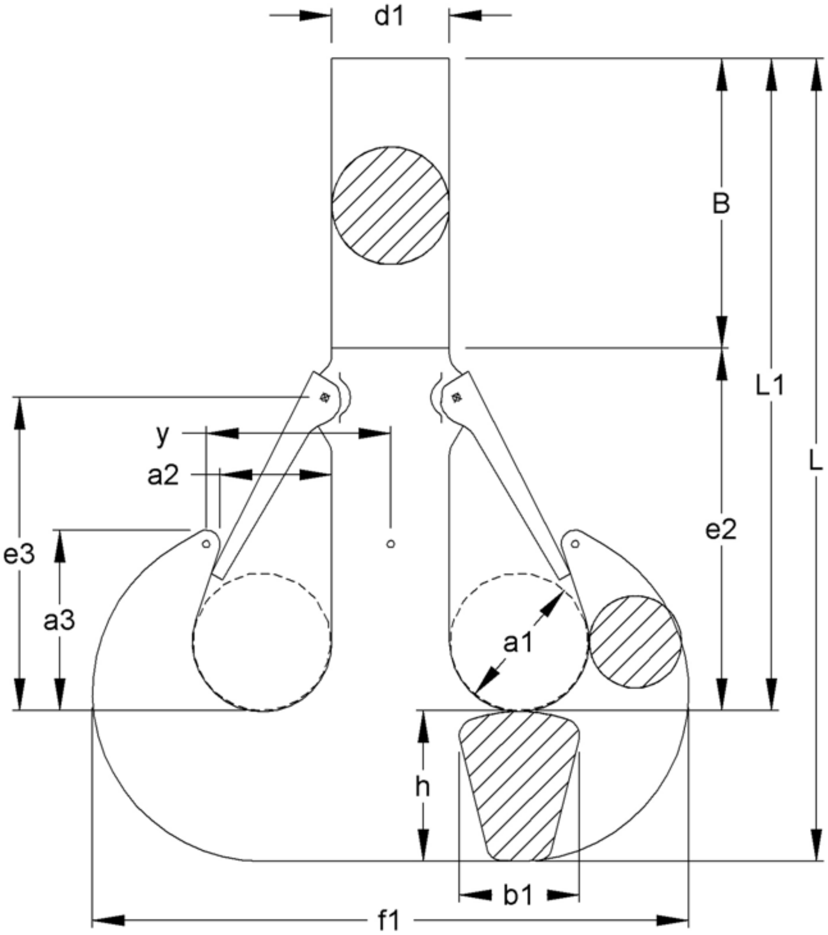


Model Number	Capacity Short Tons Carbon Class P	Capacity Short Tons Alloy Class T	Capacity Short Tons Super Alloy Class V	a1	a2	a3	B	f1	b1	e2	d1	e3	h	L	L1	y	Weight Lbs.
GD 6	14	22	28	2.80	2.21	3.62	7.21	11.86	2.36	7.56	2.36	6.30	2.96	17.73	14.78	3.66	37
GD 8	18	28	35	3.15	2.48	4.06	7.76	13.28	2.64	8.59	2.64	7.17	3.35	19.70	16.35	4.12	56
GD 10	22	35	44	3.55	2.80	4.57	8.67	14.85	2.96	9.06	2.96	7.56	3.74	21.47	17.73	4.63	80
GD 12	28	44	55	3.94	3.15	5.12	10.17	16.59	3.35	9.93	3.35	8.27	4.18	24.27	20.09	5.22	111
GD 16	35	55	69	4.41	3.55	5.75	11.66	18.56	3.74	11.19	3.74	9.34	4.65	27.50	22.85	5.85	157
GD 20	44	69	88	4.93	3.94	6.42	13.08	20.92	4.18	12.53	4.18	10.44	5.20	30.81	25.61	6.52	219
GD 25	55	88	110	5.52	4.41	7.17	13.67	23.56	4.65	14.89	4.65	12.41	5.91	34.08	28.17	7.29	304
GD 32	69	110	138	6.30	4.93	8.08	15.29	26.48	5.20	15.84	5.20	13.20	6.70	37.82	31.13	8.16	434
GD 40	88	138	176	7.09	5.52	9.06	17.14	29.71	5.91	17.73	5.91	14.78	7.49	42.36	34.87	9.18	631
GD 50	110	176	220	7.88	6.30	10.24	18.16	33.17	6.70	19.86	6.70	16.55	8.35	46.37	38.02	10.44	869
GD 63	138	220	276	8.83	7.09	11.50	21.20	37.19	7.49	21.75	7.49	18.12	9.30	52.24	42.95	11.70	1206
GD 80	176	276	353	9.85	7.88	12.81	24.31	41.84	8.35	24.35	8.35	20.29	10.44	59.10	48.66	13.04	1673
GD 100	220	353	441	11.03	8.83	14.34	26.99	46.73	9.26	27.19	9.26	22.66	11.82	66.00	54.18	14.58	2337
GD 125	276	441	551	12.41	9.85	16.08	30.57	52.40	10.44	30.50	10.44	25.41	13.20	74.27	61.07	16.33	3287
GD 160	353	551	705	13.99	11.03	18.05	34.45	59.30	11.82	34.25	11.82	28.57	14.78	83.53	68.75	18.36	4663
GD 200	441	705	882	15.76	12.41	20.29	40.82	66.39	13.20	37.83	13.20	31.52	16.75	95.47	78.72	20.59	6647
GD 250	551	882	1102	17.73	13.99	22.85	47.05	74.27	14.78	41.53	14.78	34.48	18.72	107.3	88.65	23.15	9409

•Design factor is 5:1 •For dimensional tolerances and extended shank options see page 70 •Capacities listed are per DIN 15400 drive group 1Am, which generally reflect loading conditions for mobile crane applications. The 1Am drive group can generally be approximated to CMAA Specification No. 70, Service Class B. For application-specific information consult the relevant standard or contact Miller for assistance.

FORGED HOOKS- DUPLEX HOOK DIN 15402

Metric Dimensions



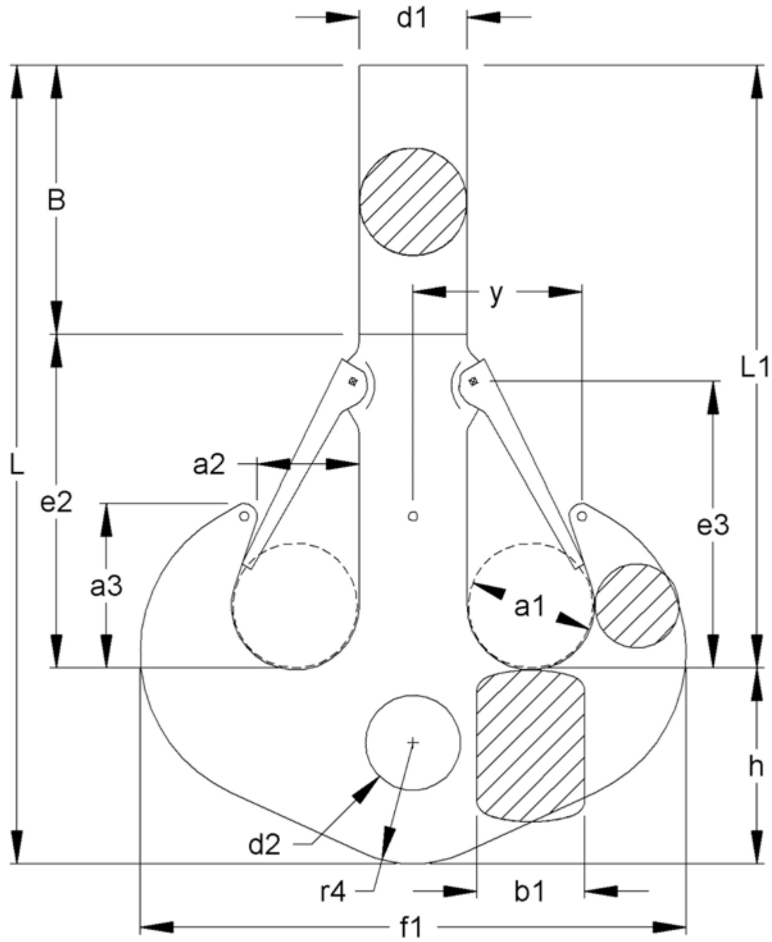
Model Number	Capacity Metric Tons Carbon Class P	Capacity Metric Tons Alloy Class T	Capacity Metric Tons Super Alloy Class V	a1	a2	a3	B	f1	b1	e2	d1	e3	h	L	L1	y	Weight Kg
GD 6	12.5	20	25	71	56	92	183	301	60	192	60	160	75	450	375	93	16.8
GD 8	16	25	32	80	63	103	197	337	67	218	67	182	85	500	415	104.5	25.3
GD 10	20	32	40	90	71	116	220	377	75	230	75	192	95	545	450	117.5	35.3
GD 12	25	40	50	100	80	130	258	421	85	252	85	210	106	616	510	132.5	50
GD 16	32	50	63	112	90	146	296	471	95	284	95	237	118	698	580	148.5	71
GD 20	40	63	80	125	100	163	332	531	106	318	106	265	132	782	650	165.5	100
GD 25	50	80	100	140	112	182	347	598	118	378	118	315	150	865	715	185	138
GD 32	63	100	125	160	125	205	388	672	132	402	132	335	170	960	790	207	197
GD 40	80	125	160	180	140	230	435	754	150	450	150	375	190	1075	885	233	286
GD 50	100	160	200	200	160	260	461	842	170	504	170	420	212	1177	965	265	394
GD 63	125	200	250	224	180	292	538	944	190	552	190	460	236	1326	1090	297	547
GD 80	160	250	320	250	200	325	617	1062	212	618	212	515	265	1500	1235	331	760
GD 100	200	320	400	280	224	364	685	1186	235	690	235	575	300	1675	1375	370	1060
GD 125	250	400	500	315	250	408	776	1330	265	774	265	645	335	1885	1550	414.5	1491
GD 160	320	500	500	355	280	458	875	1505	300	870	300	725	375	2120	1745	466	2115
GD 200	400	640	800	400	315	515	1037	1685	335	961	335	800	425	2423	1998	522.5	3015
GD 250	500	800	1000	450	355	580	1195	1885	375	1055	375	875	475	2725	2250	587.5	4268

•Design factor is 5:1 •For dimensional tolerances and extended shank options see page 70 •Capacities listed are per DIN 15400 drive group 1Am, which generally reflect loading conditions for mobile crane applications. The 1Am drive group can generally be approximated to CMAA Specification No. 70, Service Class B. For application-specific information consult the relevant standard or contact Miller for assistance.

MILLER®

FORGED HOOKS- DUPLEX HOOK DIN 15402-B

Imperial Dimensions

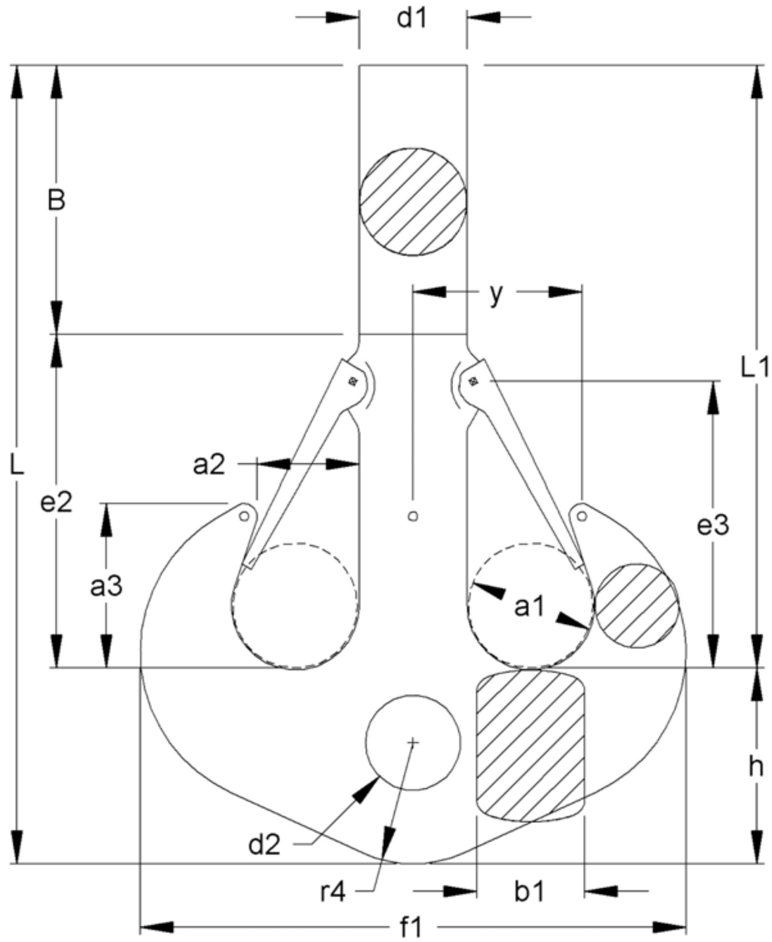


Model Number	Capacity Short Tons Carbon Class P	Capacity Short Tons Alloy Class T	Capacity Short Tons Super Alloy Class V	a1	a2	a3	B	f1	b1	d2	e2	d1	e3	h	L	L1	r4	y	Weight Lbs.
GDB 10	22	35	44	3.55	2.80	4.57	8.67	14.85	2.96	2.92	9.06	2.96	7.56	5.12	22.85	17.73	3.35	4.63	90
GDB 12	28	44	55	3.94	3.15	5.12	10.17	16.59	3.35	3.07	9.93	3.35	8.27	5.91	26.00	20.09	3.74	5.22	126
GDB 16	35	55	69	4.41	3.55	5.75	11.66	18.56	3.74	3.39	11.19	3.74	9.34	6.70	29.55	22.85	4.17	5.85	181
GDB 20	44	69	88	4.93	3.94	6.42	13.08	20.92	4.18	3.78	12.53	4.18	10.44	7.49	33.10	25.61	4.65	6.52	254
GDB 25	55	88	110	5.52	4.41	7.17	13.67	23.56	4.65	4.18	14.89	4.65	12.41	8.35	36.52	28.17	5.2	7.29	353
GDB 32	69	110	138	6.30	4.93	8.08	15.29	26.48	5.20	4.57	15.84	5.20	13.20	9.30	40.42	31.13	5.9	8.16	505
GDB 40	88	138	176	7.09	5.52	9.06	17.14	29.71	5.91	5.16	17.73	5.91	14.78	10.44	45.31	34.87	6.7	9.18	728
GDB 50	110	176	220	7.88	6.30	10.24	18.16	33.17	6.70	5.75	19.86	6.70	16.55	11.82	49.84	38.02	7.48	10.44	1010
GDB 63	138	220	276	8.83	7.09	11.50	21.20	37.19	7.49	6.62	21.75	7.49	18.12	13.20	56.15	42.95	8.35	11.70	1407
GDB 80	176	276	353	9.85	7.88	12.81	24.31	41.84	8.35	7.41	24.35	8.35	20.29	14.78	63.43	48.66	9.29	13.04	1967
GDB 100	220	353	441	11.03	8.83	14.34	26.99	46.73	9.26	8.20	27.19	9.26	22.66	16.75	70.92	54.18	10.4	14.58	2751
GDB 125	276	441	551	12.41	9.85	16.08	30.57	52.40	10.44	9.26	30.50	10.44	25.41	18.72	79.79	61.07	11.8	16.33	3873
GDB 160	353	551	705	13.99	11.03	18.05	34.45	59.30	11.82	10.24	34.25	11.82	28.57	20.88	89.64	68.75	13.2	18.36	5512
GDB 200	441	705	882	15.76	12.41	20.29	40.82	66.39	13.20	11.11	37.83	13.20	31.52	23.64	100.8	77.22	14.8	20.59	7848
GDB 250	551	882	1102	17.73	13.99	22.85	47.05	74.27	14.78	12.29	41.54	14.78	34.48	26.40	113.4	87.07	16.7	23.15	11096

•Design factor is 5:1 •For dimensional tolerances and extended shank length options see page 70. •Hole tolerance +2%/-0
•Capacities listed are per DIN 15400 drive group 1Am, which generally reflect loading conditions for mobile crane applications. The 1Am drive group can generally be approximated to CMAA Specification No. 70, Service Class B. For application-specific information consult the relevant standard or contact Miller for assistance.

FORGED HOOKS- DUPLEX HOOK DIN 15402-B

Metric Dimensions



Model Number	Capacity Metric Tons Carbon Class P	Capacity Metric Tons Alloy Class T	Capacity Metric Tons Super Alloy Class V	a1	a2	a3	B	f1	b1	d2	e2	d1	e3	h	L	L1	r4	y	Weight Kg
GDB 10	20	32	40	90	71	116	220	377	75	74	230	75	192	130	580	450	85	117.5	41
GDB 12	25	40	50	100	80	130	258	421	85	78	252	85	210	150	660	510	95	132.5	57
GDB 16	32	50	63	112	90	146	296	471	95	86	284	95	237	170	750	580	106	148.5	82
GDB 20	40	63	80	125	100	163	332	531	106	96	318	106	265	190	840	650	118	165.5	115
GDB 25	50	80	100	140	112	182	347	598	118	106	378	118	315	212	927	715	132	185	160
GDB 32	63	100	125	160	125	205	388	672	132	116	402	132	335	236	1026	790	150	207	229
GDB 40	80	125	160	180	140	230	435	754	150	131	450	150	375	265	1150	885	170	233	330
GDB 50	100	160	200	200	160	260	461	842	170	146	504	170	420	300	1265	965	190	265	458
GDB 63	125	200	250	224	180	292	538	944	190	168	552	190	460	335	1425	1090	212	297	638
GDB 80	160	250	320	250	200	325	617	1062	212	188	618	212	515	375	1610	1235	236	331	892
GDB 100	200	320	400	280	224	364	685	1186	235	208	690	235	575	425	1800	1375	265	370	1248
GDB 125	250	400	500	315	250	408	776	1330	265	235	774	265	645	475	2025	1550	300	414.5	1757
GDB 160	320	500	640	355	280	458	875	1505	300	260	875	300	725	530	2275	1745	335	466	2500
GDB 200	400	640	800	400	315	515	1037	1685	335	282	1037	335	800	600	2560	1960	375	522.5	3560
GDB 250	500	800	1000	450	355	580	1195	1885	375	312	1195	375	875	670	2880	2210	425	587.5	5030

•Design factor is 5:1 •For dimensional tolerances and extended shank length options see page 70.
•Hole tolerance is +2%/-0 •Capacities listed are per DIN 15400 drive group 1Am, which generally reflect loading conditions for mobile crane applications. The 1Am drive group can generally be approximated to CMAA Specification No. 70, Service Class B. For application-specific information consult the relevant standard or contact Miller for assistance.

DIN SHANK HOOKS- TOLERANCES & OPTIONAL LONG SHANKS

DIN SHANK HOOKS- TOLERANCES AND OPTIONAL EXTENDED LENGTH SHANKS

METRIC DIMENSIONS

Single hooks per DIN 15401

Single Hook Number	Long Shank Add Value to L1 (mm)
2.5	62
4	100
5	92
6	50
8	102
10 to 20	150
25 to 125	200
160 and 200	Per Order

Single Hook Number	Dimensional Tolerances (mm)									
	a1	a2	a3	b1	b2	d1	e3	h1	h2	L1
1.6 and 2.5	+3 / -0									
4 and 5	+4 / -0									
6 and 8	+5 / -0									
10 to 16	+6 / -0									
20	+8 / -0									
25 and 32	+12 / -0		+/- 10	+16 / -0		+12 / -0	+/- 10	+20 / -0		
40 to 63	+16 / -0		+/- 12	+20 / -0		+16 / -0	+/- 12	+24 / -0		
80 to 125	+20 / -0		+/- 16	+25 / -0		+20 / -0	+/- 16	+32 / -0		
160 to 200	+25 / -0		+/- 20	+32 / -0		+25 / -0	+/- 20	+40 / -0		

Duplex hooks per DIN 15402

Duplex Hook Number	Long Shank Add Value to L1 (mm)
6	115
8	120
10 and 12	110
16 and 20	N/A
25	150
32 to 160	200
200 and 250	Per Order

Duplex Hook Number	Dimensional Tolerances (mm)							
	a1	a2	a3	b1	d1	e	h	L1
6 and 8	+5 / -0							
10 to 16	+6 / -0							
20 and 25	+8 / -0							
32	+12 / -0		+/- 10	+16 / -0	+12 / -0	+5 / -0	+20 / -0	
40 to 63	+16 / -0		+/- 12	+20 / -0	+16 / -0	+6 / -0	+24 / -0	
80 to 125	+20 / -0		+/- 16	+25 / -0	+20 / -0	+8 / -0	+32 / -0	
160 to 250	+25 / -0		+/- 20	+32 / -0	+25 / -0	+10 / -0	+40 / -0	

IMPERIAL DIMENSIONS

Single hooks per DIN 15401

Single Hook Number	Long Shank Add Value to L1 (in)
2.5	2.44
4	3.94
5	3.62
6	1.97
8	4.02
10 to 20	5.91
25 to 125	7.88
160 and 200	Per Order

Single Hook Number	Dimensional Tolerances (inches)									
	a1	a2	a3	b1	b2	d1	e3	h1	h2	L1
1.6 and 2.5	+.12 / -0									
4 and 5	+.16 / -0									
6 and 8	+.2 / -0									
10 to 16	+.24 / -0									
20	+.32 / -0									
25 and 32	+.47 / -0		+/- .39	+.63 / -0		+.47 / -0	+/- .39	+.79 / -0		
40 to 63	+.63 / -0		+/- .47	+.79 / -0		+.63 / -0	+/- .47	+.95 / -0		
80 to 125	+.79 / -0		+/- .63	+.99 / -0		+.79 / -0	+/- .63	+1.26 / -0		
160 to 200	+.99 / -0		+/- .79	+1.26 / -0		+.99 / -0	+/- .79	+1.58 / -0		

Duplex hooks per DIN 15402

Duplex Hook Number	Long Shank Add Value to L1 (in)
6	4.53
8	4.73
10 and 12	4.33
16 and 20	N/A
25	5.91
32 to 160	7.88
200 and 250	Per Order

Duplex Hook Number	Dimensional Tolerances (inches)							
	a1	a2	a3	b1	d1	e	h	L1
6 and 8	+.2 / -0							
10 to 16	+.24 / -0							
20 and 25	+.32 / -0							
32	+.47 / -0		+/- .39	+.63 / -0	+.47 / -0	+.2 / -0	+.79 / -0	
40 to 63	+.63 / -0		+/- .47	+.79 / -0	+.63 / -0	+.24 / -0	+.95 / -0	
80 to 125	+.79 / -0		+/- .63	+.99 / -0	+.79 / -0	+.32 / -0	+1.26 / -0	
160 to 250	+.99 / -0		+/- .79	+1.26 / -0	+.99 / -0	+.39 / -0	+1.58 / -0	

LATCH KITS FOR HOOKS

Miller Lifting Products
100A Sturbridge Rd.
Charlton MA 01507 USA
800.733.7071
sales@millerproducts.net

LATCH KITS FOR STANDARDIZED EUROPEAN SHANK HOOKS

DIN 15401 SINGLE HOOKS ▪ DIN15402 DOUBLE HOOKS

STANDARD OR HEAVY-DUTY LOCKING TYPE

*Attachment hardware included

Hook Frame Number		Standard	Heavy-Duty
Single	Duplex	Flapper	Positive-Locking
1.6	2.5	M291805106	M291807106
2.5	4	M291805106	M291807205
4	5	M291805004	M291807005
5	6	M291805005	M291807007
6	8	M291805006	M291807006
8	10	M291805008	M291807008
10	12	M291805010	M291807010
12	16	M291805012	M291807012
16	20	M291805016	M291807016
20	25	M291805020	M291807020
25	32	M291805025	M291807025
32	40	M291805032	M291807032
40	50	M291805040	M291807040
50	63	M291805050	M291807050
63	80	M291805063	M291807063
80	100	M291805080	M291807080
100	125	M291805100	M291807100
125	160	M291805125	M291807125
160	200	M291805160	M291807160
200	250	M291805200	M291807200

Standard **HD Pos Lock**

*Lock Pin Not Shown

How to locate hook frame number:

See hook markings located as indicated below. Hook frame number is followed by a single letter P, T,V, S, or M

LATCH KITS FOR EUROPEAN EYE HOOKS

DIN 7540 Standard Flapper Latch Kits

*Attachment hardware included

Hook Frame Number	Part Number
34	M291991034
35	M291991035
36	M291991036
37 and 38	M291991037
39, 40 and 41	M291991039
42	M291991042

* Positive locking latches by special order

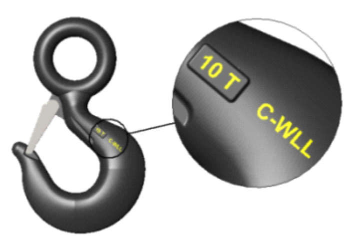
How to locate hook frame number:

See hook markings located as indicated below. Hook frame number is a two-digit number 34 through 42

LATCH KITS FOR ELD EYE HOOKS

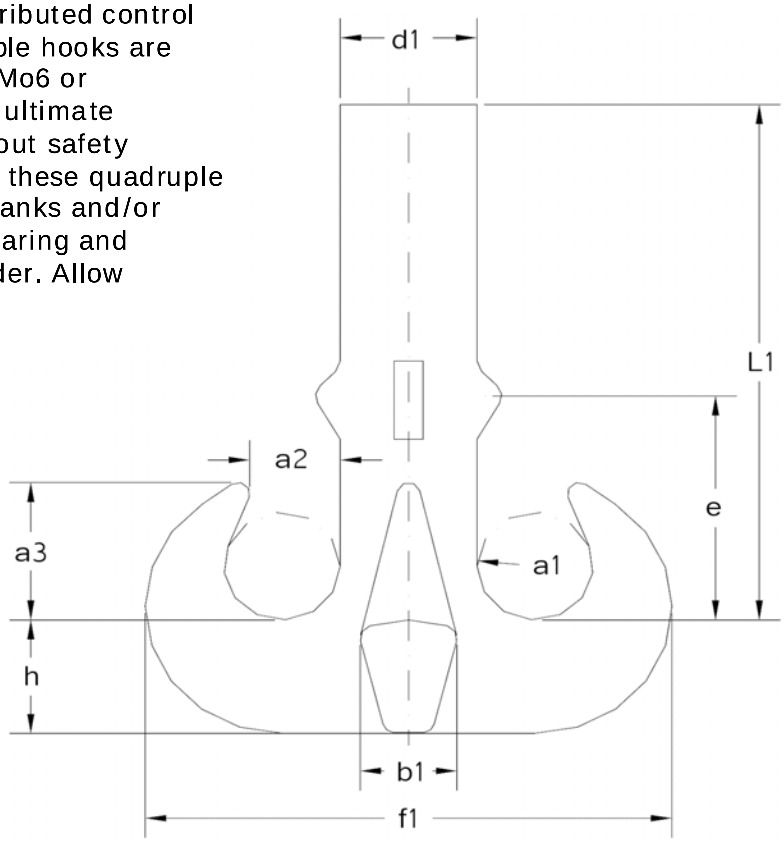
*Attachment hardware included

Working Load & Material	Part Number
3 TON CARBON	M291803000
5 TON CARBON / 7.5 TON ALLOY	M291804005
7.5 TON CARBON / 11.5 TON ALLOY	M291804007
10 TON CARBON / 16 TON ALLOY	M291804004
22 TON ALLOY	M291804009



FORGED HOOKS- QUADRUPLE HOOK - BASED ON DIN 15402

For heavy lift activity where more distributed control of the load is necessary, these quadruple hooks are forged from class V superalloy, 34CrNiMo6 or 30CrNiMo8. The static design factor to ultimate strength is 5:1. Available with or without safety latches. Shown here as a raw forging, these quadruple hooks are also available with longer shanks and/or machined with matching nut, thrust bearing and trunnion. Quad hooks are forged to order. Allow adequate time for delivery.



Metric Dimensions

Model Number	Capacity (SWL) Metric Tons*	a1	a2	a3	b1	d1	e	f1	h	L1**	Weight Kg
GQ16V	160	112	90	146	95	132	237	508	118	580	146
GQ20V	200	125	100	163	106	150	265	575	132	650	208
GQ25V	250	140	112	182	118	170	315	650	150	715	300
GQ32V	320	160	125	205	132	190	335	730	170	790	418
GQ40V	400	180	140	230	150	212	375	816	190	885	604
GQ50V	500	200	160	260	170	236	420	908	212	965	785

Imperial Dimensions

Model Number	Capacity (SWL) Short Tons*	a1	a2	a3	b1	d1	e	f1	h	L1**	Weight Lbs.
GQ16V	176	4.41	3.54	5.75	3.74	5.20	9.33	20.00	4.65	22.83	322
GQ20V	220	4.92	3.94	6.42	4.17	5.91	10.43	22.64	5.20	25.59	459
GQ25V	275	5.51	4.41	7.17	4.65	6.69	12.40	25.59	5.91	28.15	661
GQ32V	352	6.30	4.92	8.07	5.20	7.48	13.19	28.74	6.69	31.10	922
GQ40V	440	7.09	5.51	9.06	5.91	8.35	14.76	32.13	7.48	34.84	1332
GQ50V	551	7.87	6.30	10.24	6.69	9.29	16.54	35.75	8.35	37.99	1731

▪ Tolerance per DIN15402

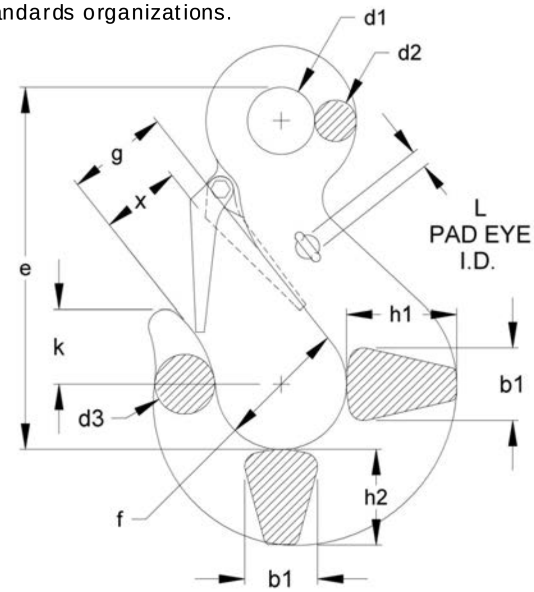
* Capacities based on FEM 1Bm/ISO M3 service class ** Additional length, L1 is available

FORGED EYE HOOKS- DIN 7540 GRADE 80

DIN

DIN is the German Institute for Standardization (Deutsches Institut für Normung) and has been based in Berlin since 1917. DIN has historically developed the detailed and exacting standards used in German engineering and is the body that represents Germany in international standards organizations.

- "**L**" suffix on model number indicates **large eye** version
- Forged from high-strength alloy steel 34CrNiMo6V
- Safe Working Loads from 40 to 400 metric tons
- Design factor 4:1 to ultimate strength
- Proof load is 2.5 times Safe Working Load
- Includes safety latch
- ROV modification (addition of pad eyes) available upon request
- Higher load capacities available upon request



Metric Dimensions

Model Number	Capacity (SWL) Metric Tons	MBL Metric ton	b1	d1	d1 Tolerance	d2	d3	e	f	g	h1	h2	K	L	X	Weight Kg
EH34	40	160	78	72	+1.9 / -3.7	44	66	388	140	109	118	103	80	19	90	31.5
EH34L	40	160	78	114	+/- 5.7	51	66	460	140	109	118	103	80	19	90	35
EH35	50	200	89	84	+1.9 / -3.7	50	74	442	158	124	135	116	90	19	103	46
EH35L	50	200	89	130	+/- 6.5	50.5	74	520	158	124	135	116	90	19	103	54.5
EH36	63	250	99	90	+2.3 / -4.7	56	83	494	176	138	151	130	101	19	114	63
EH36L	63	250	99	144	+/- 7.2	64	83	548	176	138	151	130	101	19	110	70
EH37	80	320	110	102	+2.3 / -4.7	63	93	610	198	155	168	145	113	19	131	80
EH38	100	400	125	116	+/- 5.0	74	120	650	225	175	195	172	133	19	147	125
EH39	150	600	140	130	+/- 6.5	86	140	765	250	200	225	199	160	19	166	250
EH40	200	800	160	150	+/- 7.5	102	161	850	275	225	260	237	195	25	*	365
EH41	250	1000	180	170	+/- 8.5	120	195	928	310	255	290	269	210	25	*	515
EH42	300	1200	200	190	+/- 9.5	140	223	1052	350	290	330	310	240	32	*	730
EH43	400	1600	240	210	+/- 10.5	170	240	1195	400	320	380	345	270	32	*	1055

Imperial Dimensions

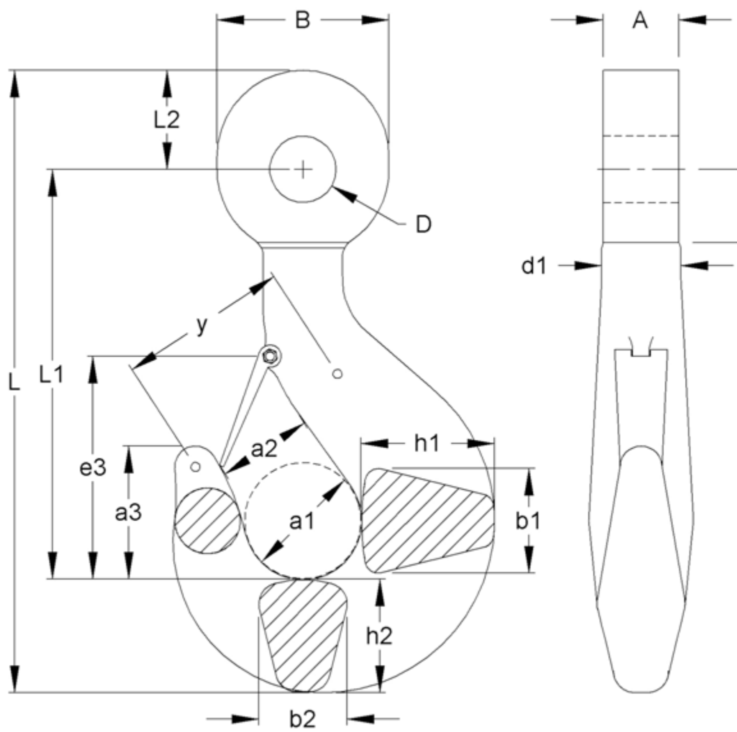
Model Number	Capacity (SWL) Short Tons	MBL Short ton	b1	d1	d1 Tolerance	d2	d3	e	f	g	h1	h2	k	L	X	Weight Lbs.
EH34	44	176	3.07	2.83	+.07 / -.14	1.73	2.60	15.28	5.51	4.29	4.65	4.06	3.15	.75	3.54	69
EH34L	44	176	3.07	4.49	+/- .22	2.01	2.60	18.11	5.51	4.29	4.65	4.06	3.15	.75	3.54	77
EH35	55	220	3.50	3.31	+.07 / -.14	1.97	2.91	17.40	6.22	4.88	5.31	4.57	3.54	.75	4.06	102
EH35L	55	220	3.50	5.12	+/- .25	1.99	2.91	20.47	6.22	4.88	5.31	4.57	3.54	.75	4.06	120
EH36	69	275	3.90	3.54	+.09 / -.18	2.20	3.27	19.45	6.93	5.43	5.94	5.12	3.98	.75	4.49	139
EH36L	69	275	3.90	5.67	+/- .28	2.52	3.27	21.57	6.93	5.43	5.94	5.12	3.98	.75	4.33	154
EH37	88	353	4.33	4.02	+.09 / -.18	2.48	3.66	24.02	7.88	6.10	6.61	5.71	4.45	.75	5.16	176
EH38	110	441	4.92	4.57	+/- .19	2.91	4.72	25.59	8.86	6.89	7.68	6.77	5.24	.75	5.79	276
EH39	165	661	5.51	5.12	+/- .25	3.39	5.51	30.12	9.84	7.87	8.86	7.83	6.30	.75	6.54	551
EH40	220	882	6.30	5.91	+/- .29	4.02	6.34	33.46	10.83	8.86	10.24	9.33	7.68	.98	*	805
EH41	276	1102	7.09	6.69	+/- .33	4.72	7.68	36.54	12.20	10.04	11.42	10.59	8.27	.98	*	1135
EH42	331	1322	7.87	7.48	+/- .37	5.51	8.78	41.42	13.78	11.42	12.99	12.20	9.45	1.26	*	1609
EH43	441	1764	8.27	8.27	+/- .41	6.69	9.45	47.05	15.75	12.60	14.96	13.58	10.63	1.26	*	2326

- Except where otherwise noted, dimensional tolerances for hooks through model EH37 are approximately ±5%, and increase somewhat for hooks larger than model EH37. Contact Miller for detailed tolerance data.
- Load capacities indicated are for operating temperatures between -40°C and 200°C (-40°F and 392°F). Outside this range check with Miller for reduced capacity limits.

***** x dimension please inquire.

MILLER®

FORGED HOOKS- HEAVY DUTY SHACKLE EYE HOOK



Heavy Duty Eye Hooks for Shackles

Miller Heavy Duty Eye Hooks are specifically intended for use with shackles or other pin-type connections. The cylindrical cross-section of the eye assures uniform pin loading while limiting relative hook motion.

Based also the DIN norms, they are typically forged from the strongest DIN "V" material class (see above) and carry a 5:1 design safety factor and a positive locking latch. Adaptable for ROV (Remotely Operated Vehicles) use, these eye hooks fit standard shackles and also are available in customized versions. Deformation indicators are also included on these heavy duty models.

Metric Dimensions

Model Number	Capacity Metric Tons	A	a1	a2	a3	B	b1	b2	D	d1	e3	h1	h2	b	L1	L2	L	y	Weight Kg
GSOJ8T	30	52	100	80	113	115	90	75	43	67	210	112	95	51	419	65	569	145	17
GSOJ8V	40	65	100	80	113	140	90	75	52.5	67	210	112	95	59	427	80	602	145	24
GSOJ12V	55	74.5	125	100	143	155	112	95	60.5	85	252	140	118	68	484	90	692	180	55
GSOJ20V	85	96.5	160	125	180	195	140	118	73	106	330	180	150	80	585	112	847	225	112
GSOJ25V	120	119	180	140	202	235	160	132	86	118	360	200	170	91	646	135	951	255	160
GSOJ32V	150	125	200	160	225	250	180	150	98.5	132	400	224	190	112	722	145	1057	290	220
GSOJ40T	175	131.5	224	180	252	285	200	170	109.5	150	447	250	212	126	793	165	1170	320	310
GSOJ40V	200	144.5	224	180	252	320	200	170	122.5	150	447	250	212	147	814	185	1211	320	310
GSOJ50V	250	179	250	200	285	340	224	190	135	170	485	280	236	168	903	195	1334	355	430
GSOJ63V	300	179	280	224	320	405	250	212	154	190	550	315	265	168	978	235	1478	400	600
GSOJ80V	400	201.5	315	250	358	460	280	236	179.5	212	598	355	300	196	1098	265	1663	450	860

•Dimensional Tolerance are, A=+0/-1%, D=+2/-0% and all others are +7/-2%

Imperial Dimensions

Model Number	Capacity Short Tons	A	a1	a2	a3	B	b1	b2	D	d1	e3	h1	h2	b	L1	L2	L	y	Weight Lbs.
GSOJ8T	33	2.05	3.94	3.15	4.45	4.53	3.55	2.96	1.69	2.64	8.27	4.41	3.74	2.01	16.51	2.56	22.42	5.71	37
GSOJ8V	44	2.56	3.94	3.15	4.45	5.52	3.55	2.96	2.07	2.64	8.27	4.41	3.74	2.32	16.82	3.15	23.72	5.71	53
GSOJ12V	61	2.94	4.93	3.94	5.63	6.11	4.41	3.74	2.38	3.35	9.93	5.52	4.65	2.68	19.07	3.55	27.26	7.09	121
GSOJ20V	94	3.80	6.30	4.93	7.09	7.68	5.52	4.65	2.88	4.18	13.00	7.09	5.91	3.15	23.05	4.41	33.37	8.87	247
GSOJ25V	132	4.69	7.09	5.52	7.96	9.26	6.30	5.20	3.39	4.65	14.18	7.88	6.70	3.59	25.45	5.32	37.47	10.05	353
GSOJ32V	165	4.93	7.88	6.30	8.87	9.85	7.09	5.91	3.88	5.20	15.76	8.83	7.49	4.41	28.45	5.71	41.65	11.43	485
GSOJ40T	193	5.18	8.83	7.09	9.93	11.23	7.88	6.70	4.31	5.91	17.61	9.85	8.35	4.96	31.24	6.50	46.10	12.61	683
GSOJ40V	220	5.69	8.83	7.09	9.93	12.61	7.88	6.70	4.83	5.91	17.61	9.85	8.35	5.79	32.07	7.29	47.71	12.61	683
GSOJ50V	276	7.05	9.85	7.88	11.23	13.40	8.83	7.49	5.32	6.70	19.11	11.03	9.30	6.62	35.58	7.68	52.56	13.99	948
GSOJ63V	331	7.05	11.03	8.83	12.61	15.96	9.85	8.35	6.07	7.49	21.67	12.41	10.44	6.62	38.53	9.26	58.23	15.76	1323
GSOJ80V	441	7.94	12.41	9.85	14.11	18.12	11.03	9.30	7.07	8.35	23.56	13.99	11.82	7.72	43.26	10.44	65.52	17.73	1896

•Dimensional Tolerance are, A=+0/-1%, D=+2/-0% and all others are +7/-2%

VAN BEEST

GREEN PIN® PRODUCT CATALOGUE

IMPERIAL

Green Pin®

LIFTING SLING FITTINGS FOR WIRE ROPE

HOOKS

Applications

Hooks are used in lifting systems as a connection between the load to be lifted and the wire rope slings.

Design

There are different types of hooks with specific designs to suit various purposes.

Hooks attachment options to sling:
- Hook with an eye at the top
- Hook with a swivel eye at the top

Hook designs:
- Self-locking hook
- Sling hook
- Sliding Choker Hook

Hooks are generally marked with:

• Working Load Limit (specific products)	- e.g. 5.4 t
• manufacturer's symbol	- e.g. GP
• traceability code	- e.g. H-AB or HA
• steel grade	- e.g. 4 or 8
• size in mm and/or inch (specific products)	- e.g. 13 and/or $^{1}/_{2}$"
• item code (specific products)	- e.g. CSO
• origin (specific products)	- e.g. France

Examples of do's and don'ts

A hook cannot be used as a clamp.

Don't use hooks without latch, as an exception is the foundry hook.

Hook body and latch may not be bent. This is a sign the hook is overloaded.

Instructions for use

Wire rope hooks, should be inspected before use to ensure that:

- all markings are legible.
- a hook with the correct WLL has been selected with respect to the sling design.
- hooks are free from nicks, gouges, and cracks.
- hooks are not heat treated, modified, repaired, or reshaped by machining or bended. (This may affect their Working Load Limit).
- hooks are not distorted or unduly worn. (Maximum allowable wear is 10 % of the original diameter)
- the latch of the hook(s) is present and functional.
- the hook is never side-, tip- or back- loaded.
- the hook is supporting the load correctly.
- the latch should not be supporting any load.
- only use the hook for in-line lifting.

Hooks must be regularly inspected in accordance with the safety standards given in the country of use. This is required because the products in use may be affected by issues such as wear, misuse and overloading, which may lead to deformation and alteration of the material structure. Inspection should take place at least every six months (follow the local rules in the country of use), and more frequently when the links are used in severe operating conditions.

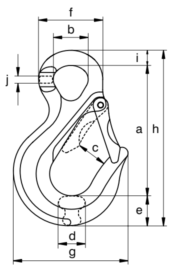
Green Pin® Sling Hook E EN 1677-2 GR8

Grade 80 eye sling hook EN 1677-2

Scan for additional product details

- **Material:** alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 4 x WLL
- **Standard:** EN1677-2 and ASTM A952/A952M
- **Finish:** painted yellow (J), red (R) or white
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 * MTC [a] MPI [b] * DGUV * CE IIB
- **Article code:** scan QR code to see article codes
- **Note:** from 8.2 t without flat part. For use of this component in a wire rope sling the safety factor must be 5 : 1, see the table on page 84

CSO



for chain diameter	working load limit	length	diameter inside eye	width opening	thickness	width	diameter eye outside	width outside	length outside	width	thickness	weight each
	metric	a	b	c	d	e	f	g	h	i	j	
inch	t	inch	inch	inch	inch	inch	inch	inch	inch	inch	inch	lbs
3/16 - 7/32	1.12	3 3/8	29/32	1 1/32	19/32	3/4	1 11/16	2 27/32	4 1/2	13/32	1/4	0.62
1/4 - 5/16	2	4 1/16	1 1/32	1 3/16	25/32	15/16	2	3 7/16	5 15/32	15/32	5/16	1.23
3/8	3.2	5 3/32	1 13/32	1 5/16	15/16	1 3/32	2 9/16	4 3/16	6 23/32	19/32	13/32	2.38
1/2	5.4	5 31/32	1 5/8	1 15/32	1 1/4	1 17/32	3 1/32	5 1/4	8 7/32	23/32	15/32	4.19
5/8	8.2	7 17/32	2 1/16	1 11/16	1 9/16	1 11/16	3 11/16	6 1/2	10 1/32	13/16	21/32	7.72
3/4	12.8	9 11/32	2 13/32	2 13/32	1 15/16	2 13/32	4 17/32	8 3/16	12 27/32	1 3/32	13/16	15.52
7/8	15.5	11 1/32	2 27/32	2 15/16	2 1/8	2 17/32	5 3/16	9 17/32	14 3/4	1 3/16	29/32	21.9
1 **	21.6	9 7/8	1 7/8	2 7/8	2 5/32	3 3/8	4 3/8	9 9/32	14 9/32	1 1/4	1 1/4	31.1
1 1/4 **	32.8	11 1/2	2 9/32	3 7/16	2 5/8	4 3/32	5 15/32	11 1/4	17 1/8	1 17/32	1 17/32	47.8

* Excluding sizes 1 inch and 1 1/4 inch
** Different design

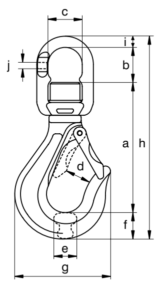
INFO CAD

Green Pin® Sling Hook SE EN 1677-2 GR8

Grade 80 swivel sling hook EN 1677-2

- **Material:** alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 4 x WLL
- **Standard:** EN1677-2 and ASTM A952/A952M
- **Finish:** painted yellow (J), red (R) or white
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC [a] MPI [b] DGUV CE IIB
- **Article code:** scan QR code to see article codes
- **Note:** equipped with thrust needle roller bearings to enable rotation under load and need to be used with rotating resistant wires. For use of this component in a wire rope sling the safety factor must be 5 : 1, see the table on page 84

CSE



for chain diameter	working load limit	length	length inside	width inside	width opening	thickness	width	width outside	length outside	diameter	thickness	weight each
	metric	a	b	c	d	e	f	g	h	i	j	
inch	t	inch	inch	inch	inch	inch	inch	inch	inch	inch	inch	lbs
3/16 - 7/32	1.12	3 15/16	1 5/16	1 1/4	1 1/32	19/32	25/32	2 27/32	6 15/32	15/32	1/4	1.19
1/4 - 5/16	2	4 31/32	1 9/16	1 15/32	1 3/16	25/32	31/32	3 7/16	8 1/32	9/16	5/16	2.2
3/8	3.2	6 7/32	1 27/32	1 7/8	1 5/16	15/16	1 5/32	4 3/16	9 27/32	5/8	7/16	4.19
1/2	5.4	7 7/16	2 5/16	2 9/32	1 15/32	1 1/4	1 17/32	5 1/4	12 1/8	13/16	9/16	7.5
5/8	8.2	8 11/16	2 5/8	2 7/8	1 23/32	1 9/16	1 23/32	6 1/2	14 1/32	31/32	21/32	14
3/4	12.8	10 11/32	3 7/16	3 7/32	2 13/32	1 15/16	2 7/16	8 7/32	17 7/32	31/32	13/16	22.9

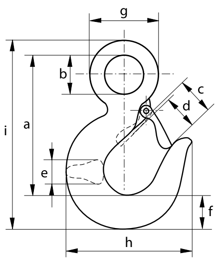
INFO CAD

Green Pin® Hook E GR8

Grade 80 large eye hook with safety latch

Scan for additional product details

P-6714A



- **Material:** alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 4 x WLL
- **Standard:** generally to EN 1677-2
- **Finish:** painted red
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 CE IIA
- **Article code:** scan QR code to see article codes
- **Note:** for use of this component in a wire rope sling the safety factor must be 5 : 1, see the table on page 84

working load limit	length	diameter eye inside	opening hook	opening hook	thickness	width	diameter eye outside	width outside	length outside	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	lbs
1.25	3 3/16	3/4	15/16	25/32	5/8	3/4	1 15/32	2 3/4	4 9/32	0.66
1.6	3 19/32	29/32	1 1/32	7/8	23/32	7/8	1 13/16	3 1/8	4 7/8	0.97
2.5	4 1/8	1 1/16	1 1/4	1 1/16	25/32	1 1/16	2 1/16	3 21/32	5 23/32	1.39
3.2	4 7/8	1 1/4	1 11/32	1 3/16	15/16	1 3/16	2 7/16	4 9/32	6 21/32	2.38
5.4	5 25/32	1 17/32	1 23/32	1 15/32	1 7/32	1 13/32	2 29/32	5	7 29/32	3.44
8.2	7 15/32	1 31/32	2 5/32	1 7/8	1 15/32	1 7/8	3 25/32	6 5/8	10 9/32	8.71
12.8	9 1/16	2 17/32	2 17/32	2 3/32	1 7/8	2 13/32	4 31/32	7 25/32	12 11/16	17.09
16	10	2 25/32	2 3/4	2 3/8	2 9/32	2 23/32	5 15/32	8 25/32	14 1/16	23.8
22	12 7/16	3 1/2	3 19/32	3 1/32	2 15/32	3 3/16	6 21/32	11 5/32	17 7/32	37.3

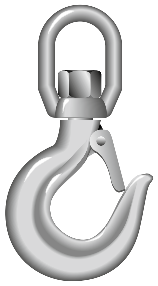
CAD

Green Pin

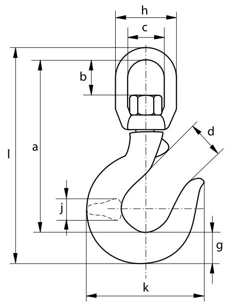
Green Pin® Hook SE GR8

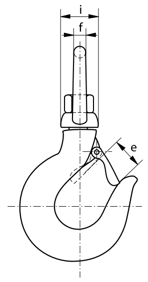
Grade 80 swivel eye hook with safety latch

Scan for additional product details



P-6703A





- **Material:** alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 4 x WLL
- **Finish:** painted red
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 CE IIA
- **Article code:** scan QR code to see article codes
- **Note:** for use of this component in a wire rope sling the safety factor must be 5 : 1, see the table on page 84

working load limit	length	length inside	width inside	opening hook	opening hook	diameter	width	width	width	thickness	width outside	length outside	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l inch	lbs
1.25	4 21/32	1 3/32	1 7/32	15/16	25/32	7/16	3/4	2 3/32	1 3/16	5/8	2 3/4	5 25/32	1.08
1.6	5 23/32	1 3/8	1 9/16	1 1/32	7/8	9/16	7/8	2 11/16	1 15/32	23/32	3 1/8	7 1/8	2.09
2.5	6 9/16	1 11/16	1 27/32	1 1/4	1 1/16	21/32	1 1/16	3 3/16	1 11/16	25/32	3 21/32	8 9/32	3.26
3.2	7 3/32	1 27/32	1 27/32	1 11/32	1 3/16	21/32	1 7/32	3 3/16	1 11/16	15/16	4 9/32	8 31/32	3.92
5.4	8 17/32	2 1/8	2 17/32	1 23/32	1 15/32	13/16	1 13/32	4 3/16	2 17/32	1 7/32	5	10 13/16	8.16
8.2	10 13/16	2 23/32	3 1/16	2 5/32	1 25/32	1 1/32	1 7/8	5 1/8	3 1/32	1 15/32	6 5/8	13 3/4	16.23
11.5	12 7/32	2 11/16	3 7/32	2 9/32	2 3/32	7/8	2 3/8	5 29/32	3 7/32	1 11/16	7 25/32	15 21/32	21.4
16	13 27/32	3 5/16	3 5/8	2 19/32	2 9/32	15/16	2 5/8	6 1/16	3 5/8	2 1/16	8 3/4	17 25/32	32.8
22	17 3/32	4 7/32	4 17/32	3 7/16	3 1/16	1 5/32	3 5/32	7 17/32	4 1/4	2 17/32	11 5/32	21 23/32	57
31.5	20 5/32	4 19/32	5 3/16	3 13/16	3 7/16	1 11/32	3 11/16	8 3/4	5 3/16	3 5/32	13 11/32	25 25/32	101

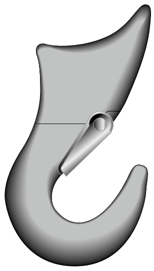
CAD

Green Pin® Alloy Sliding Choker Hook

Grade 80 sliding choker hook with safety latch

Scan for additional product details

- **Material:** alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Finish:** painted red
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 CE IIA
- **Article code:** scan QR code to see article codes



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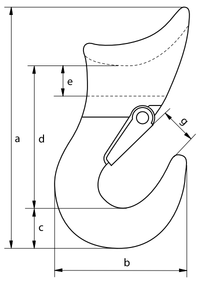
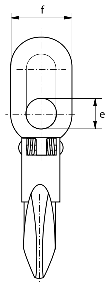


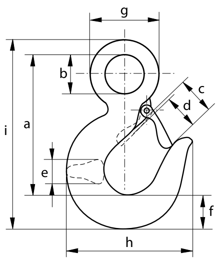


working load limit	diameter rope	length	width	thickness	length	diameter	thickness	opening hook	weight each
metric t	inch	a inch	b inch	c inch	d inch	e inch	f inch	g inch	lbs
0.8	1/4 - 7/16	4 13/32	2 15/32	3/4	2 9/16	9/16	1 3/16	5/8	0.88
1.6	3/8 - 1/2	5 5/8	3 7/32	1 1/32	3 9/32	21/32	1 3/16	23/32	1.68
2.5	9/16 - 5/8	6 11/16	3 27/32	1 3/16	3 13/16	3/4	1 5/16	15/16	2.82
3.2	5/8 - 7/8	7 23/32	4 17/32	1 13/32	4 11/32	7/8	1 9/16	1 1/16	4.21
5.4	7/8 - 1	10 1/4	5 19/32	1 13/16	5 23/32	1 13/32	2 3/8	1 3/8	9.26

CAD

C

P-6714C



Eye hook

Grade 40 large eye hook with safety latch

- **Material:** carbon steel, grade 40
- **Safety factor:** MBL equals 5 x WLL
- **Finish:** painted green
- **Certification:** 2.1 2.2 3.1 CE IIA

working load limit	length	diameter eye inside	opening hook	opening hook	thickness	width	diameter eye outside	width outside	length outside	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	lbs
0.8	3 3/16	3/4	15/16	25/32	5/8	3/4	1 15/32	2 3/4	4 9/32	0.64
1	3 19/32	29/32	1 1/32	7/8	23/32	7/8	1 13/16	3 1/8	4 7/8	0.97
1.6	4 1/8	1 1/16	1 1/4	1 1/16	25/32	1 1/16	2 1/16	3 21/32	5 23/32	1.41
2	4 7/8	1 1/4	1 11/32	1 3/16	15/16	1 3/16	2 7/16	4 9/32	6 21/32	2.38
3.2	5 25/32	1 17/32	1 23/32	1 15/32	1 7/32	1 13/32	2 29/32	5	7 29/32	3.99
5	7 15/32	1 31/32	2 5/32	1 7/8	1 15/32	1 7/8	3 25/32	6 5/8	10 9/32	8.27

CAD

CHAIN & RIGGING ATTACHMENTS

SHACKLES

CHAIN & SYNTHETIC SLING COMPONENTS

RIGGING HARDWARE

CLAMPS

TRANSPORTATION CHAIN & ATTACHMENTS

SPECIALTY PRODUCTS

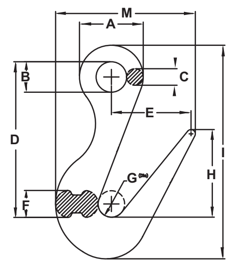
COLUMBUS McKINNON CORPORATION

SORTING HOOK HERC-ALLOY 800®

WORKING LOAD LIMIT: 7-1/2 TONS

BENEFITS & FEATURES

- Quenched and tempered alloy steel
- Long tapered point designed for easy grab in rings, pear links, eye bolts or lifting holes
- Most useful for efficient handling of cylindrical shapes
- Durable orange powder coated finish
- Design factor 5:1
- Meets ASME B30.10
- May be loaded within 1 in. of the tip. The working load limit with the load sitting in the bowl (or saddle of the the hook is 7-1/2 tons, whereas the working load limit is 2 tons if the hook is being loaded 1 in. from the tip. **DO NOT LOAD THE LAST ONE INCH OF THE TIP.**



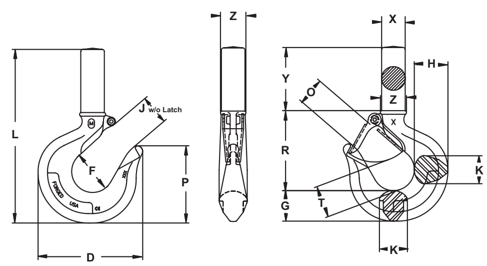
Working Load Limit (lbs.)		With Handle		Without Handle		Dimensions (in.)									
At Tip (ton)	At bottom of Hook (ton)	Product Code	Weight (lbs.)	Product Code	Weight (lbs.)	A	B	C	D	E	F	G	H	I	M
2	7-1/2	M129H	7.0	M129	6.8	3	1.44	0.78	7.34	3.75	1.28	1.25	3.93	10.09	6.58

SHANK HOOK HERC-ALLOY 800®

WORKING LOAD LIMIT: 1 TO 7 TONS

BENEFITS & FEATURES

- Heat treated alloy steel provides strength without bulk or weight
- Pre-drilled boss allows for latch
- Shank hooks are supplied unthreaded, but can be supplied threaded as a special order item
- Shank hooks made from other material (bronze, stainless steel, etc.) available upon request
- Design factor 4:1
- Meets ASME B30.10



Alloy Shank Hook		Weight (lbs.)	Latch Kit Product Code	Max Shank Diameter (in.)	Latch Hole (in.)	Dimensions (in.)												
Working Load Limit* (ton)	Product Code					D	F	G	H	J	K	L	O	P	R	X	Y	Z
1	M1302AV	0.65	4X1302	0.72	0.14	3.09	1.25	0.87	1.01	0.94	0.63	5.44	0.88	2.13	2.35	0.68	2.22	0.69
1-1/2	M1303AV	0.80	4X1303	0.79	0.19	3.37	1.38	0.94	1.11	0.97	0.71	5.94	0.91	2.27	2.59	0.76	2.41	0.80
2	M1304AV	1.47	4X1304	0.86	0.18	3.75	1.50	1.04	1.21	1.06	0.88	6.45	1.00	2.58	2.75	0.82	2.66	1.00
3	M1305AV	1.85	4X1305	1.11	0.18	4.23	1.63	1.25	1.43	1.21	0.94	7.42	1.16	2.84	3.16	0.99	3.01	1.03
5	M1307AV	4.04	4X1307	1.30	0.20	5.16	2.00	1.44	1.63	1.51	1.38	8.80	1.41	3.52	3.85	1.27	3.52	1.30
7	M1309AV	7.25	4X1309	1.56	0.20	6.28	2.50	1.82	2.01	1.76	1.68	10.72	1.69	4.63	4.70	1.53	4.01	1.56

* Working Load Limits in Metric Tons.

⚠ WARNING

The following should be observed:

- Shanks are not intended for internal threading or swaging.
- To obtain maximum strength threads should be class 1 or 2.
- Thread engagement in nut or object must be a minimum of 1-1/2 times the thread diameter. Insufficient thread engagement can result in loss of load.

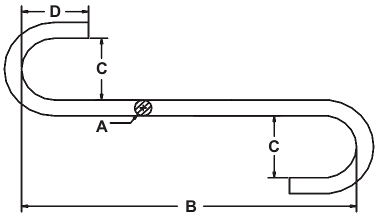
ALLOY "S" HOOK HERC-ALLOY 800®

MADE IN THE USA

WORKING LOAD LIMIT: 210 TO 6,250 LBS.

BENEFITS & FEATURES

- Made from alloy material
- Proof tested at two times WLL
- Durable orange powder coated finish
- WLL embossed on hooks
- Custom sizes available upon request
- Serialized upon request
- Design factor 4:1



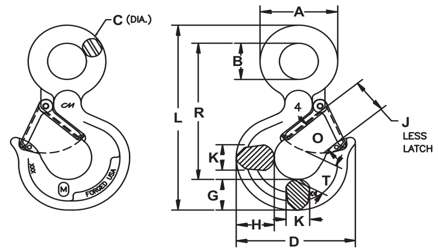
Size (in.)	Working Load Limit (lbs.)	Product Code	Dimensions (in.)				Weight (lbs.)
			A	B	C	D	
9/32	210	562228	0.28	4.50	1.13	1.13	0.15
3/8	410	562237	0.38	6.00	1.50	1.50	0.35
1/2	870	562250	0.56	7.50	2.00	2.00	1.04
5/8	1,120	562262	0.63	9.00	2.50	2.50	1.66
3/4	1,730	562275	0.75	10.50	3.00	3.00	2.60
7/8	2,370	562287	0.88	12.00	3.50	3.50	4.20
1	2,920	562300	1.00	13.00	4.00	4.00	6.00
1-5/32	3,150	562310	1.13	15.00	4.50	4.50	9.30
1-1/4	4,450	562325	1.25	16.00	5.00	5.00	11.70
1-3/8	6,100	562337	1.38	17.00	5.50	5.50	15.40
1-1/2	6,250	562350	1.50	18.00	6.00	6.00	19.50

RIGGING HOOK

WORKING LOAD LIMIT: 3/4 TO 22 TONS

BENEFITS & FEATURES

- Load rating marked on each hook body
- Design factor 5:1
- Hook tips are pre-drilled to allow securement of the latch
- Powder coated orange
- Meets ASME B30.10
- Pre-drilled for latches
- Hooks furnished with or without latches
- Hooks are heat treated, quenched and tempered



Standard Package	Carbon Painted Orange WLL* (tons)	Carbon Product Code without Latch	Carbon Product Code with Latch	Alloy Painted Orange WLL* (tons)	Alloy Product Code without Latch	Alloy Product Code with Latch	Zinc Plated LatchKit (optional) Product Code	A	B	C	D	G	H	J	K	L	O	R	T	Weight (lbs.)
10	3/4	M6402C	M6502C	1	M6402A	M6502A	4X1302	1.50	0.75	0.38	3.12	0.87	1.01	0.93	0.63	4.37	0.93	3.13	0.87	0.66
10	1	M6403C	M6503C	1-1/2	M6403A	M6503A	4X1303	1.75	0.88	0.44	3.37	0.94	1.11	0.97	0.71	5.04	0.97	3.66	0.97	1.12
5	1-1/2	M6404C	M6504C	2	M6404A	M6504A	4X1304	2.13	1.10	0.50	3.80	1.06	1.21	1.02	0.74	5.65	1.02	4.09	1.03	1.46
5	2	M6405CB	M6505CB	3	M6405A	M6505A	4X1305	2.50	1.25	0.64	4.20	1.26	1.43	1.19	0.94	6.55	1.16	4.67	1.16	2.42
5	3	M6407C	M6507C	5	M6407A	M6507A	4X1307	3.08	1.56	0.77	5.11	1.44	1.63	1.50	1.38	7.97	1.41	5.78	1.53	4.10
5	5	M6409C	M6509C	7	M6409A	M6509A	4X1309	3.88	1.98	0.94	6.24	1.82	2.01	1.78	1.68	10.07	1.69	7.31	1.94	8.16
–	7-1/2	–	–	11	M6411A	M6511A	4X1311	4.69	2.44	1.13	7.89	2.25	2.63	2.38	1.88	12.41	2.19	9.03	2.52	15.60
–	10	–	–	15	M6415A	M6515A	4X1315	5.34	2.84	1.25	8.53	2.75	3.10	2.50	2.03	14.05	2.30	10.21	2.54	21.58

Other finishes are available as special orders.

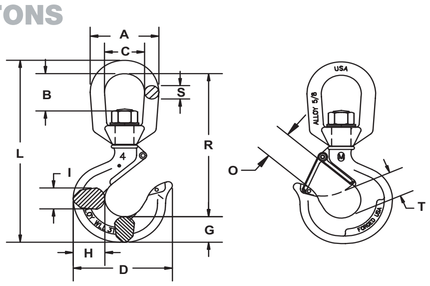
* Working Load Limits in Metric Tons.

SWIVEL RIGGING HOOK

WORKING LOAD LIMIT: 1 TO 22 TONS

BENEFITS & FEATURES

- Design factor 5:1
- Pre-drilled for latches
- Hooks furnished with or without latches
- Powder coated orange
- Hooks are heat treated, quenched and tempered
- Load rating marked on each hook body
- Hook tips are pre-drilled to allow securement of the latch
- Meets ASME B30.10



Standard Package	Alloy WLL* (tons)	Alloy Product Code without Latch	Alloy Product Code with Latch	Zinc Plated Latch Kit (optional) Product Code	A	B	C	D	G	H	I	L	R	S	T	O	Latch Kit	Weight (lbs.)
10	1	M3402A	M3502A	4X1302	2.00	1.11	1.31	3.06	0.87	1.05	0.63	5.83	4.63	0.38	0.87	0.93	4X1302	1.05
10	1-1/2	M3403A	M3503A	4X1303	2.50	1.38	1.50	3.33	0.94	1.11	0.71	6.83	5.44	0.50	0.97	0.97	4X1303	1.56
10	2	M3404A	M3504A	4X1304	3.00	1.65	1.75	3.67	1.06	1.21	0.88	7.76	6.25	0.63	1.03	1.06	4X1304	2.50
10	3	M3405A	M3505A	4X1305	3.00	1.65	1.75	4.20	1.27	1.43	0.94	8.40	6.49	0.63	1.16	1.16	4X1305	3.20
10	5	M3407A	M3507A	4X1307	3.50	1.77	2.00	5.11	1.44	1.63	1.31	9.76	7.53	0.75	1.53	1.41	4X1307	5.36
Bulk	7	M3409A	M3509A	4X1309	4.75	2.39	2.75	6.24	1.82	2.01	1.68	12.42	9.67	1.00	1.94	1.69	4X1309	10.56
Bulk	11	M3411A	M3511A	4X1311	5.50	2.55	3.25	7.69	2.25	2.63	1.88	14.89	12.06	1.13	2.46	2.22	4X1311	19.00
Bulk	15	M3415A	M3515A	4X1315	6.00	2.47	3.50	8.37	2.59	2.94	2.19	15.79	11.95	1.25	2.62	2.23	4X1315	26.75
Bulk	22	M3422A	M3522A	4X1322	7.75	3.82	4.75	10.19	3.00	3.50	2.69	21.18	16.68	1.50	2.74	3.05	4X1322	51.80

* Working Load Limits in Metric Tons

REPLACEMENT LATCHES FOR SWIVEL, RIGGING & SHANK HOOKS

Working Load Limit (tons)		Product Code			
		Old Style	New Style		
Alloy	Carbon	Stainless	Zinc	Stainless	Locking
1	3/4	4X405	4X1302	4X1302S	4X1302M
1-1/2	1	4X405	4X1303	4X1303S	4X1303M
2	1-1/2	4X406	4X1304	4X1304S	4X1304M
3	2	4X406	4X1305	4X1305S	4X1305M
5	3	–	4X1307	4X1307S	4X1307M
7	5	4X412	4X1309	4X1309S	4X1309M
11	7-1/2	4X414	4X1311	4X1311S	4X1311M
15	10	595530	4X1315	–	4X1315M
22	15	595533	4X1322	–	4X1322M


Old Style

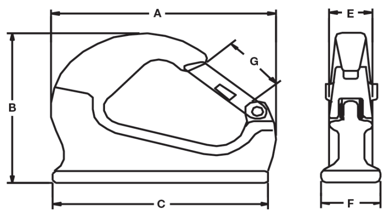

New Style

YALE® IMPORT WELD-ON LIFTING HOOK

WORKING LOAD LIMIT: 2,205 TO 17,500 LBS.

BENEFITS & FEATURES

- Heavy steel construction with a durable finish
- Designed for use with excavating equipment – can be welded directly onto the bucket
- Latch prevents chain from accidentally unhooking
- Design factor 4:1



Product Code	Working Load Limit (lbs.)	Dimensions (in.)						Weight (lbs.)	Replacement Latch Product Code
		A	B	C	E	F	G		
48211	2,205	4.63	4.13	2.78	1.00	1.25	0.84	0.88	4X48211
48212	6,615	5.81	4.72	3.74	1.25	1.30	0.98	2.88	4X48212
48213	11,025	7.97	6.45	5.12	1.16	1.72	1.79	5.31	4X48213
48214	17,500	7.61	5.43	6.69	1.58	1.97	1.85	8.50	4X48214

Imported

RG4761_Rigging Catalog Final.indd 101 7/2/25 12:02 PM

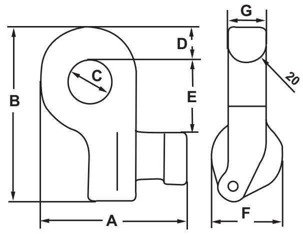
CLB CONTAINER LIFTING LUGS

WORKING LOAD LIMIT: 88,100 LBS. (PER SET OF 4)

Supplied in sets of 4, CLB Lifting Lugs serve as flexible lashing points for transporting containers from the sides.

BENEFITS & FEATURES

- Improved robust design ensures a secure hold and allows for quick and easy replacement of the locking pin in the field
- Mounted horizontally at the side of the container in either upper or lower holes
- Easy installation and removal – simply insert and turn to install
- Designed to eliminate the dangerous use of standa hooks
- Lugs cannot drop out when slings become slack
- The set of (4) lugs has (2) right-hand and (2) left-hand units.
- For maximum capacity, use a lifting beam in conjunction with CLB lifting lugs
- Design factor 4:1
- Mounted vertically at the top
- Designed to meet U.S. military specifications



Product Code	Working Load Limit (lbs. per set of 4)	Dimensions							Weight (lbs.)
		A	B	C	D	E	F	G	
CLB40	88,100	5.98	7.13	1.77	1.46	2.87	2.95	1.58	39.7

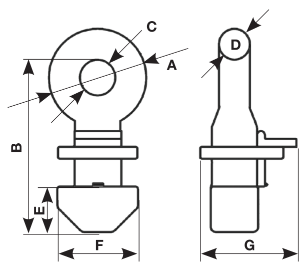
CLT CONTAINER LIFTING LUGS

WORKING LOAD LIMIT: 123,480 LBS. (PER SET OF 4)

Supplied in sets of 4, CLT Lifting Lugs serve as flexible lashing points for transporting containers from the top.

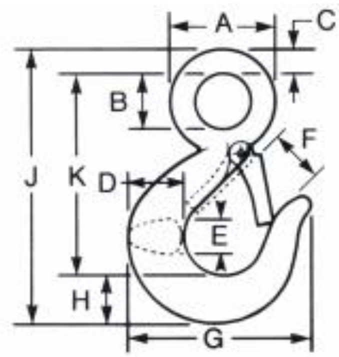
BENEFITS & FEATURES

- Mounted vertically at the top of the container
- Easy installation and removal – simply insert and turn to install
- Designed to eliminate the dangerous use of standard hooks
- Lugs lock into place by simply turning the lug 90°. This configuration allows for transportation via the use of a lifting frame in conjunction with cables, chains or slings.
- Design factor 4:1



Product Code	Working Load Limit at 90° (lbs. per set of 4)	Dimensions (in.)							Weight (lbs.)
		A	B	C	D	E	F	G	
CLT56	123,480	4.84	8.54	1.77	1.54	2.24	3.98	4.76	61.7

EYE HOOKS

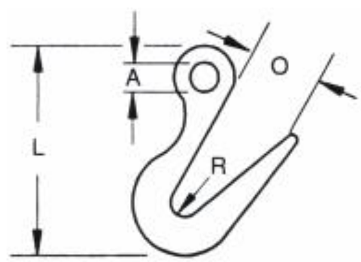


EYE HOOKS

Size (WLL)*		Dimensions in inches										Approximate Weight Each with Latch in Pounds
Carbon	**Alloy**	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**J**	**K**	
3/4 Ton**	**1 Ton**	1.50	0.75	0.38	0.88	0.63	0.94	2.88	0.75	4.38	3.25	0.60
1**	**1.1/2**	1.75	0.88	0.44	1.00	0.69	1.06	3.13	0.81	4.88	3.63	0.82
1.1/2**	**2**	2.00	1.13	0.44	1.19	0.81	1.06	3.50	1.00	5.50	4.13	1.44
2	**3**	2.38	1.25	0.59	1.38	0.94	1.22	3.94	1.19	6.31	4.56	1.94
3	**5**	3.00	1.56	0.69	1.63	1.19	1.50	5.00	1.50	7.94	5.75	3.94
5	**7**	3.81	2.00	0.88	2.06	1.50	1.88	6.25	1.75	10.00	7.38	7.75
7.1/2	**11**	4.69	2.44	1.13	2.63	1.63	2.25	7.56	2.25	12.44	9.06	15.40
—	**15**	5.37	2.84	1.26	2.94	2.19	2.51	8.30	2.59	13.93	10.07	22.20
—	**22**	6.64	3.50	1.58	3.50	2.69	3.30	10.30	3.00	17.06	12.50	37.60

Carbon or Alloy steel, forged, heat-treated, painted, with latch installed or without.
Some sizes also available in DOMESTIC.
* Working Load Limit applies only when the load is applied to the center of the saddle of the hook.
** 3/4, 1, 1.1/2, and 2 Ton Carbon Steel Hooks also available in Hot Galvanized.

SORTING HOOKS

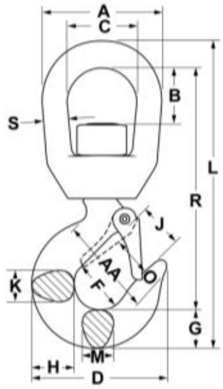


SORTING HOOKS

Working Load Limit 2.1/2" from tip		2 Tons
Working Load Limit, at bottom		7.1/2 Tons
Approximate Weight Each		7.75 Pounds
Overall Length in Inches	**L**	9.69
I.D. of Eye in Inches	**A**	1.38
Opening at top of Hook in Inches	**O**	2.81
Radius at Bottom of Hook in Inches	**R**	0.63

Forged alloy steel, heat treated, painted.

ALLOY SWIVEL HOOKS

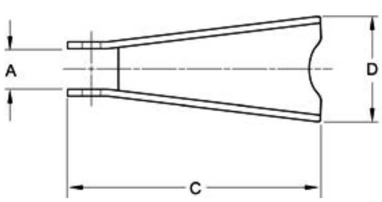


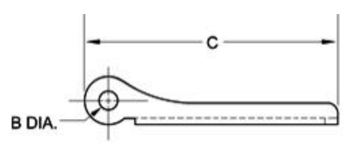
ALLOY SWIVEL HOOKS

Size (WLL) (t)*	Dimensions in Inches															Approximate Weight Each with Latch in Pounds
	A	B	C	D	F	G	H	J	K	L	M	O	R	S	AA	
1	2.00	0.82	1.25	2.86	1.25	0.73	0.81	0.93	0.63	2.66	0.63	0.89	4.55	0.38	1.50	0.75
2	3.00	1.50	1.75	3.59	1.50	1.00	1.16	1.06	0.88	7.75	0.88	1.00	6.12	0.63	2.00	2.25
3	3.00	1.50	1.75	4.00	1.62	1.13	1.31	1.19	0.94	8.25	0.94	1.09	6.50	0.63	2.00	2.30
5	3.50	1.64	2.00	4.84	2.00	1.44	1.63	1.50	1.31	9.69	1.13	1.36	7.50	0.75	2.50	4.96
7	4.56	2.29	2.50	6.28	2.50	1.81	2.06	1.78	1.66	12.47	1.44	1.61	9.63	1.00	3.00	10.29
11	5.00	2.53	2.75	7.54	3.00	2.25	2.63	2.41	1.88	14.75	1.63	2.08	11.37	1.13	4.00	19.40
15	5.62	2.48	3.12	8.34	3.25	2.59	2.94	2.62	2.19	16.40	1.94	2.27	12.25	1.25	4.00	23.25

* Working Load Limit applies only when the load is applied to the center of the saddle of the hook.
Also available in domestic.

LATCH KITS





STAINLESS LATCH KITS

Hook Size		Approximate Weight Each in Pounds	Dimensions (in)			
Carbon	Alloy		A	B	C	D
3/4 Ton	**1 Ton**	0.02	0.45	0.14	1.50	0.62
1	**1.1/2**	0.03	0.42	0.14	1.85	0.72
1.1/2	**2**	0.03	0.46	0.18	1.85	0.76
2	**3**	0.03	0.48	0.18	2.04	0.84
3	**5**	0.06	0.64	0.17	2.40	1.02
5	**7**	0.11	0.74	0.19	2.96	1.47
7.1/2	**11**	0.15	0.81	0.27	3.66	1.59
10	**15**	0.17	0.68	0.26	3.67	1.47
—	**22**	0.34	0.85	0.39	5.09	1.89

WARNING: NEVER EXCEED WORKING LOAD LIMIT
Read important warnings and information on pages 2 - 9 and page 33.